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# Genomics-informed species delimitation to support morphological identification of anglewing butterflies (Lepidoptera: Nymphalidae: *Polygonia*)

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Species delimitation and identification are integral to virtually all biological disciplines, but are far from straightforward tasks. Taxonomy has recently focused on integrative approaches that consider multiple types of data to resolve species boundaries, yet methodologies to that end are still being developed. Here, we assess species limits in an area of wide distributional overlap between several species of anglewing butterflies in western Canada. Focusing on an area of sympatry provides a rich system to test species boundaries in the face of potential gene flow between morphologically variable yet similar species. Mitochondrial DNA and genome-wide single nucleotide polymorphisms provided clear species delimitation, although previously identified cytonuclear discordance was also apparent. Analysis of two morphological data sets, based on characters commonly used as diagnostic characters in the literature and field guides, was variably successful at separating species. Using analyses that were guided by the results of the genetic data increased successful species identification for traditionally used characters, but not for the morphological data set based on digital colour analysis. Our application of genetics to guide morphological analysis demonstrates a useful approach for implementing iterative methodologies as part of integrative taxonomy.

ADDITIONAL KEYWORDS: cytonuclear discordance – genome-wide SNPs – genotyping-by-sequencing – mitochondrial phylogeny – *Polygonia* – species identification.

## INTRODUCTION

Species delimitation and identification are essential to virtually all biological disciplines, but are far from straightforward tasks. Morphological variation (Lumley & Sperling, 2010; McKay *et al.*, 2014; Proshek *et al.*, 2015), gene discordance (Maddison, 1997) and complex evolutionary histories (Dejaco *et al.*, 2016) can obfuscate species limits, creating phylogenetic and taxonomic uncertainty. Such uncertainty is particularly undesirable in systems where incorrect delimitation of units can have large economic (e.g. Frey *et al.*, 2013; Dumas *et al.*, 2015), health-related (Sperling & Sperling, 2009; Müller *et al.*, 2013) or conservation consequences (Bickford *et al.*, 2007; Bortolus, 2008; Hedin, 2015; Proshek *et al.*, 2015). Integrative taxonomy (Dayrat, 2005; Schlick-Steiner *et al.*, 2010), which delimits species using a consensus of multiple data types (morphological, molecular, ecological, chemical, etc.), provides an appealing solution to cases of difficult species delimitation, particularly compared to approaches that use single data types (Dupuis, Roe & Sperling, 2012). It is open to discussion whether integrative taxonomy approaches are strictly analytically integrative and objective (e.g. with analyses that consider multiple data types simultaneously, and allowing results to be recreated) or whether they

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are iterative, considering data sources successively and independently to arrive at a qualitative consensus (Yeates *et al.*, 2011). Regardless, assimilating variable data types into broadly integrative taxonomic consideration has yielded unprecedented resolution in many historically difficult groups (Lumley & Sperling, 2010; Wachter *et al.*, 2015; Dejaco *et al.*, 2016; Gratton *et al.*, 2016).

Genomic data are particularly powerful for resolving species boundaries due to the sheer size of genomic data sets. Reduced-representation genomic libraries generated from, for example, restriction site-associated DNA sequencing methods (RAD-seq), can, in similar timeframes, create data sets that have hundreds of times more phylogenetic information than traditional Sanger sequencing-based approaches (Cruaud *et al.*, 2014). Such RAD-seq data sets have been used to resolve phylogenetic relationships (Wagner *et al.*, 2013; Cruaud *et al.*, 2014; DaCosta & Sorenson, 2016; Dupuis *et al.*, 2017), define species limits (Leaché *et al.*, 2014; Pante *et al.*, 2015; Gratton *et al.*, 2016; Razkin *et al.*, 2016) and elucidate evolutionary history (Jones *et al.*, 2013; Nadeau *et al.*, 2013; Stervander *et al.*, 2015). Although the various iterations of these methods have technical advantages and disadvantages (Andrews *et al.*, 2016), and particular considerations for phylogenetic analysis (DaCosta & Sorenson, 2016; Dupuis *et al.*, 2017), they have become one of the mainstay methodologies for systematic and evolutionary biology studies.

Here, we assess species limits between anglewing butterflies (*Polygonia* Hübner, 1819) in western Canada. Of the 15 species described worldwide (Wahlberg *et al.*, 2009), four are widespread in Alberta and British Columbia: *Polygonia faunus* (Edwards, 1862), *P. gracilis* (Grote & Robinson, 1867), *P. progne* (Cramer, 1775) and *P. satyrus* (Edwards, 1869). Widely variable yet shared morphological characteristics and similar ecologies (Nylin *et al.*, 2015) make identification of these species challenging in the field (Bird *et al.*, 1995; Layberry, Hall & Lafontaine, 1998; Acorn & Sheldon, 2006). Although previous phylogenetic studies have found these species to form monophyletic groups, discordance between mitochondrial and nuclear genes, morphology and ecology make for tenuous hypotheses of evolutionary relationships (Fig. 1; Wahlberg *et al.*, 2009). Taken together, the complex situation across much of Alberta and British Columbia makes this an ideal system for an integrative taxonomic approach, a point highlighted by Wahlberg *et al.* (2009): 'careful study of populations in Alberta seems warranted'.

We use mitochondrial DNA (mtDNA) gene sequences, genome-wide single nucleotide polymorphisms (SNPs) generated by genotyping-by-sequencing (GBS, a form of RAD-seq) and two morphological data sets (visually and digitally scored wing characters based on commonly used field diagnostic characters) to test species

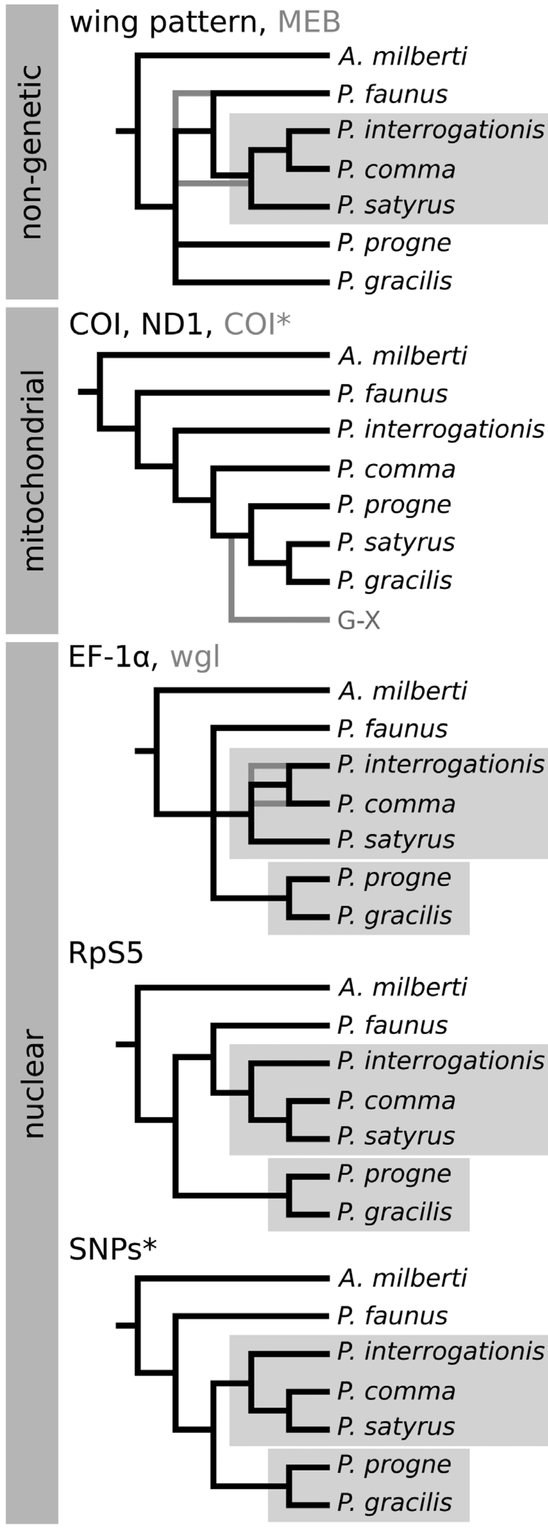



**Figure 1.** Alternate topologies of pertinent New World *Polygonia* species, generated from non-genetic, mitochondrial DNA and nuclear DNA data sets. Some data sets of similar types from different studies have been combined into a single tree, and differences in topologies between

limits between the four widespread species (and two subspecies of *P. gracilis*) in this region. Using phylogenetic and population genetic analyses, we first ask whether genetic data sets clearly separate species or whether there are signs of admixture or introgression between species. We then use multivariate analyses of the morphological data sets to test whether quantitative analysis of morphology leads to similar delimitation of species. Both genetic data sets predict distinct species limits, despite comparable cytonuclear discordance to previous studies, but only one morphological data set conforms to the genetic clusters. Given the clear species limits from genome-wide SNPs and mtDNA, we use the genetic data to inform subsequent analyses of the morphological data sets and, from these genetically guided morphological analyses, determine which morphological characters best delimit species. These results provide an iterative methodological approach to taxonomy, informing more accurate diagnoses of these species in the field. With the popularization of citizen science (Silvertown, 2009; Bonney *et al.*, 2014), particularly in the realm of butterflies (Howard & Davis, 2009; Devictor, Whittaker & Beltrame, 2010; Larrivee *et al.*, 2014), a framework for integrative taxonomy to guide the development of better diagnostic identification can have wide-reaching applications.

## MATERIAL AND METHODS

### Specimen collection, DNA extraction and photography

Specimens were collected primarily as adults with an aerial net or bait trap (Leptraps, baited with banana) and frozen alive at −80 °C. Larvae were also collected and reared to adulthood on host clippings of the same species from which they were collected (Supporting Information, Table S1). We focused sampling in central and southern Alberta, southeast British Columbia and adjacent states, where six species/subspecies are regularly found: *Polygonia faunus*, *P. gracilis gracilis* (Grote and Robinson, 1867), *P. gracilis zephyrus* (Edwards, 1870), *P. oreas* (Edwards, 1869), *P. progne* and *P. satyrus*. Adequate sampling was achieved for five of these lineages, but we were unable to collect *P. oreas*, which is uncommon in the southwest corner of Alberta. Several additional specimens were included to represent other North American lineages: *P. comma* (Harris, 1841) and *P. interrogationis* (Fabricius, 1798); the latter is also occasionally found in Alberta during irruptive years (Layberry *et al.*, 1998). We identified specimens using several field guides (Bird *et al.*, 1995; Brock & Kaufman, 2003; Acorn & Sheldon, 2006) and the consensus opinion of multiple coauthors (CMM, JHA and FAHS), and follow the nomenclature of Pelham (2008). A total of 258 *Polygonia* specimens were collected [one *P. comma*, 106 *P. faunus*, 21 *P. gracilis* (10 *P. g. gracilis* and 11 *P. g. zephyrus*), one *P. interrogationis*, 71 *P. progne* and 58 *P. satyrus*] as well as seven specimens of *Aglais milberti* and *Nymphalis* spp. for use as outgroups. While we delimited subspecies for *P. gracilis*, we did not do so for *P. faunus* due to the lack of empirical evidence for those subspecies (Kodandaramaiah *et al.*, 2012).

Genomic DNA was extracted from legs and the ventral portion of the thorax using DNeasy Blood and Tissue kits (Qiagen). RNase A was used during the extraction (following Qiagen protocol), and the resulting DNA was ethanol precipitated and eluted in millipore water. DNA quality and quantity were assessed using a Nanodrop ND-1000 (Thermo Scientific) and a Qubit 1.0 dsDNA BR assay kit (Invitrogen), respectively. After DNA extraction, specimens were pinned and spread, and missing thoracic tissue was fortified with Archival Quality Neutral pH adhesive (Lineco). Some specimens lacked adequate thoracic tissue for pinning and were stored in glassine envelopes. Images of dorsal and ventral wing surfaces of all specimens were captured using a Canon Powershot A650 IS camera in macro setting, using aperture priority mode at f 22 to optimize depth of focus, and an exposure compensation of +12/3 to optimize the brightness of the images. The camera was manually white-balanced between photography sessions, with the same lighting used throughout the imaging process. The camera was mounted on a copy stand, and we maintained a consistent distance from specimen to camera lens using a modified plastazote foam pedestal. Image files were converted from JPEG to TIFF format in Adobe Photoshop to preserve resolution. All specimen information and photographs are databased in the University of Alberta E. H. Strickland Entomological Museum (accessions UASM370001-UASM370265; Supporting Information, Table S1).

### mtDNA sequencing and analysis

The entire mitochondrial cytochrome oxidase subunit I gene (*COI*) was amplified in two fragments by polymerase chain reaction (PCR). We targeted the first

those data sets are indicated with grey text and branches. Commonly observed monophyletic groups are highlighted with grey boxes when present, and data sets generated in the current study are noted with an asterisk. Wing pattern: Nylin *et al.* (2001); MEB (synthesis of morphology, ecology and behaviour): Nylin *et al.* (2001), Wahlberg & Nylin (2003); cytochrome oxidase subunit I gene (*COI*): Wahlberg & Nylin (2003), Wahlberg *et al.* (2009); ND1: Nylin *et al.* (2001), Wahlberg & Nylin (2003), Wahlberg *et al.* (2009); EF-1α: Wahlberg & Nylin (2003), Wahlberg *et al.* (2009); wgl: Nylin *et al.* (2001), Wahlberg & Nylin (2003), Wahlberg *et al.* (2009); RpS5: Wahlberg *et al.* (2009).

~650 bp using the primers LepF1 (5′-ATT CAA CCA ATC ATA AAG ATA TTG G-3′) and HCO (5′-TGA TTT TTT GGT CAC CCT GAA GTT TA-3′), and the remaining ~800 bp with JerryI (5′-CAA CAT TTA TTT TGA TTT TTT GG-3′) and PatII (5′-AGG TAA TGT ATA TTA GAC GGT ATA ACT-3′). Some samples were not successfully amplified using the latter pair, so JerryI and MilaIV (5′-AAA TTA GGA CAT TTA TTA CC-3′) were used to produce a shorter fragment. PCR cycling conditions consisted of a 5-min denaturation step at 94 °C, 35 cycles of 94 ° C for 30 s, 45 °C for 30 s and 72 °C for 2 min, and a final 5-min elongation at 74 °C. PCR products were cleaned using Exonuclease I and shrimp alkaline phosphatase (New England Biolabs) before dye-termination using BigDye sequencing pre-mix v2.1 (Life Technologies). Fragments were sequenced using the 3730 DNA analyzer (Applied Biosystems) at the Molecular Biology Service Unit at the University of Alberta.

Forward and reverse *COI* reads were merged and manually inspected for sequencing call errors in Geneious v7.0.6 (Kearse *et al.*, 2012). We then used Mesquite v2.75 (Maddison & Maddison, 2008) to align individual genes with the ClustalW algorithm (Thompson, Gibson & Higgins, 2003), before manually concatenating sequencing products into a single *COI* haplotype for every individual. *COI* sequences generated in our study are accessioned on GenBank (KY318516–KY318762; Supporting Information, Table S1). We also incorporated 89 *COI* sequences from five *Polygonia* taxa that were used by Wahlberg *et al.* (2009) (Supporting Information, Table S1). Finally, we manually trimmed the ends of the sequences and ~100 bp at the junction between the two amplified regions, both of which contained high amounts of missing data. This resulted in a final alignment of 1348 bp for 315 individuals.

Parsimony-based haplotype networks were built in TCS (Clement, Posada & Crandall, 2000), using default settings. Gaps were treated as missing data to accommodate fragments that varied slightly in length. The genetic identity of these species was unambiguous (see 'Results'), so haplotype networks were constructed for *P. faunus* and *P. progne* separately, while *P. gracilis* and *P. satyrus* were combined due to their similarity. We conducted maximum likelihood (ML) tree searches in IQ-TREE v1.4.2 (Nguyen *et al.*, 2015) with 1000 replicates each of ultra-fast bootstrapping (Minh, Nguyen & von Haeseler, 2013) and the Shimodaira/Hasegawa approximate likelihood ratio test (SH-aLRT – Guindon *et al.*, 2010). IQ-TREE's model selection procedure using the Bayesian information criterion predicted a transition model (TIM2) + Γ. We conducted Bayesian inference (BI) analysis in ExaBayes v1.5 (Aberer, Kobert & Stamatakis, 2014) with four Metropolis-coupled replicates for one million generations (three heated chains), sampling every 1000 generations, and otherwise default parameters and models. We used Tracer v1.6 (Rambaut *et al.*, 2014) to assess convergence and sampling of parameter values' posterior distributions, ensuring that all effective sample sizes were >200, and also assessed the average SD of split frequencies and potential scale reduction factors across runs, which should approach zero and one, respectively. A consensus tree was built using TreeAnnotator (Drummond *et al.*, 2012) after manually removing the first 25% of trees as burn-in. Both ML and Bayesian analyses of *COI* were conducted on a smaller data set of unique haplotypes only (172 haplotypes, including outgroups). Preliminary trees were viewed in FigTree v1.3.1 (Rambaut & Drummond, 2010).

### GBS library preparation, sequencing and data filtering

GBS library preparation following Poland *et al.* (2012) was conducted at the Institut de Biologie Intégrative et des Systèmes at Université Laval. PstI and MspI restriction enzymes were used with a duplex-specific nuclease treatment (Zhulidov *et al.*, 2004), and complexity reduction (a single C appended to the reverse primer) as described in Sonah *et al.* (2013). Two to eight base pair barcodes were used to identify individuals and four lanes (including individuals not included in the present study) of 96-plex; 100 bp single-end sequencing was performed on an Illumina HiSeq 2000 at McGill University's Génome Québec Innovation Centre.

*Process_radtags* of *Stacks* v1.35 (Catchen *et al.*, 2011) was use to demultiplex and rename raw FASTQ files, allowing no ambiguities in the individual-specific barcodes. As sequencing errors are more probably at the ends of sequencing reads, 5 bp were trimmed from the 3′ end of the reads using an in-house Perl script. We then used *Ustacks*, *Cstacks* and *Sstacks* (Catchen *et al.*, 2011) to assemble *de novo* catalogs and map reads to these catalogs. Default parameters were used for these utilities, except a minimum depth of five reads was required for initial catalog creation, and three mismatches were allowed in matching secondary reads to primary stacks. Finally, *populations* (Catchen *et al.*, 2011) was used to call SNPs with a population map specifying a single population of all individuals; loci had to be present in one population and one individual to be processed, and a minimum stack depth of six was required for individuals at each locus. All *Stacks* utilities and associated filtering were run using Perl wrappers, which also made use of the BioPerl toolkit (Stajich *et al.*, 2002), and these are available at https://github.com/muirheadk/GBS_analysis_pipeline.

Two main types of input files were constructed from *populations* vcf and fasta outputs: one for phylogenetic and the second for population genetic analyses. First, we used vcftools v0.1.12 (Danecek *et al.*, 2011) to calculate a preliminary measure of missing data per individual and removed individuals missing >50% genotype data (calculated after removing loci missing >50% data per individual); these individuals were removed from both phylogenetic and population genetic input files in the following steps. For phylogenetic analyses, we used an in-house Perl script that ingests fasta output from *populations*, removes loci with high missing data (here a threshold of 50% was used, but preliminary analyses showed little difference when using alternative missing data thresholds) and outputs DNA sequence alignment with missing data symbols for missing loci per individual. This alignment contained SNPs as well as their flanking sequences from the variable length loci of the *de novo* catalog construction and resulted in an alignment of 121 513 bp (2572 SNPs) for 248 individuals. Including flanking sequence around SNPs allows the use of more realistic models of DNA evolution, such as those that take into account invariant sites (e.g. Steel & Penny, 2000, and references therein). We also used vcftools and an in-house Perl script to construct a SNP-only alignment with the same set of loci, where SNPs were coded using IUPAC ambiguity codes (e.g. an A/G heterozygote would be coded R). For population genetic analyses, we removed loci that had >20% missing data from vcf output of *populations* using vcftools and used PGDSpider v2.0.1.9 (Lischer & Excoffier, 2012) to convert to structure format. We also removed outgroup specimens, leading to a final data set of 241 individuals and 961 SNPs.

### GBS ANALYSES

ML tree searches of the GBS alignments were conducted identically to analyses of the *COI* alignment. IQ-TREE's model selection procedure selected the transversion model (TVM) + I + Γ and TVM + Γ for the 121 513 bp sequence-based alignment and the 2572 SNP-based alignment, respectively. Bayesian analyses of the GBS alignments were also conducted the same as was done for *COI*, except that we ran the analyses for five million generations and sampled every 2500 generations.

We used two methods to assess population structure in the 961 SNP data sets. First, we estimated the number of genetic clusters ($K$) with STRUCTURE v2.3.4 (Pritchard, Stephens & Donnelly, 2000; Falush, Stephens & Pritchard, 2003). Ten iterations of $K = 1$–10 were run with the admixture model, correlated allele frequencies, and without any prior information about geographic origin or species. A burn-in of 50 000 Markov chain Monte Carlo generations preceded 150 000 generations that were summarized using CLUMPAK (Kopelman *et al.*, 2015). We considered $\mathrm{LnPr}(X\,|\,K)$ (Pritchard *et al.*, 2000) and $\Delta K$ (Evanno, Regnaut & Goudet, 2005) to select the most likely value of $K$ and conducted hierarchical structure analysis (Pritchard & Wen, 2004; Vähä *et al.*, 2007) with the same parameters to address substructure in the main genetic clusters. Second, discriminant analysis of principal components (DAPC – Jombart, Devillard & Balloux, 2010) was implemented the adegenet library (Jombart, 2008) in R v3.3.1 (R Core Team, 2016). We used *find.clusters* to estimate $K$ with default parameters and retaining all principal components (PCs), and cross-validation (*xvaldapc*) to determine the optimal number of PCs to retain in the discriminant analysis (testing 100 maximum PCs and 100 replicates). Hierarchical analysis was also conducted with DAPC.

### MORPHOLOGICAL CHARACTER SCORING

Two main categories of morphological wing characters are used to delimit *Polygonia* species in field guides: those based on discrete or ordinal characteristics of the wings (spots, shapes, etc.) and those based on colour. We assembled a data matrix from both and, given the inherent differences in data type, conducted independent analyses for each category. First, we reviewed the literature and field guides to summarize characters commonly used to diagnose and delimit the five lineages of *Polygonia* considered here (Supporting Information, Table S2). From this list, we selected characters that were considered highly diagnostic, and those with states that were unique to individual species. Characters that were present in similar states in multiple species were excluded (e.g. submarginal green spots on the ventral hindwing are considered highly diagnostic of *P. faunus*, but similar spots, although more yellow-green and slightly smaller, are found in *P. satyrus* and *P. progne*, respectively). We selected and scored ten discrete/ordinal characters, each with between two and five character states (Fig. 2), and refer to these as 'visually scored characters'.

Second, we assembled a set of colour luminance characters with insight from field guides and the literature, using a digital scoring approach developed for tortricid moths by Lumley & Sperling (2010). These characters were defined by shapes on the wings, and bounded by either scale patterns or veins so that they were reliably scored. Additionally, we required that these wing areas were present across all taxa (i.e. the colour of a spot could not be measured if the spot was absent in one species). Luminance values were measured from the standardized photographs of specimens in ImageJ (Schneider, Rasband & Eliceiri, 2012) using the Colour Histogram package (Prodanov, 2010). Red, green and blue (RGB) values were measured for all pixels within each wing

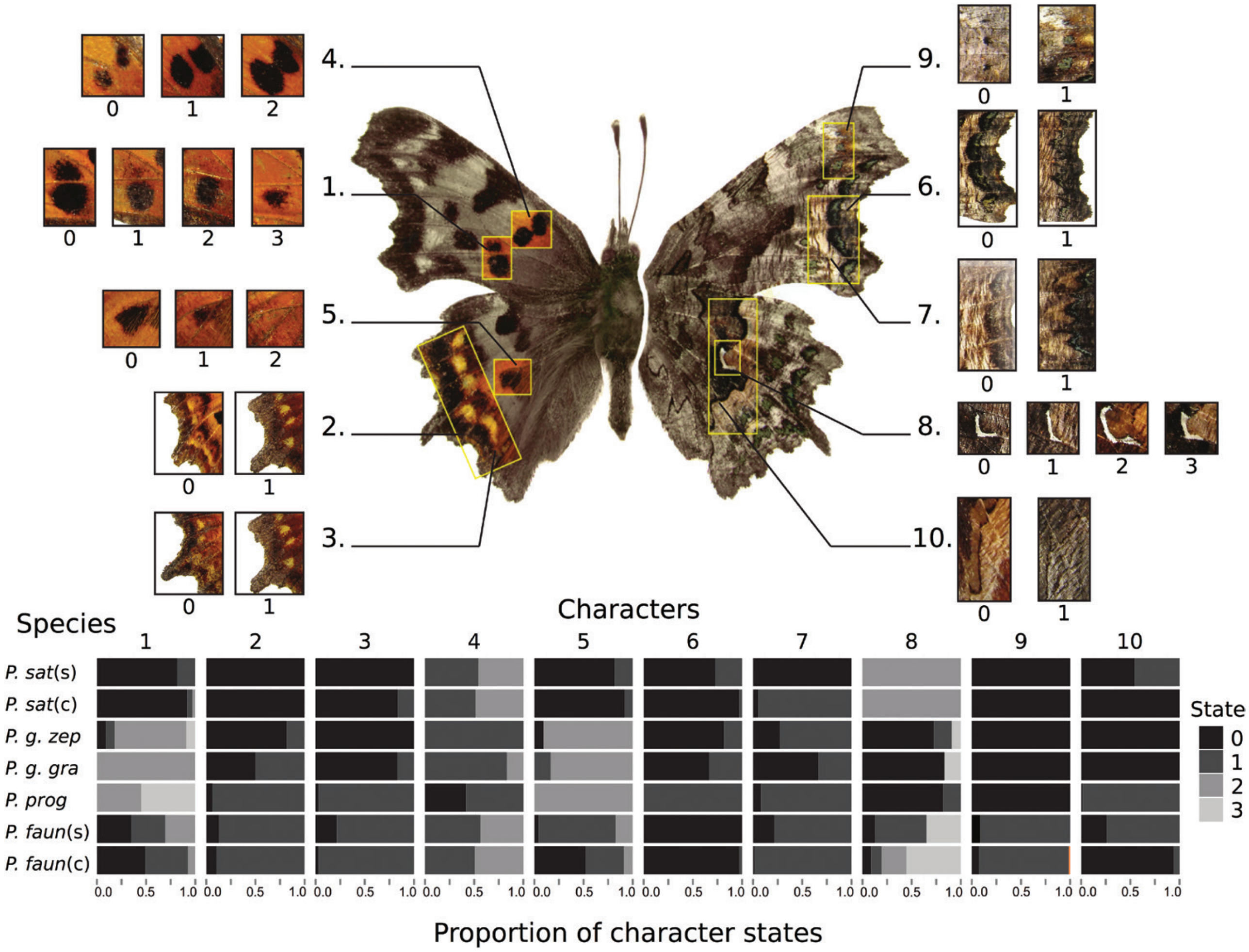



**Figure 2.** Visually scored characters on dorsal and ventral wing surfaces, and proportion of each character state across species – (c) and (s) denote contrasted and smeared forms of *Polygonia faunus* and *P. satyrus*, respectively.

area and averaged to create a set of three individual characters for each wing area. In total, six wing areas were measured, creating a final matrix of 18 characters (Fig. 3), and we refer to these as 'RGB characters'.

All morphological character scoring was conducted by CMM. Female *P. faunus* and *P. satyrus* exhibit two morphotypes, referred to as 'smeared' and 'contrasted' in this study (see Supporting Information, Fig. S1). Smeared morph females generally lack striations and contrast on the ventral side of the wings, whereas the contrasted morph females appear striated with alternating bands of light and dark coloration, much like their conspecific males. These morphs were treated as separate taxa for morphological analysis.

### Morphological analysis

We used eigenvector-based multivariate analyses to maximize the variance among individuals for morphological characters and visualize morphological similarity between species. Because of the inherent differences between the visually scored and RGB characters (the former were discrete/ordinal and the latter continuous), separate multivariate analyses were used for each data set. Multiple correspondence analysis (MCA) and principal component analysis (PCA) were used for the visually scored and RGB characters, respectively. In some cases, these analyses did not provide resolution of species boundaries. In an effort to increase this resolution, we also conducted discriminant correspondence analysis (DCA) and linear discriminant analysis (LDA) (for visually scored and RGB characters, respectively), which incorporate *a priori* defined groups into the analysis. The results of the genetic analyses were unambiguous as to the species-level identifications, so we used SNP genetic clusters as the *a priori* species identity of individuals. *Polygonia faunus* and *P. satyrus* were further divided into morphological types (smeared vs. contrasted morphs), and *P. gracilis* was divided into putative subspecies *P. g. gracilis* and

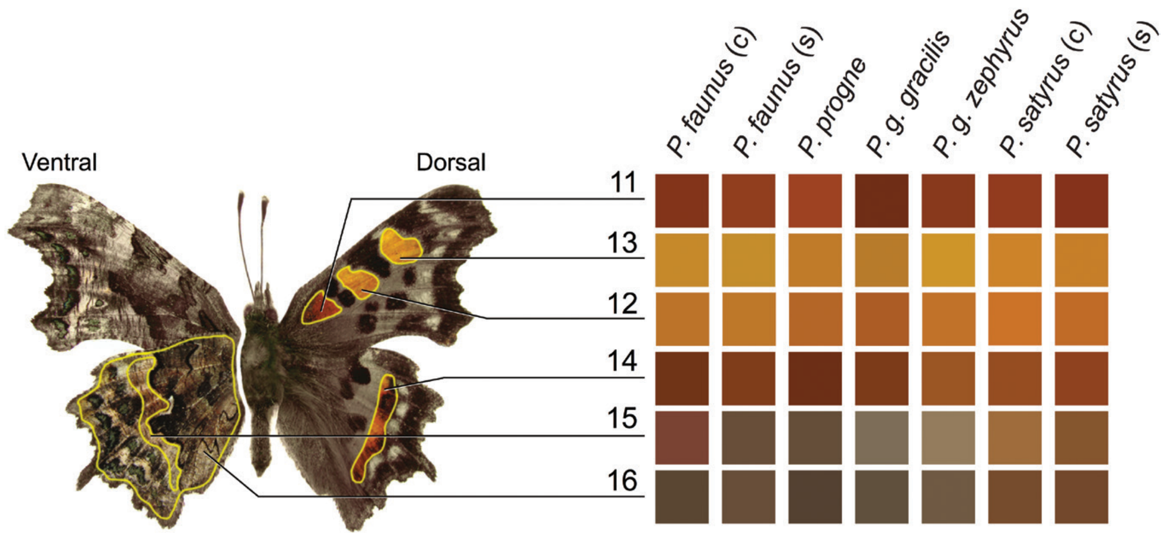



**Figure 3.** Red, green and blue (RGB) characters on the dorsal and ventral wing surfaces, and the average RGB luminance values (combined red, green and blue) visualized as colour swatches for each species – (c) and (s) denote contrasted and smeared forms of *Polygonia faunus* and *P. satyrus*, respectively.

*P. g. zephyrus* based on diagnostic descriptions in field guides and collection location (Bird *et al.*, 1995; Pyle, 2002; Shepard & Guppy, 2011). All analyses were conducted using the *ade4* library (Dray & Dufour, 2007) in R. Individuals containing missing data (due to wing wear, missing characters, etc.) were removed prior to the analysis, resulting in the following sample sizes: Visually scored characters: *P. faunus* (smeared $n$ = 23, contrasted $n$ = 67), *P. g. gracilis* ($n$ = 6), *P. g. zephyrus* ($n$ = 11), *P. progne* ($n$ = 62) and *P. satyrus* (smeared $n$ = 11, contrasted $n$ = 37) and RGB characters: *P. faunus* (smeared $n$ = 28, contrasted $n$ = 67), *P. g. gracilis* ($n$ = 8), *P. g. zephyrus* ($n$ = 11), *P. progne* ($n$ = 68) and *P. satyrus* (smeared $n$ = 17, contrasted $n$ = 38). Character states for all individuals are provided in Supporting Information, Table S1.

## RESULTS

### Mitochondrial DNA

All analyses of *COI* (parsimony-based haplotype networks and ML and BI tree searches) gave the same broad relationships between species that have been identified previously using mtDNA (Fig. 1; Supporting Information, Fig. S2). The only exception was a divergent mitochondrial clade (G-X) found in three individuals collected for this study. This clade is positioned basally to *P. gracilis*, *P. progne* and *P. satyrus*, but morphological and GBS data identified these individuals as *P. gracilis* (both subspecies). The low number of non-synonymous mutations (two in total) in this lineage supported a mitochondrial origin, rather than a nuclear insertion of a mitochondrial gene, and repeated extraction and sequencing confirmed that these sequences were not chimeras or the result of wet lab error. Apart from this divergent lineage, all species conformed to well-supported monophyletic clades (Supporting Information, Fig. S2). Within these clades, there was little structured genetic variation, and many individuals from disparate geographic locations and ecozones shared identical haplotypes (Fig. 4). Additionally, *P. g. gracilis* and *P. g. zephyrus* were not distinct in either phylogenetic or haplotype network analyses (Supporting Information, Fig. S2).

### Genotyping-by-sequencing

In total, 452.8 million reads were generated in four lanes of sequencing for the specimens included in this study. Filtering with *process_radtags* reduced this to 345.0 million reads, and 315.5 million reads (average 1.3 million per individual) were incorporated into the *de novo* assembly for SNP calling, leading to 127 661 catalog loci. All ML and BI phylogenetic analyses of filtered GBS data (including flanking sequence or SNPs only) agreed on a single set of relationships between species (Figs 1, 5; Supporting Information, Figs S3–S5). As with mtDNA, *P. faunus* was basal to the remaining *Polygonia* species, but the remainder of the tree showed a sister relationship between the clades containing *P. interrogationis* + (*P. comma* + *P. satyrus*) and *P. gracilis* + *P. progne*. All species were reciprocally monophyletic and little geographic structure was apparent within species (Fig. 5). Bayesian runs converged by all metrics for the sequence-based

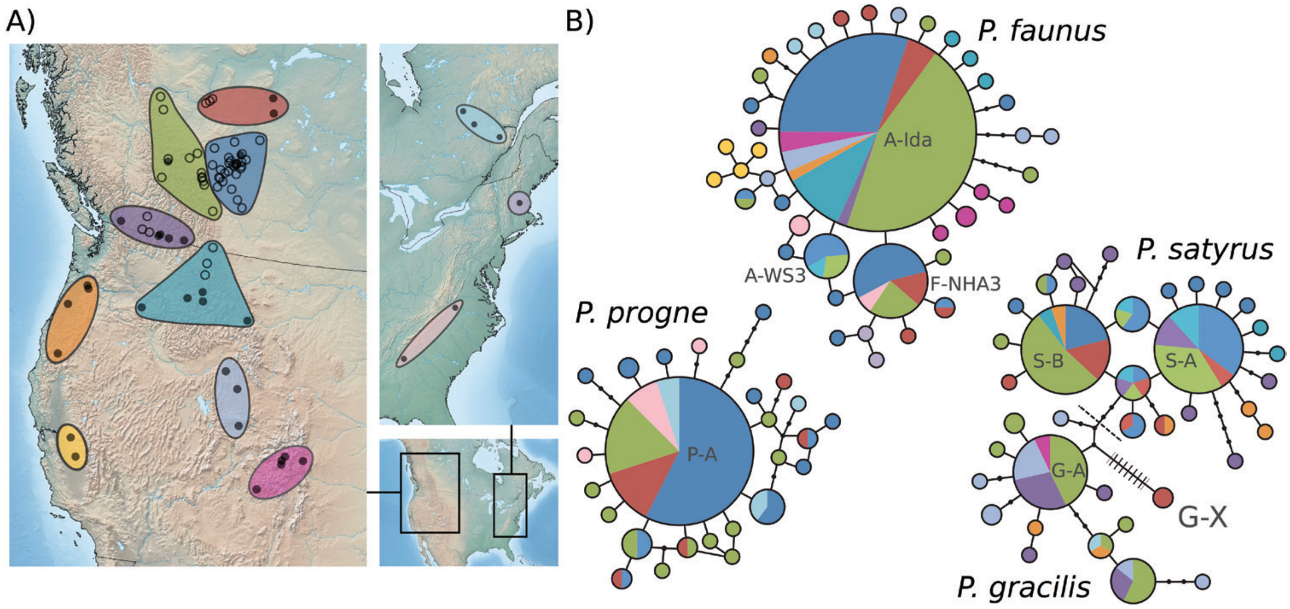


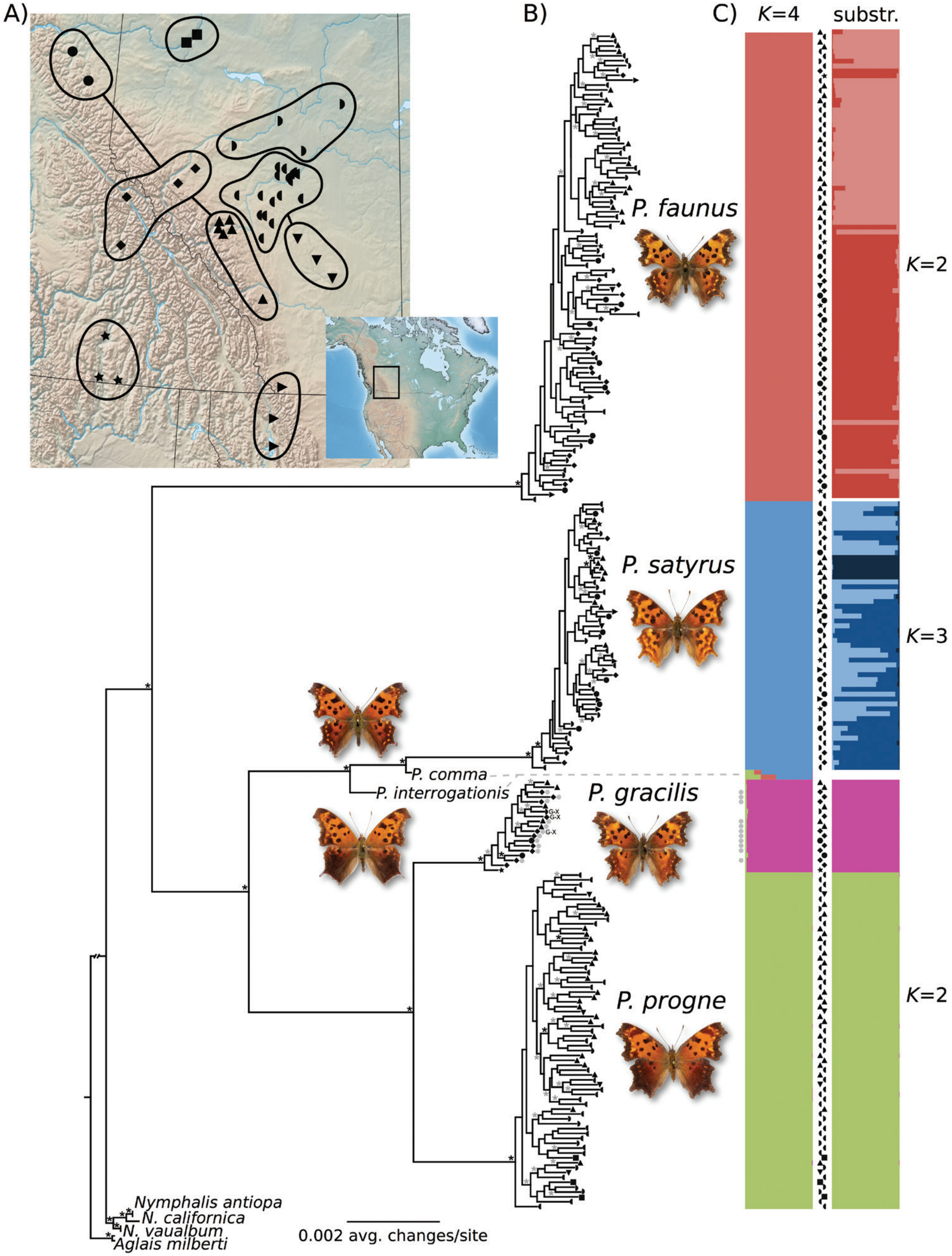

**Figure 4.** A, cytochrome oxidase subunit I gene (*COI*) sampling localities divided by general ecozones and geographic proximity. Open circles represent specimens collected for this study [for which genotyping-by-sequencing (GBS) and morphological data may be available], and closed circles represent sequences from GenBank. B, Haplotype networks constructed from *COI* data set for each species (the dashed line between *Polygonia gracilis* and *P. satyrus* represent the break between those two species). Circles represent individual haplotypes, circle diameter is proportional to the number of individuals for each haplotype. Small dots represent missing haplotypes, and the cross-hatched line extending to haplotype G-X represents 22 substitutions. Some major haplotypes for *P. gracilis*, *P. progne* and *P. satyrus* were named in this study, those for *P. faunus* follow Kodandaramaiah *et al.* (2012).

analysis; however, several effective sample size values for the SNPs only analysis were below 200; regardless, the same main topology was found for both data sets.

Population genetic analysis also supported these divisions. To evaluate the overall STRUCTURE analysis, we used two methods: Ln Pr($X|K$) supported values of $K$ between three and five, and $\Delta K$ supported $K$ = 5. However, it was apparent that $K$ = 5 only introduced a partial cluster to *P. comma* and *P. interrogationis* (each represented by a single individual), and $K$ = 3 simply combined *P. gracilis* and *P. progne*. For these reasons, we chose $K$ = 4 as the optimal value of $K$, which corresponded to the four species considered here (barplots of all values of $K$ provided in Supporting Information, Fig. S6). Hierarchical analysis of these four clusters identified substructure in *P. faunus* ($K$ = 2) and *P. satyrus* ($K$ = 3) (Fig. 5; barplots of all values of $K$ for hierarchical analysis provided in Supporting Information, Fig. S6). Although this substructure partially corresponds to distinct clades identified with the phylogenetic analyses, there are no clear geographic or habitat-related patterns except a distinct cluster of five individuals of *P. satyrus* from a single population in the foothills of the Rocky Mountains (Fig. 5). DAPC corroborated the overall pattern of genetic structure, with *find.clusters* predicting values of $K$ from four to six, and the first and second discriminant functions strongly separating the four species (Supporting Information, Fig. S7). Some substructure was also supported by DAPC, notably the two main clusters of *P. faunus* and the distinct cluster of five individuals in *P. satyrus* (right most cluster in *P. satyrus* portion of Supporting Information, Fig. S7). Individuals of *P. gracilis* exhibiting the divergent G-X mitochondrial type were not distinct in any of the SNP-based analyses, and there was no support for separation of *P. g. gracilis* and *P. g. zephyrus* (Fig. 5).

### Morphological characters

The plot of visually scored characters on the first two MCA axes produced several clusters with relatively little overlap that corresponded to the species clusters identified using SNP data (Fig. 6A). MCA axis 1 separated *P. progne* from the rest of the species and *P. satyrus* from *P. gracilis*, and axis 2 separated *P. faunus* from *P. satyrus* and *P. gracilis*. The only notable overlap between these clusters was between specimens of *P. faunus* and *P. gracilis*; although some of these individuals were worn, others showed


A)
B)
C)
K=4
substr.
P. faunus
K=2
P. satyrus
K=3
P. comma
P. interrogationis
P. gracilis
G-X
G-X
G-X
P. progne
K=2
Nymphalis antiopa
N. californica
N. vaualbum
Aglais milberti
0.002 avg. changes/site

morphological similarity in relatively fresh specimens. Gradients were evident between smeared and contrasted forms of *P. faunus* and *P. satyrus*, but no clustering was apparent for subspecies of *P. gracilis*. Based on the relative magnitudes of the canonical weights (loadings) of the MCA, several characters (notably characters 1, 5 and 6) were equally effective for separating *P. progne* from the rest of the species (Table 1), and a different set (characters 9, 2, 3 and 8) separated *P. gracilis* and *P. satyrus* from *P. faunus*. DCA of the visually scored characters (using *a priori* groupings based on SNP data) also separated the same main clusters but showed more overlap between species and less distinction between smeared/contrasted morphs (Fig. 6B). DCA axis 2 mainly separated *P. satyrus* from the other species, while DCA axis 1 separated *P. faunus* from *P. progne*, with *P. gracilis* as intermediate. Character 9 was most effective at separating species along the first axis (*P. faunus* from *P. progne*), and characters 9 and 10 were most effective along the second axis for the clear separation of *P. satyrus* from the rest of the species (Table 1). Third axes of MCA and DCA provided no additional separation of any species clusters (Supporting Information, Fig. S8).

PCA of the RGB characters did not separate any species (Fig. 6C). Despite the fact that the first PC axis explained a relatively large amount of variation (54.8%), all species overlapped in all major axes and relative intraspecific variation was high. LDA of the same characters (using prior groupings) resulted in more overall separation between species than in the PCA; however, overlap was still present in some individuals. *Polygonia satyrus* was the only species that was more-or-less completely separated, along axis 1, and aspects of characters 15 and 16 (both ventral characters) were most effective at separating this species. There was no apparent clustering of any smeared vs. contrasted morphs, nor of *P. gracilis* subspecies. Based on pairwise comparisons of their luminance means, *P. progne* and *P. g. zephyrus* were most different (Supporting Information, Table S3, Fig. S9). Again, the third axes of PCA and LDA provided no additional separation (Supporting Information, Fig. S8).

## DISCUSSION

### Species limits and relationships

All genetic data sets and analyses supported clear species limits between the four widespread species considered here: *P. faunus*, *P. gracilis*, *P. progne* and *P. satyrus*. Within these lineages, there was little or no intraspecific geographic clustering with genome-wide SNPs or mtDNA and no signatures of admixture or introgression between species. The only geographically coherent clustering we observed was with SNP data in individuals of *P. satyrus* from a single population in the foothills of the Rocky Mountains, which formed a monophyletic clade in phylogenetic analyses and a distinct genetic cluster using STRUCTURE (Fig. 5). These individuals were not distinctive with *COI* and exhibited one of the most common mitochondrial haplotypes (Supporting Information, Fig. S2). Additionally, they were all collected from the same locality on the same day, so siblingship may easily explain their genetic similarity.

The simplest explanation for the broad patterns of genetic differentiation among these four *Polygonia* species is that reproductive isolation between species is strong, and evolutionary forces (retained ancestral polymorphism, high levels of gene flow, etc.) are maintaining homogeneity within these lineages. At least some *Polygonia* species undergo two-way migrations (*P. interrogationis*: Layberry *et al.*, 1998), and all *Polygonia* have high dispersal capability (Braschler & Hill, 2007; Burke, Fitzsimmons & Kerr, 2011). Given their longevity (*Polygonia* overwinter as adults, with individual lifespans of 10–12 months; Bird *et al.*, 1995), high dispersal rates could explain the low interspecific variation we observed here, particularly in combination with high levels of retained ancestral polymorphism.

**Figure 5.** A, Sampling across the zone of contact with broad ecozones indicated by different shapes and black outlines. Black lines connecting shapes indicate comparable ecozones in Figure 4 that were subdivided here to provide more resolution. B, Consensus tree from maximum likelihood (ML) analysis of 121 513 bp alignment of genotyping-by-sequencing (GBS) data (including flanking sequence). Black asterisks indicate >95% ultra-fast bootstrap + >80% Shimodaira/Hasegawa approximate likelihood ratio test (SH-aLRT) from ML analysis and >0.9 posterior probability from Bayesian inference (BI); grey asterisks indicate highly supported nodes (using the aforementioned criteria) in only one analysis (generally nodes were well-supported in ML but not with BI). Grey circles for *Polygonia gracilis* individuals indicate *P. g. zephyrus*, and individuals with the G-X mitochondrial haplotype indicated with ‘G-X’. C, STRUCTURE barplots from 961 single nucleotide polymorphisms (SNPs) for $K = 4$ (overall clustering) and the finest level of substructure for each main cluster (excluding *P. comma* and *P. interrogationis*); no substructure was observed within *P. gracilis* or *P. progne*. Photographs, used with permission, by Kim Davis, Mike Strangeland and Andrew Warren.

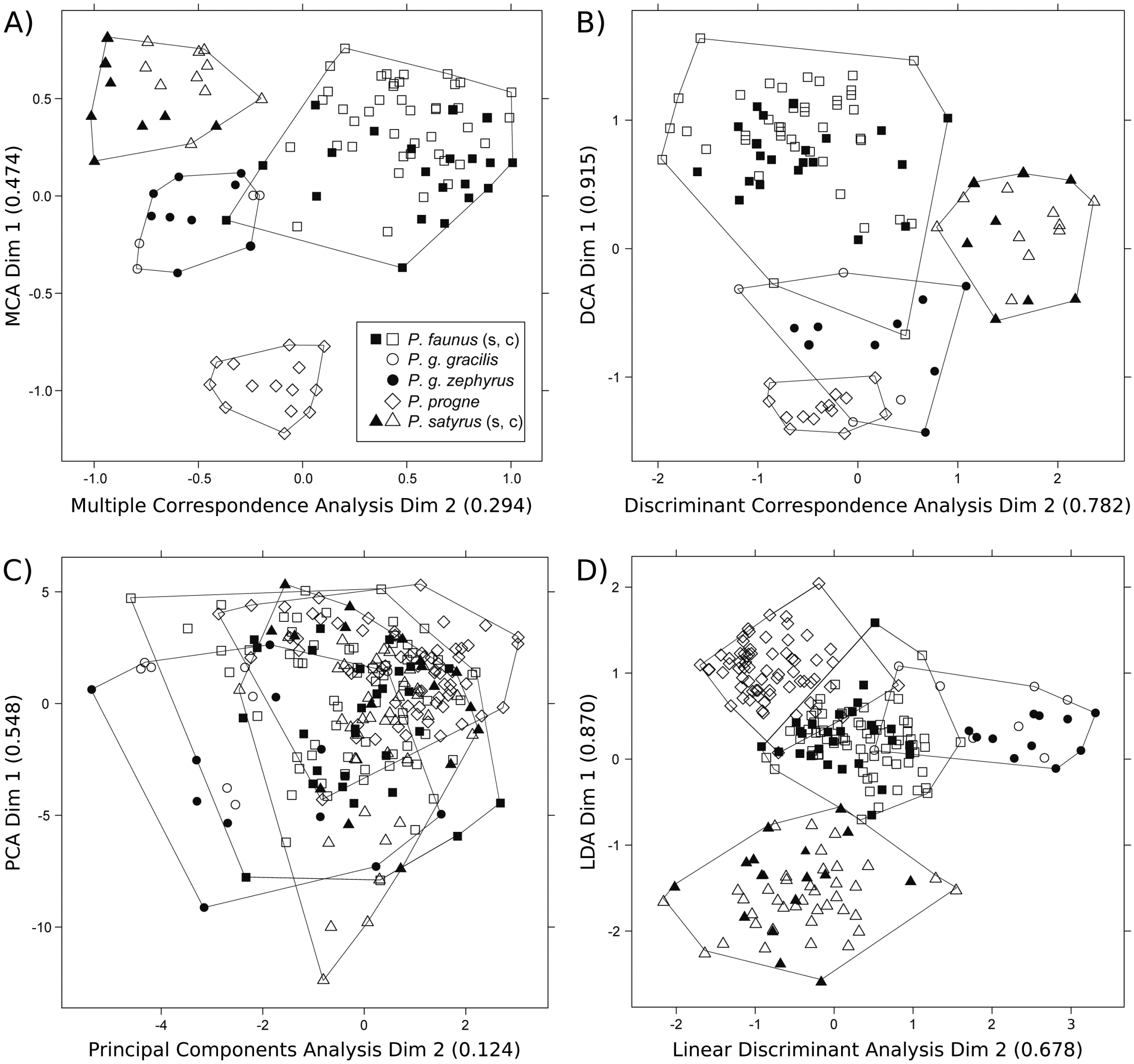



**Figure 6.** Multivariate analyses of wing characters. Multiple correspondence analysis (A) and discriminant correspondence analysis (B) of visually scored characters, and principal components analysis (C) and linear discriminant analysis (D) of red, green and blue (RGB) characters. Decimal values on axis labels indicate proportion of total variation explained [multiple correspondence analysis (MCA) and principal component analysis (PCA)] and the remaining proportion of variation [discriminant correspondence analysis (DCA) and linear discriminant analysis (LDA)]. Species inclusion for each specimen was determined by membership in single nucleotide polymorphism (SNP) clusters.

Another consideration is that the geographic scale of our sampling was relatively small compared to these species' full distributions (Layberry *et al.*, 1998). *Polygonia faunus*, for example, is found from Alaska south to California and New Mexico, and across Canada and the northern United States to the Atlantic coast; Kodandaramaiah *et al.* (2012) found two mitochondrial haplotype groups and around five nuclear genetic clusters using microsatellites across the range of *P. faunus*. A similar pattern has been found in *P. c-album* (Linnaeus, 1758), a species found across Europe, North Africa and Asia, where microsatellites only show patterns of strong differentiation at the largest geographic scales (Kodandaramaiah *et al.*, 2011). However, the smaller area of distributional overlap that we sampled creates an opportunity to test the

**Table 1.** Canonical loadings for three axes from multiple correspondence analysis (MCA) and discriminant correspondence analysis (DCA) of ten visually scored characters and principal component analysis (PCA) and linear discriminant analysis (LDA) of 18 mean red, green and blue luminance values from six RGB characters

| Char. | MCA1 | MCA2 | MCA3 | DCA1 | DCA2 | DCA3 |
|---|---|---|---|---|---|---|
| 1 | 0.807 | 0.295 | 0.095 | −2.146 | −6.032 | −0.398 |
| 2 | 0.190 | 0.528 | 0.006 | 4.870 | −6.046 | 8.963 |
| 3 | 0.149 | 0.512 | 0.073 | 7.210 | −4.617 | −5.972 |
| 4 | 0.526 | 0.006 | 0.004 | −1.540 | 8.894 | 9.558 |
| 5 | 0.776 | 0.270 | 0.219 | −1.729 | −7.054 | 7.608 |
| 6 | 0.774 | 0.030 | 0.003 | −14.391 | 4.310 | 8.249 |
| 7 | 0.002 | 0.134 | 0.426 | −9.374 | 1.571 | −42.971 |
| 8 | 0.706 | 0.506 | 0.516 | −0.203 | 6.762 | 4.765 |
| 9 | 0.199 | 0.659 | 0.009 | 22.324 | −15.677 | 3.823 |
| 10 | 0.607 | 0.002 | 0.090 | −1.859 | 16.252 | 8.434 |
| | PCA1 | PCA2 | PCA3 | LDA1 | LDA2 | LDA3 |
| 11R | −0.158 | 0.408 | 0.017 | 0.047 | 0.137 | 0.012 |
| 11G | −0.193 | 0.251 | 0.259 | 0.324 | 0.418 | −0.528 |
| 11B | −0.158 | −0.176 | 0.307 | −0.141 | −0.350 | 0.416 |
| 12R | −0.224 | 0.374 | −0.156 | 0.047 | −0.427 | −0.196 |
| 12G | −0.265 | 0.219 | 0.161 | −0.398 | 0.135 | 0.932 |
| 12B | −0.232 | −0.029 | 0.392 | 0.201 | 0.052 | −0.408 |
| 13R | −0.222 | 0.361 | −0.128 | 0.107 | 0.198 | −0.435 |
| 13G | −0.270 | 0.177 | 0.132 | 0.221 | −0.239 | 0.556 |
| 13B | −0.237 | −0.043 | 0.356 | −0.027 | −0.046 | 0.077 |
| 14R | −0.247 | 0.069 | −0.276 | 0.261 | −0.057 | −0.109 |
| 14G | −0.267 | −0.054 | −0.147 | −0.452 | −0.360 | −0.754 |
| 14B | −0.209 | −0.136 | 0.310 | 0.168 | 0.250 | 0.151 |
| 15R | −0.265 | −0.072 | −0.342 | 0.323 | −0.660 | 0.419 |
| 15G | −0.268 | −0.235 | −0.241 | −1.163 | 0.685 | 1.668 |
| 15B | −0.217 | −0.372 | −0.049 | 0.614 | −1.024 | −2.617 |
| 16R | −0.275 | −0.064 | −0.269 | −0.630 | 1.105 | −1.276 |
| 16G | −0.268 | −0.230 | −0.130 | 0.001 | −0.154 | 0.673 |
| 16B | −0.219 | −0.324 | 0.118 | 0.564 | −0.204 | 0.994 |

genomic integrity of these species. Despite their morphological variation and similarity, we found no signatures of admixture or introgression between these species, further supporting the hypothesis of strong reproductive isolation. This lack of introgression is an interesting contrast to some other Nymphalidae, where introgression and reticulate evolution has played an important evolutionary role [e.g. *Heliconius* spp. (Heliconius Genome Consortium, 2012; Pardo-Diaz *et al.*, 2012; Zhang *et al.*, 2016)].

In keeping with the observed lack of geographic patterns within species, we found no genetic support for differentiation of the subspecies of *P. gracilis*: *P. g. gracilis* and *P. g. zephyrus*. There is substantial disagreement in the literature regarding these lineages, with some considering them to be separate species (Guppy & Shephard, 2001; Wahlberg *et al.*, 2009) and others conspecifics (Layberry *et al.*, 1998; Pelham, 2008). Morphologically, there is a continuum between *P. g. gracilis*-like and *P. g. zephyrus*-like characteristics, and while the extremes of this continuum are morphologically diagnosable, many individuals from sympatric regions fall in the middle (Wahlberg *et al.*, 2009). Genetically, they share haplotypes for several mitochondrial and nuclear genes, as well as with other close relatives – *P. progne*, *P. oreas*, *P. haroldii* (Dewitz, 1877) (Wahlberg *et al.*, 2009). Although our sample size is small for these lineages, genome-wide data support their conspecific status.

Given the clear species limits between the lineages considered here, a logical step is to consider their phylogenetic relationships. However, our phylogenetic sampling is not comprehensive; we are missing five species that are thought to be interspersed among the taxa considered here [*P. c-album*, *P. g-argenteum* (Doubleday, 1848), *P. haroldii*, *P. interposita* (Staudinger, 1881) and

*P. oreas*] and four basal species [*P. c-aureum* Linnaeus, 1758, *P. egea* (Cramer, 1775), *P. gigantea* (Leech, 1883) and *P. undina* (Grum-Grshimailo, 1890)]. For this reason, we will only briefly address phylogenetic considerations, and for more comprehensive phylogenetic discussions direct readers to Nylin *et al.* (2001), Wahlberg & Nylin (2003) and Wahlberg *et al.* (2009).

Most notably, we observed similar cytonuclear discordance between mitochondrial and nuclear markers as has been observed using traditional gene sequencing approaches (Fig. 1; Nylin *et al.*, 2001; Wahlberg & Nylin, 2003; Wahlberg *et al.*, 2009). Discordance between relationships based on mtDNA and nuclear markers is well documented (Funk & Omland, 2003; Chan & Levin, 2005; Dupuis *et al.*, 2012), and in *Polygonia* has been explained by ancestral hybridization creating shared haplotypes between species (Wahlberg *et al.*, 2009) or infection with the maternally inherited bacterium *Wolbachia* (Kodandaramaiah *et al.*, 2011, 2012). Both phenomena may have acted to obfuscate the relationships dictated by mtDNA, and we consider the true species relationships to lie closer to those predicted using more extensive nuclear markers and non-genetic data (to the extent that it can be illustrated using a bifurcating tree given the reticulate history of the group [Wahlberg *et al.*, 2009; see Soucy, Huang & Gogarten (2015)]. Our discovery of the divergent G-X mitochondrial lineage in specimens of *P. gracilis* (Figs 1, 4) that are indistinguishable from other *P. gracilis* using SNPs further complicates the reticulate history of the group (Wahlberg *et al.*, 2009). Whether this is the result of ancient hybridization or retention of an ancestral mitochondrial haplotype (Melo-Ferreira *et al.*, 2014; Dupuis & Sperling, 2015; Naciri & Linder, 2015) is difficult to determine given the current sampling. It is clear that genome-wide assessments of genetic diversity with comprehensive range-wide sampling of each species will be essential to further understand these evolutionary histories.

### Genetic data informing morphological diagnostics

For each of the morphological data sets (visually scored and RGB characters), we conducted two sets of analyses: one without prior information on species designations and one with prior information based on species identity from mtDNA and SNP data. While the visually scored characters separated species without guidance from genetic data (Fig. 6A), RGB characters did not (Fig. 6C). Interestingly, with the guidance of genetic data, the separation of species using visually scored characters actually decreased slightly (Fig. 6B), whereas delimitations using RGB characters improved but only to the point of distinguishing *P. satyrus* (Fig. 6D). While the respective performance of visually scored and RGB characters is likely due to the inherent information content of each of these character sets, there are also some species-specific differences. For instance, overall, *P. satyrus* was the most consistently separated species from the rest in analyses with prior species designations (DCA and LDA), despite *P. progne* having the largest degree of separation using designation-free MCA (Fig. 6A).

We considered the relative loadings of characters for the axes from each analysis to infer which characters were most effectively separating, or failing to separate, species. For example, the division of the dorsal forewing spot across vein CuA2 (character 1, Fig. 2), contributed most to separation along the first axis of the MCA, which notably separated *P. progne* from the rest of the species (Fig. 6; Table 1). Given the prevalence of this character as a delimiting feature in the literature, this is unsurprising. However, other characters that are equally prevalent in the literature were not as diagnostic as expected, such as the shape of the silver comma (character 8). This may be because diagnostic characters in field guides are often written to allow users to distinguish between pairs of similar species, rather than all species in a group; a single character with overlapping states between *P. satyrus* and *P. progne* would not affect their species identification since they are otherwise easily distinguished. In fact, many of these characters would be diagnostic if *P. faunus* were removed from the taxon pool, as the variation in character states of this species was broad (Fig. 2). For example, every state of the aforementioned shape of the silver comma (character 8) was found in at least some specimens of *P. faunus* included in our study, several individuals did not fit neatly into any of the character states, and one specimen exhibited a marking that was upside-down.

Surprisingly, we found few cases of visually scored characters being equally diagnostic across analyses (i.e. the same character contributing most to the separation of a given species in both MCA and DCA). The prominence of ventral forewing spots basal to the submarginal chevrons (character 9) was an exception to this trend, as it was the most explanatory character across axis 2 in the MCA and axis 1 in the DCA (Table 1). Neither of these axes clearly separated one species from the rest, but both displayed gradients from a large group of *P. faunus* to the other species, albeit with some overlap (Fig. 6). Overall, the following visually scored characters provided the most discrimination in these analyses: discal forewing spots across CuA2 (character 1), ventral forewing spots basal to the submarginal chevrons (character 9) and contrast between basal and distal halves of the ventral hindwing (character 10).

Average RGB luminance measurements did not perform well in separating the species in our study. When prior species identifications were employed in an LDA, *P. satyrus* was the only species distinguished without substantial overlap (Fig. 6). Some characters that are frequently used in species diagnoses in the literature were informative for this separation, such as two ventral hindwing characters: the submarginal region of the ventral hindwing near the comma (character 15) and the overall colour of the ventral hind wing (character 16). Others were not as informative despite similar use in the literature, such as the shade of orange on the dorsal forewing (characters 11, 12 and 13) (Bird *et al.*, 1995; Pyle, 2002). One character in particular, the overall colour of the ventral hind wing (character 16), encompassed a large portion of the wing with highly contrasting patterns. While we expected that averaging the luminance values across this contrast might obfuscate the overall colour of the character state, this was one of the more informative RGB characters. Qualitatively, many wing areas appear very similar in average colour across the species (Fig. 3), which may be due to relative amounts of reddish and black pigments (melanin) producing the same hue but different tones (Hovanitz, 1941).

Overall, the utility of colour or luminance-based morphological characters may depend on group-specific characteristics, as they have been used to successfully delimit species in other groups (Lumley & Sperling, 2010; Watanabe, Nakazono & Ono, 2012; McKay *et al.*, 2014). *Polygonia* may be a particularly difficult group for quantifying colour differences, or alternatively our focus on several species at once could be overcomplicating delimitation of these species. Although it is beyond the scope of this paper, combining quantitative methods such as those used here with species pair sampling across the morphological diversity of each species should more comprehensively address the transition from species delimitation to species identification.

## CONCLUSION

Here, we assessed species limits in an area of widespread overlap between four species of *Polygonia* butterflies in western Canada. mtDNA and genome-wide SNPs provide clear species delimitation, despite well-documented cytonuclear discordance. Further, there was little evidence for population structure or subspecific distinctions. Visually and digitally scored morphological characters were variably successful for separating species, and the application of genetic data to guide morphological analyses improved species delimitation for the digitally scored data set. We identified several morphological characters that are highly informative for separating species, and more geographically comprehensive sampling is now needed to test the efficacy of the most informative diagnostic characters that we identified for field identifications of this morphologically difficult group. This application of genetic data to guide morphological analysis provides a framework for iterative approaches within broader integrative taxonomy and will undoubtedly continue to help taxonomists to resolve complicated groups.

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## SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

**Table S1.** Specimen collection information, mtDNA GenBank accessions and morphological data. 'mtDNA only' indicates specimens used from Wahlberg *et al.* (2009).

**Table S2.** Diagnostic characters of *Polygonia* butterflies published in North American field guides, with an emphasis on western North America. If an author designated a character as being primary, or highly diagnostic, it is indicated with a grey background. DFW, dorsal forewing; DHW, dorsal hindwing; VFW, ventral forewing; VHW, ventral hindwing.

**Table S3.** Summary of pairwise luminance mean values with non-overlapping confidence intervals in pairwise comparisons.

**Figure S1.** Examples of smeared and contrasted morphs of *P. faunus* and *P. satyrus* used in this study.

**Figure S2.** Consensus tree from ML analysis of unique *COI* haplotypes. Dots on nodes indicate high support: >80% SH-aLRT and >95% ultra-fast bootstrap in ML analysis, and >0.8 posterior probability in Bayesian analysis. Major clades are named (as in Fig. 4) in grey text, and grey circles indicate *P. g. zephyrus* individuals.

**Figure S3.** Consensus tree from ML analysis of 2572 SNP data set. Node support values are SH-aLRT/ultra-fast bootstrap.
**Figure S4.** Fifty per cent majority rule consensus from BI analysis of 121 513 bp alignment of GBS data (including flanking sequence) with posterior probability values for node support.
**Figure S5.** Fifty per cent majority rule consensus from BI analysis of 2572 SNPs from GBS data with posterior probability values for node support.
**Figure S6.** $\Delta K$ and $\mathrm{LnPr}(X|K)$ results, and STRUCTURE results for all values of $K$, for all individuals, *P. faunus*, *P. satyrus*, *P. gracilis* & *P. progne*, *P. gracilis* and *P. progne*.
**Figure S7.** Discriminant analysis of principal components (DAPC) on 961 SNPs. Top left shows overall DAPC, and remaining panels are individual species coloured by localities. *Polygonia gracilis* did not have adequate sampling across localities to analyze by itself. Insets show relative explanatory power of individual discriminant functions.
**Figure S8.** Multivariate analyses of wing characters, displaying third axes of each analysis. Multiple correspondence analysis (A) and discriminant correspondence analysis (B) of visually scored characters, and principal components analysis (C) and linear discriminant analysis (D) of RGB characters. Decimal values on axis labels indicate proportion of total variation explained (MCA, PCA) and the remaining proportion of variation (DCA, LDA). Species inclusion for each specimen was determined by membership in SNP clusters.
**Figure S9.** Central tendency statistics of mean R, G and B values of six wing characters across 248 *Polygonia* butterflies. The upper and lower boundaries of each box are third and first quartiles, respectively; vertical span of the notches correspond to the upper and lower 95% confidence bounds; the mean is represented by black lines bisecting each box. The upper and lower fences (whiskers) indicate the maximum and minimum values in the data set, while outliers – those outside 1.5 times the interquartile range above the fourth quartile or below the first quartile – are shown as black points.

ONLINE SUPPLEMENTARY MATERIAL

Table S1. Specimen collection information, mtDNA GenBank accessions, and morphological data. "mtDNA only" indicates specimens used from Wahlberg *et al.* (2009).

| specimen identifier, UASM accession | species | collection locality, date collected, collector, and method | mtDNA GenBank accession; haplotype name | morphological data and contrasted vs. smeared (when applicable) |
|---|---|---|---|---|
| 8301, UASM370001 | *Polygonia faunus* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318516; pf_8301 | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 107.67, 49.73, 17.7; 159.34, 105.67, 24.43; 168.71, 116.31, 26.65; 95.23, 49.2, 22.92; 95.13, 81.25, 65.13; 69.57, 61.48, 49.34; contrasted |
| 8302, UASM370002 | *Polygonia faunus* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2011-05-16; collector(s): Sperling, F. A. H., Aerial net | KY318517; pf_A-Ida | visually scored: 1, 1, 1, 1, 0, 1, 1, 3, 1, 0; RGB: 111.3, 51.86, 19.38; 151.62, 91.75, 12.01; 167.63, 124.57, 33.41; 101.19, 59.09, 27.1; 109.9, 95.2, 75.85; 93.68, 81.2, 63.65; contrasted |
| 8303, UASM370003 | *Polygonia faunus* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2011-05-16; collector(s): Sperling, F. A. H., Aerial net | KY318518; pf_F-NHa3 | visually scored: 1, 1, 1, 2, 1, 0, 1, 3, 1, 0; RGB: 87.95, 46.3, 23.4; 147.06, 96.51, 23.72; 156.18, 108.11, 25.32; 91.75, 49.74, 21.35; 110.98, 95.96, 77.99; 84.7, 74.33, 59.9; contrasted |
| 8304, UASM370004 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-13; collector(s): Sperling, F. A. H., Aerial net | KY318519; pf_A-Ida | visually scored: 1, 1, 1, 2, 0, 0, 1, 2, 1, 0; RGB: 137.7, 76.57, 12.15; 164.94, 129.76, 34.33; 176.344, 147.56, 56.02; 105.59, 70.66, 25.53; 117.55, 112.14, 96.31; 91.52, 89.28, 75.19; contrasted |
| 8305, UASM370005 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-13; collector(s): Sperling, F. A. H., Aerial net | NA | visually scored: 1, 1, 1, 1, 1, 1, 1, 2, 1, 0; RGB: NA |
| 8306, UASM370006 | *Polygonia faunus* | CANADA: Alberta, Edmonton nr Emily Murphy Park (53.533, -113.536); date collected: 2012-05-10; collector(s): Soderlind, L., Aerial net | KY318520; pf_F-NHa3 | visually scored: 2, 1, 1, 2, 1, 0, 1, 0, 1, 1; RGB: 113.25, 54.63, 15.15; 163.09, 102.59, 6.96; 168.32, 127.49, 48.02; 87.39, 46.21, 23.98; 102.3, 87.31, 70.34; 95.69, 82.69, 65.79; smeared |
| 8307, UASM370007 | *Polygonia faunus* | CANADA: Alberta, Edmonton nr Emily Murphy Park (53.533, -113.536); date collected: 2012-05-07; collector(s): Soderlind, L., Aerial net | KY318521; pf_8307 | visually scored: 1, 1, 1, 2, 0, 0, 1, 2, 1, 0; RGB: 140.68, 64.104, 11.12; 198.97, 120.36, 23.53; 193.34, 131.77, 37.26; 105.77, 50.98, 21; 131.96, 99.9, 71.32; 93.04, 73.66, 54.04; contrasted |
| 8308, UASM370008 | *Polygonia faunus* | CANADA: Alberta, Aylmer Recreation Area (52.401, -116.081); date collected: 2012-05-15; collector(s): Soderlind, L., Aerial net | KY318522; pf_A-Ida | visually scored: 1, 1, 1, 2, 1, 0, 0, 1, 1, 1; RGB: 158.44, 87.07, 28.69; 189.36, 127.1, 47.13; 182.02, 129.71, 46.11; 139.86, 83.12, 35; 125.22, 88.71, 60.25; 115.77, 85.63, 63.73; smeared |
| 8309, UASM370009 | *Polygonia faunus* | CANADA: Alberta, North Saskatchewan River Valley, near University of Alberta (53.529, -113.52); date collected: 2012-05-05; collector(s): Soderlind, L., Aerial net | KY318523; pf_8309 | visually scored: 0, 1, 1, 2, 0, 0, 1, 2, 1, 0; RGB: 121.93, 57.14, 23.64; 210.39, 145.3, 61.52; 221.39, 158.21, 53.32; 122.53, 65.97, 26.3; 151.86, 115.99, 83.39; 110.82, 87.14, 62.32; contrasted |
| 8310, UASM370010 | *Polygonia faunus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Dupuis, J. R., Aerial net | KY318524; pf_A-Ida | visually scored: 2, 1, 1, 2, 1, 0, 1, 3, 1, 0; RGB: 166.12, 91.02, 12.86; 197.31, 129.57, 13.74; 207.54, 157.67, 50.19; 148.29, 84.49, 21.24; 132.51, 96.12, 68.05; 111.29, 84.21, 62.51; contrasted |
| 8311, UASM370011 | *Polygonia faunus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Soderlind, L., Aerial net | KY318525; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, 0, 1, 2, 1, 1; RGB: 152.65, 67.47, 19.17; 208.63, 139.58, 33.15; 212.71, 153.94, 37.84; 170.72, 87.86, 21.93; 133.11, 102.29, 71.79; 111.79, 85.92, 60.08; contrasted |
| 8312, UASM370012 | *Polygonia faunus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Dupuis, J. R., Aerial net | KY318526; pf_A-Ida | visually scored: 2, 0, 1, 1, 2, 0, 0, 1, 0, 1; RGB: 146.09, 75.52, 23.37; 190.58, 126.69, 22.04; 195.59, 148.2, 56.87; 147.59, 91.19, 37.12; 126.34, 96.47, 67.18; 108.07, 84.6, 61.27; smeared |

| | | | | |
|---|---|---|---|---|
| 8313, UASM370013 | *Polygonia faunus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Dupuis, J. R., Aerial net | KY318527; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, ?, 0, 1, 1, 1; RGB: 180.4, 97.43, 27.67; 226.72, 161.75, 63.89; 236.77, 184.37, 85.99; 157.94, 81.24, 25.19; 130, 95.7, 68.32; 118.26, 89.22, 63.56; smeared |
| 8314, UASM370014 | *Polygonia faunus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Dupuis, J. R., Aerial net | KY318528; pf_A-Ida | visually scored: 1, 1, 1, 1, 2, 0, 1, 1, 1, 0; RGB: 156.84, 80.69, 22.19; 212.93, 114.47, 34.13; 203.98, 155.78, 62.87; 155.51, 85.87, 27.17; 143.11, 103.39, 67.04; 123.69, 92.34, 62.74; smeared |
| 8315, UASM370015 | *Polygonia faunus* | CANADA: Alberta, North Ram River (52.269, -115.915); date collected: 2012-08-13; collector(s): Dupuis, J. R., Aerial net | KY318529; pf_A-Ida | visually scored: NA; RGB: NA |
| 8316, UASM370016 | *Polygonia faunus* | CANADA: Alberta, North Saskatchewan River Valley, near University of Alberta (53.529, -113.52); date collected: 2012-04-18; collector(s): Soderlind, L., Aerial net | KY318530; pf_8316 | visually scored: 0, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 155.54, 73.54, 16.84; 207.64, 132.28, 14.6; 288.39, 162.38, 39.38; 128.22, 62.55, 24.99; 147.28, 116.79, 87.16; 108.38, 85.24, 61.72; contrasted |
| 8317, UASM370017 | *Polygonia faunus* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.079); date collected: 2012-06-03; collector(s): Sperling, F. A. H., Aerial net | KY318531; pf_F-NHa3 | visually scored: 2, 1, 0, 2, 1, 0, 1, 3, 1, 1; RGB: 134.82, 77.74, 26.38; 167.84, 116.65, 34.49; 169.31, 128.19, 55.28; 124.04, 79.13, 29.05; 118.84, 97.27, 75.533; 116.21, 95.6, 78.3; smeared |
| 8318, UASM370018 | *Polygonia faunus* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.08); date collected: 2012-06-03; collector(s): Sperling, F. A. H., Aerial net | KY318532; pf_A-WS4 | visually scored: 1, 1, 1, 1, 1, 0, 1, ?, 1, 0; RGB: 159.39, 68.51, 11.45; 212.77, 143.75, 35.24; 226.45, 186.52, 106.53; 146.51, 73.67, 28.39; 150.41, 118.18, 86.93; 121.73, 95.15, 67.85; smeared |
| 8319, UASM370019 | *Polygonia progne* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2012-05-28; collector(s): Dupuis, J. R., Aerial net | KY318612; NA | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 168.31, 77.75, 11.31; 189.45, 117.45, 23.81; 210.01, 149.42, 52.49; 119.83, 51.54, 21.08; 119.15, 92.82, 68.65; 105.71, 83.46, 62.32 |
| 8320, UASM370020 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-21; collector(s): Sperling, F. A. H., Aerial net | KY318533; NA | visually scored: 0, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 135.13, 60.12, 20.07; 189.85, 134.69, 57.31; 190.26, 138.1, 50.47; 135.41, 69.13, 26.39; 138.43, 105.68, 80.79; 112.75, 88.09, 67.37; smeared |
| 8321, UASM370021 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-21; collector(s): Sperling, F. A. H., Aerial net | KY318534; pf_A-WS5 | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 151.76, 75.1, 19.49; 195.47, 123.27, 29.04; 190.25, 139.61, 55.99; 137.92, 90.66, 52.14; 150.95, 118.99, 87.46; 111.55, 87.06, 64.38; contrasted |
| 8322, UASM370022 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-21; collector(s): Sperling, F. A. H., Aerial net | KY318535; pf_A-Ida | visually scored: 2, 1, 1, 1, 2, ?, 0, 1, 1, 0; RGB: 151, 92.22, 44.67; 180.64, 126.41, 49.79; 203.63, 158.53, 93.46; 149.56, 84.07, 34.21; 145.77, 105.57, 72.54; 132.19, 98.36, 68.67; smeared |
| 8323, UASM370023 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-20; collector(s): Sperling, F. A. H., Aerial net | KY318536; pf_F-NHa3 | visually scored: 0, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 122.82, 54.71, 14.23; 158.03, 106.09, 18.1; 164.84, 119.5, 29.94; 91.53, 51.98, 22.96; 114.86, 98.31, 77.62; 91.42, 79.38, 61.45; contrasted |
| 8324, UASM370024 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-21; collector(s): Sperling, F. A. H., Aerial net | KY318537; NA | visually scored: 1, 1, 1, 1, 1, 0, 1, 2, 1, 0; RGB: 134.93, 74.61, 33.08; 195.41, 142.99, 72.1; 223.26, 177.46, 101.37; 126.9, 72.45, 35.79; 131.75, 98.95, 70.42; 101.44, 78, 56.71; contrasted |
| 8325, UASM370025 | *Polygonia faunus* | CANADA: Alberta, Medicine Lodge Hills (52.494, -114.323); date collected: 2012-05-20; collector(s): Acorn, B., Aerial net | KY318538; NA | visually scored: 1, 1, 1, 1, 2, ?, 0, 1, 0, 1; RGB: 137.52, 54.76, 13.42; 199.23, 139.01, 49.8; 212.56, 174.48, 113.69; 123.17, 71.39, 36.28; 117.47, 85.47, 59.1; 98.94, 73.52, 51.98; smeared |
| 8326, UASM370026 | *Polygonia faunus* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2012-05-28; collector(s): Dupuis, J. R., Aerial net | KY318539; pf_A-Ida | visually scored: 0, 1, 1, 1, 0, 0, 1, 1, 0, 1; RGB: 111.51, 59.54, 20.77; 147.6, 104.16, 35.07; 163.64, 118.14, 36.19; 94.32, 48.99, 19.57; 98.94, 84.9, 64.42; 90.89, 79.09, 60.22; smeared |

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| 8327, UASM370027 | *Polygonia faunus* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.081); date collected: 2012-05-27; collector(s): Sperling, F. A. H., Aerial net | KY318540; pf_A-WS6 | visually scored: 2, 1, 1, 1, 1, 1, 1, 3, 0, 0; RGB: 118.72, 56.66, 17.14; 154.57, 87.89, 9.16; 163.57, 113, 25.93; 100.62, 56.12, 24.2; 96.7, 80.92, 61.09; 79.81, 68.44, 52.54; contrasted |
| 8328, UASM370028 | *Polygonia faunus* | CANADA: Alberta, Whitecourt Mountain (54.031, -115.721); date collected: 2012-06-07; collector(s): Dupuis, J. R., Aerial net | KY318541; pf_8328 | visually scored: 1, 1, 1, 1, 2, ?, 0, 1, 0, 1; RGB: 160, 89.13, 37.84; 205.11, 143.45, 56.13; 213.6, 166.73, 86.3; 184.99, 125.23, 65.52; 171.14, 133.95, 97.03; 152.84, 121.23, 90.35; smeared |
| 8329, UASM370029 | *Polygonia faunus* | CANADA: Alberta, Whitecourt Mountain (54.031, -115.721); date collected: 2012-06-07; collector(s): Dupuis, J. R., Aerial net | KY318542; NA | visually scored: 1, 1, 1, 2, 0, 0, 1, 2, 0, 0; RGB: 170.78, 92.39, 27; 227.83, 160.92, 56.84; 228.7, 174.62, 77.05; 163.46, 90.81, 28.84; 168.12, 125.29, 84.08; 131.3, 100.09, 70.01; contrasted |
| 8330, UASM370030 | *Polygonia faunus* | CANADA: Alberta, Whitecourt Mountain (54.031, -115.721); date collected: 2012-06-07; collector(s): Dupuis, J. R., Aerial net | NA | visually scored: 1, 1, 1, 2, 1, 0, 1, 2, 0, 0; RGB: NA |
| 8331, UASM370031 | *Polygonia faunus* | CANADA: Alberta, Whitecourt Mountain (54.031, -115.721); date collected: 2012-06-07; collector(s): Dupuis, J. R., Aerial net | KY318543; NA | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 0, 0; RGB: 156.96, 85.59, 28.63; 211.18, 143.2, 35.58; 218.4, 170.39, 83.39; 155.85, 98.34, 43.45; 186.13, 150.58, 110.7; 136.29, 105.48, 75.56; contrasted |
| 8332, UASM370032 | *Polygonia faunus* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.082); date collected: 2012-06-16; collector(s): Sperling, F. A. H., Aerial net | KY318544; pf_8332 | visually scored: 1, 1, 1, 2, 2, 0, 1, 2, 1, 1; RGB: 152.33, 85.24, 35.65; 198.74, 124.12, 26.61; 213.6, 160.49, 75.88; 160.29, 90.86, 34.77; 144.41, 105.65, 74.63; 116.98, 87.17, 62.26; contrasted |
| 8333, UASM370033 | *Polygonia faunus* | CANADA: Alberta, Edith Ravine (53.506, -113.613); date collected: 2012-08-07; collector(s): Acorn, J., Bait trap | KY318545; pf_A-Ida | visually scored: 0, 0, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 125.64, 55.82, 15.94; 197.11, 128.25, 16.99; 203.71, 138.88, 17.92; 121.52, 61.07, 11.13; 139.61, 112.93, 83.33; 102.59, 84.2, 62.04; contrasted |
| 8334, UASM370034 | *Polygonia faunus* | CANADA: Alberta, Edith Ravine (53.506, -113.613); date collected: 2012-08-06; collector(s): Acorn, J., Bait trap | KY318546; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, 0, 1, 1, 1, 0; RGB: 135.57, 59.55, 12.32; 164.41, 99.64, 4.58; 175.23, 122.66, 13.59; 76.52, 35.02, 16.72; 106.21, 96.98, 78.86; 82.25, 76.16, 60.33; contrasted |
| 8335, UASM370035 | *Polygonia faunus* | CANADA: Alberta, Edith Ravine (53.506, -113.613); date collected: 2012-08-04; collector(s): Acorn, J., Bait trap | KY318547; pf_F-NHa3 | visually scored: 2, 1, 1, 1, 2, 0, 1, 1, 1, 1; RGB: 142.77, 68.78, 16.39; 186.5, 108.92, 2.47; 192.1, 131.82, 18.03; 108.93, 53.72, 19.36; 119.53, 92.69, 68.25; 101.88, 78.98, 57.29; smeared |
| 8336, UASM370036 | *Polygonia faunus* | CANADA: Alberta, Edith Ravine (53.506, -113.613); date collected: 2012-08-09; collector(s): Acorn, J., Bait trap | KY318548; pf_A-Ida | visually scored: 1, 1, 1, 2, 1, 0, 1, 2, 2, 0; RGB: 68.23, 32.61, 20.66; 140.24, 79.05, 11.08; 157.25, 90.48, 7.1; 72.92, 34.4, 17.38; 114.25, 102.74, 85.88; 83.31, 75.55, 60.44; contrasted |
| 8337, UASM370037 | *Polygonia faunus* | CANADA: Alberta, North Ram River (52.269, -115.915); date collected: 2012-08-13; collector(s): Dupuis, J. R., Aerial net | KY318549; pf_A-Ida | visually scored: 0, 1, 1, 2, 0, 0, 1, 2, 1, 0; RGB: 135.67, 59.02, 14.99; 191.86, 113.57, 7.75; 212.8, 147.99, 35.2; 123.06, 57.79, 20.31; 111.19, 84.54, 57.01; 77.77, 62.17, 43.35; contrasted |
| 8338, UASM370038 | *Polygonia faunus* | CANADA: Alberta, North Ram River (52.269, -115.915); date collected: 2012-08-13; collector(s): Dupuis, J. R., Aerial net | KY318550; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, 0, 1, 3, 1, 1; RGB: 141.96, 69.4, 18.3; 199.02, 127.02, 15.34; 205.57, 146.45, 38.91; 114.85, 52.78, 20.41; 115.14, 86.55, 59.7; 104.74, 81.44, 57.59; smeared |
| 8339, UASM370039 | *Polygonia faunus* | CANADA: Alberta, North Ram River (52.269, -115.915); date collected: 2012-08-13; collector(s): Dupuis, J. R., Aerial net | KY318551; pf_8339 | visually scored: 2, 1, 1, 1, 1, 0, 1, 1, 1, 1; RGB: 158.94, 83.56, 15.82; 189.02, 112.47, 7.65; 199.21, 134.46, 20.81; 137.48, 67.52, 13.68; 111.07, 77.67, 50.63; 97.55, 72.16, 50.23; smeared |
| 8340, UASM370040 | *Polygonia faunus* | CANADA: Alberta, North Ram River (52.269, -115.915); date collected: 2012-08-13; collector(s): Dupuis, J. R., Aerial net | KY318552; NA | visually scored: 0, 1, 1, 2, 0, 0, 1, 2, 1, 0; RGB: 144.04, 68.41, 22.28; 207.56, 139.33, 28.99; 212.74, 144.38, 26.15; 131.14, 66.54, 17.99; 141.94, 106.58, 74.29; 101.59, 78, 54.75; contrasted |

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| 8341, UASM370041 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318613; NA | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 149.97, 65.64, 11.53; 187.54, 99.34, 5.26; 203.8, 127.2, 19.05; 117.03, 52.74, 17.35; 121.22, 92.54, 70.38; 96.45, 76.13, 58.86 |
| 8342, UASM370042 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318614; NA | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 162.64, 67.9, 6.12; 189.13, 102.86, 6.68; 195.26, 117.95, 5.39; 149.05, 70.37, 17.94; 123.71, 96.04, 70.47; 101.3, 80.99, 61.68 |
| 8343, UASM370043 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318615; pp_8343 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 167.72, 84.32, 8.32; 194.68, 125.07, 24.36; 213.69, 145.44, 28.42; 112.1, 50.45, 16.82; 113.43, 84.89, 62.63; 92.66, 71.59, 53.92 |
| 8344, UASM370044 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318616; pp_8344 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 164.82, 81.63, 9.16; 191.94, 109.78, 3.8; 204.11, 126.52, 9.66; 109.99, 53.83, 25.22; 98.31, 74.58, 55.07; 83.31, 64.66, 48.18 |
| 8345, UASM370045 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318617; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 1, 1, 0, 1; RGB: 166.49, 99.25, 30.79; 188.22, 125.36, 39.31; 204.76, 137.14, 33.14; 139.49, 71.25, 29.85; 112.26, 86.2, 65.83; 93.33, 73.23, 56.82 |
| 8346, UASM370046 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318618; pp_P-C | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 172.32, 75.97, 8.13; 197.15, 106.34, 4.31; 203.04, 120.72, 10.3; 88.72, 36.55, 18.55; 90.38, 69.73, 51.26; 78.07, 60.72, 44.84 |
| 8347, UASM370047 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318619; pp_8347 | visually scored: 3, 1, 1, 0, 2, 1, 1, 1, 0, 1; RGB: 180.41, 74.33, 5.94; 203.36, 104.88, 4.7; 225.27, 150.73, 24.95; 94.78, 38.84, 18.95; 117.98, 94.86, 72.72; 97.97, 77.69, 58.9 |
| 8348, UASM370048 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318620; pp_8348 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 155.57, 62.82, 10.32; 168.63, 80.98, 4.2; 180.38, 109.42, 12.98; 116.67, 49.72, 15.3; 102.6, 78.76, 58.98; 85.74, 65.89, 48.9 |
| 8349, UASM370049 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318621; pp_P-A | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 165.7, 71.15, 8.55; 194.5, 113.24, 19.26; 199.19, 126.55, 23.6; 107.43, 43.3, 18.66; 110.78, 84.92, 64.46; 96.14, 74.23, 56.41 |
| 8350, UASM370050 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318622; pp_8350 | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 169.81, 66.17, 5.7; 180.76, 87.64, 2.47; 191.4, 114.08, 8.46; 104.58, 43.58, 20.76; 103.75, 81.86, 63.17; 86.65, 69.1, 53.39 |
| 8351, UASM370051 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318623; pp_8351 | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 183.82, 98.49, 16.42; 199.89, 112.05, 3.47; 214.58, 142, 19.02; 122.97, 51.98, 20.08; 118.48, 96.89, 75.94; 97.42, 80.28, 62.94 |
| 8352, UASM370052 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318624; pp_8352 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 171.33, 70.4, 4.92; 204.95, 119.02, 15.57; 215.23, 132.61, 16.69; 110.81, 41.73, 20.47; 86.62, 68.16, 49.72; 76.2, 60.81, 44.95 |
| 8353, UASM370053 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318625; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 168.06, 79.61, 11.38; 198.16, 111.45, 6.33; 212.72, 137.58, 17.84; 108.54, 43.85, 18.54; 86.9, 65.33, 47.39; 75.47, 57.4, 41.73 |
| 8354, UASM370054 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318626; pp_8354 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 176.83, 78.07, 3.64; 200.1, 110.04, 6.05; 208.55, 128.92, 10.76; 128.29, 49.94, 20.36; 112.99, 87.84, 65.5; 92.85, 73.56, 55.06 |

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| 8355, UASM370055 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318627; pp_8355 | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 177.97, 82.46, 8.79; 188.14, 107.32, 14.58; 201.19, 115.46, 7.7; 116.1, 47.22, 13.73; 109.19, 86.41, 64.92; 90.35, 72.59, 55.37 |
| 8356, UASM370056 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318628; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 136.17, 69.7, 17.53; 158.53, 81.56, 5.06; 163.63, 97.97, 5.83; 98.54, 48.53, 18.86; 83.12, 73.46, 59.87; 72, 64.69, 52.71 |
| 8357, UASM370057 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318629; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 186.52, 83.08, 2.3; 208.77, 125.84, 15.01; 214.24, 145.76, 28; 95.9, 39.02, 20.34; 126.22, 99.9, 76.59; 108, 84.79, 64.86 |
| 8358, UASM370058 | *Polygonia satyrus* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.078); date collected: 2012-07-02; collector(s): Sperling, F. A. H., Aerial net | KY318701; ps_8358 | visually scored: 0, 0, 0, 1, 0, 1, 0, 2, 0, 0; RGB: 135.5, 47.83, 8.5; 194.72, 101.48, 1.34; 207.76, 127.62, 3.83; 164.13, 78.31, 9.17; 140.36, 90.09, 48.17; 120.44, 74.51, 41.67; smeared |
| 8359, UASM370059 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2011-05-16; collector(s): Sperling, F. A. H., Aerial net | KY318630; pp_P-C | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 181.81, 99.84, 26.18; 203, 116.86, 15.04; 212.45, 136.13, 16.92; 100.95, 44.01, 22.92; 113.08, 89.25, 66.18; 98.53, 78.17, 58.69 |
| 8360, UASM370060 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2011-05-16; collector(s): Sperling, F. A. H., Aerial net | KY318631; pp_P-A | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 158.32, 65.55, 11.35; 198.71, 105.66, 5.05; 205.73, 120.95, 6.6; 117.03, 45.34, 17.64; 106.01, 84.39, 62.67; 89.3, 72.69, 54.34 |
| 8361, UASM370061 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2011-05-16; collector(s): Sperling, F. A. H., Aerial net | KY318632; pp_8361 | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 180.04, 68.11, 6.05; 195.8, 106.1, 12.69; 211.79, 143.72, 38.58; 122.42, 56, 25.37; 139.19, 110.98, 84.71; 118.85, 94.16, 70.72 |
| 8362, UASM370062 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2011-05-16; collector(s): Sperling, F. A. H., Aerial net | KY318633; pp_8362 | visually scored: 2, 1, 1, 1, 2, 1, -, 0, -, 1; RGB: 147.8, 66.58, 11.13; 174.6, 98.25, 7.74; 189.71, 121.26, 26.7; 112.3, 56.53, 26.93; 97.99, 73.62, 54.99; 82.86, 63.43, 47.79 |
| 8363, UASM370063 | *Polygonia progne* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.083); date collected: 2011-07-02; collector(s): Sperling, F. A. H., Aerial net | KY318634; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 163.33, 74.6, 14.66; 185.52, 109.81, 22.9; 214.06, 150.82, 59.02; 157.52, 87.31, 33.92; 136.43, 96.98, 66.78; 104.81, 77.11, 55.05 |
| 8364, UASM370064 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-13; collector(s): Sperling, F. A. H., Aerial net | KY318635; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 159.99, 67.65, 10.83; 186.11, 107.97, 14.11; 200.43, 124.25, 17.7; 131.81, 55.87, 21.89; 106.91, 78.97, 57.33; 89.31, 68.25, 50.84 |
| 8365, UASM370065 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-13; collector(s): Sperling, F. A. H., Aerial net | KY318636; pp_P-A | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 145.96, 66.3, 13.62; 172.76, 111.68, 33.83; 193.13, 127.77, 28.61; 109.36, 57.63, 27.48; 114.93, 89.08, 65.64; 96.28, 75.4, 56.57 |
| 8366, UASM370066 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-13; collector(s): Sperling, F. A. H., Aerial net | KY318637; pp_P-C | visually scored: 3, 0, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 141.48, 62.25, 13.86; 182.13, 98.34, 8.24; 195.7, 122.92, 16.14; 145.56, 66.71, 19.8; 119.85, 94.74, 70.13; 95.05, 75.81, 56.75 |
| 8367, UASM370067 | *Polygonia progne* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2010-06-12; collector(s): Sperling, F. A. H., Aerial net | KY318638; pp_P-C | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 158.12, 86.24, 28.42; 191.07, 125.86, 46.47; 205.38, 155.2, 82.21; 140.21, 82.18, 39.73; 118.04, 90.57, 65.13; 92.02, 71.98, 53.11 |
| 8368, UASM370068 | *Polygonia progne* | CANADA: Alberta, Peace Park near Wetaskewin (53.017, -113.365); date collected: 2012-05-06; collector(s): Bird, H., Aerial net | KY318639; pp_P-A | visually scored: 3, 0, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 170.45, 76.36, 9.56; 196.33, 115.53, 14.45; 205.86, 129.55, 37.86; 144.83, 61.46, 18.59; 114.19, 86.49, 61.76; 94.54, 72.88, 53.24 |

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| 8369, UASM370069 | *Polygonia progne* | CANADA: Alberta, Peace Park near Wetaskewin (53.017, -113.365); date collected: 2012-05-06; collector(s): Bird, H., Aerial net | KY318640; pp_P-A | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 155.65, 70.52, 8.68; 185.27, 103.67, 10.39; 197.4, 121.09, 26.49; 121.9, 55.27, 23.96; 115.61, 88.74, 66.16; 94.9, 73.59, 55.08 |
| 8370, UASM370070 | *Polygonia progne* | CANADA: Alberta, Hand Hills (51.418, -112.285); date collected: 2012-05-20; collector(s): Dupuis, J. R., Aerial net | KY318641; pp_P-D | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 143.78, 65.51, 9.22; 152.01, 96.37, 16.23; 161.93, 113.73, 26.3; 102.38, 47.72, 20.89; 92.25, 79.5, 61.36; 78.27, 68.5, 53.94 |
| 8371, UASM370071 | *Polygonia progne* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2012-05-28; collector(s): Dupuis, J. R., Aerial net | KY318642; pp_8371 | visually scored: 3, 1, 1, 1, 2, -, 1, 0, 0, 1; RGB: 173.83, 93.76, 16.3; 203.25, 135.53, 34.27; 203.08, 148.5, 53.4; 144.17, 78.32, 32.96; 134.87, 99.65, 69.95; 97.89, 73.12, 53.57 |
| 8372, UASM370072 | *Polygonia progne* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2012-05-28; collector(s): Dupuis, J. R., Aerial net | KY318643; pp_P-A | visually scored: 3, 0, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 130.28, 62.25, 14.14; 155.71, 97.75, 16.61; 165.31, 110.95, 22.96; 113.71, 68.99, 30.61; 98.03, 80.07, 63.64; 84.16, 71.14, 57.21 |
| 8373, UASM370073 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-06-03; collector(s): Sperling, F. A. H., Aerial net | KY318644; pp_P-A | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 152.06, 89.31, 11.02; 165.91, 119.08, 32.73; 180.7, 136.88, 39.91; 116.18, 75.23, 29.79; 91.62, 78.56, 63.29; 81.87, 70.59, 56.67 |
| 8374, UASM370074 | *Polygonia progne* | CANADA: Alberta, Hand Hills (51.418, -112.285); date collected: 2012-05-20; collector(s): Dupuis, J. R., Aerial net | KY318645; pp_P-A | visually scored: 2, 1, 1, 1, 2, -, 0, 0, 0, 1; RGB: 143.93, 76.24, 15.59; 144.34, 95.5, 27.15; 156.53, 109.43, 24.68; 98.65, 53.33, 26.77; 93.11, 76.9, 60.32; 84.43, 71.06, 55.81 |
| 8375, UASM370075 | *Polygonia satyrus* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.536); date collected: 2012-07-13; collector(s): Sperling, F. A. H., Aerial net | KY318699; ps_S-B | visually scored: 0, 0, 0, 2, 0, -, 0, 2, 0, 0; RGB: 163.82, 65.59, 6.71; 202, 98.8, 1.88; 208.25, 121.91, 13.28; 146.22, 66.02, 13.92; 141.76, 90.03, 48.43; 124.36, 75.58, 41.16; smeared |
| 8376, UASM370076 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-26; collector(s): Sperling, F. A. H., Aerial net | KY318646; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 149.11, 69.77, 6.33; 160.91, 112.37, 21.85; 170.8, 121.63, 26.9; 95.9, 50.57, 21.16; 95.41, 81.12, 62.79; 75.01, 64.77, 51.16 |
| 8377, UASM370077 | *Polygonia progne* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2012-05-28; collector(s): Dupuis, J. R., Aerial net | KY318647; pp_8377 | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 134.96, 72.55, 18.52; 149.54, 105.53, 31.07; 155.25, 116.7, 42.57; 101.41, 55.93, 25.37; 104.22, 92.86, 74.75; 88.2, 79.33, 64.13 |
| 8378, UASM370078 | *Polygonia progne* | CANADA: Alberta, Hand Hills (51.418, -112.285); date collected: 2012-05-20; collector(s): Dupuis, J. R., Aerial net | KY318648; pp_8378 | visually scored: 2, 1, 1, 1, 2, 1, 1, 1, 0, 1; RGB: 144.25, 68.49, 7.94; 158.77, 95.22, 9.29; 160.48, 118.31, 40.79; 114.07, 58.3, 18.4; 86.86, 75.16, 59.74; 76.69, 67.81, 54.34 |
| 8379, UASM370079 | *Polygonia progne* | CANADA: Alberta, Medicine Lodge Hills (52.494, -114.323); date collected: 2012-05-20; collector(s): Acorn, J., Aerial net | KY318649; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 136.43, 75.09, 16.93; 152.12, 108.34, 35.94; 161.36, 115.36, 35.35; 86.33, 41.26, 22.01; 90.61, 78.98, 63.67; 88.21, 76.59, 60.88 |
| 8380, UASM370080 | *Polygonia progne* | CANADA: Alberta, Edith Ravine (53.506, -113.613); date collected: 2012-08-07; collector(s): Acorn, J., Bait trap | KY318650; pp_8380 | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 146.69, 71.42, 4.55; 157.82, 93.03, 5.09; 165.73, 110.05, 10.32; 78.38, 38.65, 22.41; 87.9, 83.12, 69.34; 77.06, 72.8, 60.78 |
| 8381, UASM370081 | *Polygonia progne* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318651; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 0; RGB: 137.64, 66.31, 14.26; 141.95, 82.56, 20.85; 156.28, 84.59, 6.01; 85.34, 44.42, 23.55; 94.93, 83.91, 69.75; 80.29, 71.13, 58.84 |
| 8382, UASM370082 | *Polygonia gracilis gracilis* | CANADA: Alberta, Nordegg (52.468, -116.082); date collected: 2012-04-30; collector(s): Soderlind, L., Aerial net | KY318681; pg_8382 | visually scored: 2, 0, 0, 1, 2, 1, 0, 0, 0, 0; RGB: 104.14, 46, 19.28; 148.47, 85.62, 13.02; 157.5, 116.08, 30.86; 121.77, 75.33, 26.4; 114.33, 110.45, 99; 86.67, 82.26, 70.93 |

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| 8383, UASM370083 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-06-03; collector(s): Sperling, F. A. H., Aerial net | KY318553; NA | visually scored: 0, 1, 1, 2, 0, 0, 1, 0, 0, 0; RGB: 141.43, 73.42, 28.88; 198.98, 119.21, 24.26; 215.27, 161.17, 75.18; 121.91, 72.83, 38.52; 138.75, 103.28, 71.84; 109.28, 82.11, 57.8; contrasted |
| 8384, UASM370084 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2012-05-26; collector(s): Sperling, F. A. H., Aerial net | KY318652; pp_P-A | visually scored: 3, 0, 1, 1, 2, -, 1, 0, 0, 0; RGB: 165.54, 96.02, 22.06; 177.77, 119.51, 33.35; 196.4, 139.15, 40.7; 157.26, 83.49, 19.07; 142.51, 107.22, 74.23; 118.66, 89.74, 63.66 |
| 8385, UASM370085 | *Polygonia satyrus* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.084); date collected: 2011-08-21; collector(s): Sperling, F. A. H., Aerial net | KY318700; ps_S-A | visually scored: 0, 0, 0, 2, 0, -, 0, 2, 0, 0; RGB: 74.52, 41.06, 27.41; 159.7, 88.68, 6.77; 169.4, 103.88, 10.76; 91.44, 45.34, 19.07; 97.12, 69.06, 48.99; 81.49, 57.96, 41.97; smeared |
| 8386, UASM370086 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, east side (52.939, -114.288); date collected: 2011-06-12; collector(s): Sperling, F. A. H., Aerial net | KY318611; pf_A-Ida | visually scored: 0, 1, 1, 1, 2, 0, 1, 0, 0, 0; RGB: 99.12, 62.14, 30.28; 150.43, 107.19, 30.51; 169.15, 134.27, 64.49; 112.55, 68.27, 28.05; 100.83, 90.6, 72.87; 88.15, 79.12, 62.38; contrasted |
| 8387, UASM370087 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, east side (52.939, -114.288); date collected: 2011-06-12; collector(s): Sperling, F. A. H., Aerial net | KY318653; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 139.54, 67.05, 17.72; 166.01, 97.72, 15.19; 175.85, 121.22, 25.07; 101.18, 51.24, 23.51; 96.68, 86.26, 70.35; 89.11, 79.23, 64.57 |
| 8388, UASM370088 | *Polygonia gracilis zephyrus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Dupuis, J. R., Aerial net | KY318682; pg_G-A | visually scored: 2, 0, 0, 1, 2, 1, 1, 0, 0, 0; RGB: 108.01, 47.66, 19.66; 150.1, 81.95, 8.22; 160.17, 107.94, 18.27; 133.64, 84.74, 20.7; 133.55, 125.77, 113.39; 106.79, 97.1, 83.09 |
| 8389, UASM370089 | *Polygonia gracilis zephyrus* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Dupuis, J. R., Aerial net | KY318683; pg_G-A | visually scored: 1, 0, 0, 1, 2, -, 1, 0, 0, 0; RGB: 127.22, 59.47, 26.2; 191.49, 112.19, 29.13; 206.7, 156.8, 88.37; 165.59, 99.53, 31.3; 169.96, 142, 114.77; 131.04, 104.59, 81.27 |
| 8390, UASM370090 | *Polygonia gracilis gracilis* | CANADA: Alberta, Shunda viewpoint (52.483, -115.738); date collected: 2012-05-15; collector(s): Soderlind, L., Aerial net | KY318684; pg_G-B | visually scored: 2, 1, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 90.88, 52.82, 26.83; 152.79, 92.96, 14.09; 155.57, 113.99, 45.14; 106.31, 67.18, 29.71; 124.38, 111.55, 94.31; 94.71, 82.48, 67.25 |
| 8391, UASM370091 | *Polygonia gracilis gracilis* | CANADA: Alberta, Edith Ravine (53.506, -113.613); date collected: 2012-08-07; collector(s): Acorn, J., Bait trap | KY318654; pp_8391 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 141.01, 61.22, 9.29; 157.15, 83.2, 3.85; 169.47, 98.6, 5.03; 84.12, 37.68, 18.05; 91.46, 86.09, 74.16; 76.99, 72.59, 61.17 |
| 8392, UASM370092 | *Polygonia gracilis gracilis* | CANADA: Alberta, Ram Lookout (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Aerial net | KY318685; pg_G-B | visually scored: 2, 0, 1, 1, 1, 0, 0, 0, 0, 0; RGB: 85.53, 43.43, 25.5; 161.75, 96.76, 4.72; 176.45, 126.01, 17.36; 101.61, 53.68, 18.88; 121.83, 117.78, 104.65; 90.8, 86.44, 73.72 |
| 8393, UASM370093 | *Polygonia satyrus* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-07-13; collector(s): Sperling, F. A. H., Aerial net | KY318702; ps_S-B | visually scored: 0, 0, 0, 2, 0, 0, 0, 2, 0, 0; RGB: 122.22, 45.09, 13.36; 183.63, 91.09, 0.79; 185.76, 117.51, 5.49; 139.79, 75.86, 13.4; 150.56, 125.26, 90.97; 121.33, 92.9, 61.82; contrasted |
| 8394, UASM370094 | *Polygonia satyrus* | CANADA: Alberta, Waterton area (49.002, -113.907); date collected: 2012-08-08; collector(s): Sperling, F. A. H., Aerial net | KY318703; ps_S-C | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 122.01, 46.05, 12.44; 165.91, 95.41, 6.99; 162.17, 108.3, 26.45; 121.27, 68.87, 14.37; 126.28, 94.04, 62.26; 104.43, 74.85, 47.64; contrasted |
| 8395, UASM370095 | *Polygonia satyrus* | CANADA: Alberta, Ram Lookout (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Aerial net | KY318704; ps_S-A | visually scored: 1, 0, 0, 1, 1, 0, 0, 2, 0, 1; RGB: 125.57, 57, 12.48; 170.09, 88.74, 0.71; 180.24, 112.72, 4.55; 121.09, 61.83, 11.2; 116.93, 83.27, 50.67; 107.21, 75.81, 47.06; smeared |
| 8396, UASM370096 | *Polygonia satyrus* | CANADA: Alberta, Edith ravine (53.506, -113.613); date collected: 2012-08-29; collector(s): Acorn, J., Bait trap | KY318705; ps_S-B | visually scored: 0, 0, 0, 2, 0, 0, 0, 2, 0, 1; RGB: 108.7, 44.01, 15.42; 168.32, 96.72, 4.07; 170.85, 105.56, 4.07; 116.3, 57.19, 15.5; 116.89, 91.26, 66.6; 102.9, 76.91, 55.02; smeared |

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| 8397, UASM370097 | *Polygonia satyrus* | CANADA: Alberta, Rowley (51.766, -112.788); date collected: 2012-07-22; collector(s): Dupuis, J. R., Aerial net | KY318706; ps_8397 | visually scored: 0, 0, 0, 2, 0, 0, 0, 2, 0, 1; RGB: 117.18, 48.07, 17.22; 167.36, 88.24, 3.56; 174.25, 108.88, 6.8; 121.35, 60.38, 19.86; 123.01, 88.08, 55.39; 113.8, 79.42, 49.62; smeared |
| 9201, UASM370098 | *Polygonia satyrus* | CANADA: Alberta, Bragg Creek, Sperling farm (50.921, -114.527); date collected: 2012-05-07; collector(s): Sperling, F. A. H., Aerial net | KY318707; ps_S-A | visually scored: 0, 0, 0, 1, 0, 0, 0, ?, 0, 1; RGB: 170.04, 82.24, 21; 204.73, 135.53, 37.68; 220.97, 164.37, 59.61; 165.06, 88.72, 19; 155.24, 105.63, 60.65; 131.52, 87.92, 52.39; smeared |
| 9202, UASM370099 | *Polygonia satyrus* | CANADA: Alberta, Edmonton, Glenora (53.547, -113.558); date collected: 2012-08-20; collector(s): Acorn, J., Bait trap | NA | visually scored: 0, 0, 0, 1, 0, 0, 0, 2, 0, 0; RGB: NA |
| 9203, UASM370100 | *Polygonia satyrus* | CANADA: Alberta, Edmonton, Glenora (53.547, -113.558); date collected: 2012-08-20; collector(s): Acorn, J., Bait trap | KY318708; ps_9203 | visually scored: 0, 0, 0, 2, 0, 0, 0, 2, 0, 0; RGB: 175.07, 93.96, 23.92; 226.1, 171.45, 73.97; 229.3, 179.5, 84.16; 184.41, 105.39, 24.9; 186.92, 134.01, 82.6; 159.03, 115.48, 75.74; smeared |
| 9204, UASM370101 | *Polygonia satyrus* | CANADA: Alberta, Edmonton, Glenora (53.547, -113.558); date collected: 2012-08-20; collector(s): Acorn, J., Bait trap | KY318709; ps_S-A | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 143.49, 61.32, 15.65; 195.13, 111.64, 14.03; 195.56, 117.94, 15.32; 149.88, 71, 14.12; 164.45, 107.72, 60.18; 132.93, 84.25, 48.17; contrasted |
| 9205, UASM370102 | *Polygonia satyrus* | CANADA: Alberta, Ram Lookout (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Aerial net | KY318710; ps_S-A | visually scored: 0, 0, 0, 2, 0, 0, 0, 2, 0, 0; RGB: 166.13, 79.17, 10.22; 212.01, 127.79, 1.99; 217.28, 149.04, 6.86; 182.76, 99.06, 4.01; 158.68, 104.72, 55.96; 134.71, 87.54, 48; smeared |
| 9206, UASM370103 | *Polygonia satyrus* | CANADA: Alberta, Ram Lookout (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Aerial net | KY318711; ps_S-A | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 132.14, 57.77, 17.65; 204.04, 110.07, 2.91; 206.18, 114.73, 2.36; 134.94, 60.48, 11.08; 161.48, 107.33, 54.23; 113.25, 71.74, 38.18; contrasted |
| 9207, UASM370104 | *Polygonia satyrus* | CANADA: Alberta, Ram Lookout (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Aerial net | KY318712; ps_S-A | visually scored: 0, 0, 0, 1, 0, 0, 0, ?, 0, 1; RGB: 132.88, 49.11, 10.08; 194.99, 96.71, 1.76; 209.13, 120.13, 4.04; 169.01, 72.26, 9.13; 137.92, 87.19, 45.73; 121.77, 76.07, 41.03; smeared |
| 9208, UASM370105 | *Polygonia satyrus* | CANADA: Alberta, Ram Lookout (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Aerial net | KY318713; ps_S-A | visually scored: 1, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 127.02, 48.16, 11.07; 197.43, 96.75, 0.99; 200.1, 112.8, 6.76; 141.48, 65.79, 13.65; 148.66, 98.65, 57.95; 114.83, 73.22, 42.88; contrasted |
| 9209, UASM370106 | *Polygonia satyrus* | CANADA: Alberta, Waterton area (49.14, -113.851); date collected: 2012-08-08; collector(s): Sperling, F. A. H., Aerial net | KY318714; ps_S-A | visually scored: 0, 0, 1, 1, 0, 1, 1, 2, 0, 0; RGB: 118.53, 48.17, 22.58; 187.36, 101.62, 8.25; 186.95, 113.4, 19.25; 151.56, 88.41, 39.37; 131.39, 87.14, 54.92; 96.81, 62.71, 41.38; contrasted |
| 9210, UASM370107 | *Polygonia satyrus* | CANADA: Alberta, Waterton area (49.14, -113.851); date collected: 2012-08-08; collector(s): Sperling, F. A. H., Aerial net | KY318715; ps_S-D | visually scored: 1, 0, 0, 1, 0, 0, 0, 2, 0, 1; RGB: 126.49, 52.31, 10.6; 165.63, 91.25, 2.46; 167.14, 105.04, 6.17; 123.23, 62.15, 15.21; 105.39, 73.16, 49.12; 94.92, 63.49, 42.67; smeared |
| 9211, UASM370108 | *Polygonia satyrus* | CANADA: Alberta, Whitecourt Mountain (54.031, -115.721); date collected: 2012-06-07; collector(s): Dupuis, J. R., Aerial net | KY318716; ps_8358 | visually scored: 0, 0, 0, 2, 1, 0, 1, 2, 0, 0; RGB: 192.66, 109.36, 23.14; 233.63, 171.86, 58.66; 234.75, 178.7, 73.5; 202.17, 141.43, 63.3; 202.73, 157.28, 102.15; 162.21, 119.1, 76.62; contrasted |
| 9212, UASM370109 | *Polygonia satyrus* | CANADA: Alberta, Whitecourt Mountain (54.031, -115.721); date collected: 2012-06-07; collector(s): Dupuis, J. R., Aerial net | KY318717; ps_9212 | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 185.18, 105.79, 34.28; 225.85, 166.77, 73.25; 232.79, 186.49, 108.16; 189.73, 126.29, 61.88; 198.98, 157.31, 107.85; 160, 118.99, 76.92; contrasted |
| 9213, UASM370110 | *Polygonia satyrus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-04-28; collector(s): Sperling, F. A. H., Aerial net | KY318718; ps_S-A | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 155.78, 75.72, 17.81; 202.83, 122.27, 7.54; 210.74, 145.97, 24.44; 160.86, 90.25, 17.41; 177.89, 131.93, 82.62; 141.14, 97.95, 57.11; contrasted |

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| 9214, UASM370111 | *Polygonia satyrus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318719; ps_S-A | visually scored: 0, 0, 0, 1, 0, 0, 0, 2, 0, 0; RGB: 153.29, 69.19, 20.68; 209.09, 123.63, 15.05; 215.67, 140.73, 19.65; 180.58, 99.48, 18.68; 187.06, 133.63, 79.19; 165.44, 116.57, 70.07; smeared |
| 9215, UASM370112 | *Polygonia satyrus* | CANADA: British Columbia, Oliver (49.092, -119.538); date collected: 2013-05-13; collector(s): McDonald, C., Aerial net | KY318720; ps_9215 | visually scored: 0, 0, 0, 1, 0, -, 0, ?, 0, 0; RGB: 164.77, 78.28, 18.27; 215.63, 147, 41.62; 218.26, 159.26, 51.87; 174.01, 106.75, 41.11; 170.03, 125.21, 80.08; 155.83, 115.13, 74.47; smeared |
| 9216, UASM370113 | *Polygonia satyrus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-05-19; collector(s): Sperling, F. A. H., Aerial net | KY318721; ps_8358 | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 180.67, 102.58, 34.61; 227.58, 169.4, 72.22; 230.22, 167.29, 54.13; 180.31, 110.86, 40.32; 204.69, 151.78, 93.57; 159.35, 109.71, 66.73; contrasted |
| 9217, UASM370114 | *Polygonia satyrus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 53 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318722; ps_S-B | visually scored: 0, 0, 0, 2, 0, -, 1, 2, 0, 0; RGB: 174.69, 78.01, 9.53; 214.8, 143.78, 40.24; 221.81, 157.41, 54.81; 187.11, 116.48, 38.29; 192.45, 149.65, 101.88; 155.69, 112.91, 71.33; contrasted |
| 9218, UASM370115 | *Polygonia satyrus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 54 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318723; ps_9218 | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 168.39, 77.44, 15.18; 213.35, 136.02, 21.47; 221.79, 160.11, 53.43; 178.36, 107.47, 38.77; 185.41, 136.18, 83.99; 147.07, 103.95, 63.57; contrasted |
| 9219, UASM370116 | *Polygonia satyrus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 55 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318724; ps_9219 | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 191.5, 87.83, 8.6; 222.83, 148.48, 33.5; 232.71, 168.08, 54.7; 169.21, 88.85, 20.6; 201.09, 150.19, 98.52; 163.17, 116.12, 73.59; contrasted |
| 9220, UASM370117 | *Polygonia satyrus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-07-30; collector(s): Sperling, F. A. H., Aerial net | KY318725; ps_S-B | visually scored: 0, 0, 0, 1, 1, 0, 1, 2, 0, 0; RGB: 205.09, 129.83, 48.85; 233.03, 172.83, 70.53; 235.95, 194.15, 119.72; 207.76, 141.7, 58.14; 208.97, 165.65, 112.65; 183.11, 137.88, 88.37; contrasted |
| 9221, UASM370118 | *Polygonia satyrus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | KY318726; ps_S-B | visually scored: 0, 0, 1, 1, 0, 0, 1, 2, 0, 0; RGB: 169.81, 91.9, 17.06; 213.3, 142.94, 21.5; 207.79, 141.39, 26.4; 152.3, 90.33, 16.4; 155.64, 106.8, 61.35; 116.08, 77.93, 46.36; contrasted |
| 9222, UASM370119 | *Polygonia satyrus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | NA | visually scored: 1, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: NA |
| 9223, UASM370120 | *Polygonia satyrus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | KY318727; ps_S-B | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 167.96, 82.5, 17; 196.88, 112.18, 6.1; 195.8, 113.81, 3.47; 136.46, 63.7, 15.5; 166.45, 114.38, 66.92; 124.84, 79.29, 45.62; contrasted |
| 9224, UASM370121 | *Polygonia satyrus* | CANADA: Alberta, South of Devonian Botanical Garden (53.396, -113.738); date collected: 2013-08-01; collector(s): McDonald, C., Aerial net | KY318728; ps_S-A | visually scored: 0, 0, 1, 1, 0, 0, 1, 2, 0, 0; RGB: 126.86, 51.19, 16.37; 195.56, 105.82, 2.6; 198.32, 115.51, 3.38; 133.25, 67.08, 11.7; 137.29, 91.94, 59.51; 104.6, 66.59, 41.18; contrasted |
| 9225, UASM370122 | *Polygonia satyrus* | CANADA: Alberta, South of Devonian Botanical Garden (53.396, -113.738); date collected: 2013-08-01; collector(s): McDonald, C., Aerial net | KY318729; ps_S-B | visually scored: 0, 0, 0, 2, 0, 0, 0, 2, 0, 0; RGB: 86.12, 42.16, 24.58; 203.9, 131.28, 14.64; 214.55, 155.1, 40.23; 145.71, 86.36, 35.69; 135.14, 91.2, 57.87; 104.07, 68.87, 45.39; smeared |
| 9226, UASM370123 | *Polygonia satyrus* | CANADA: Alberta, South of Devonian Botanical Garden (53.396, -113.738); date collected: 2013-08-01; collector(s): McDonald, C., Aerial net | KY318730; ps_S-D | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 126.28, 49.46, 14.75; 202.18, 123.93, 17.86; 213.43, 144.82, 24.51; 153.3, 72.58, 20.6; 165.41, 119.2, 73.84; 118.3, 78.11, 46.22; contrasted |

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| 9227, UASM370124 | *Polygonia satyrus* | CANADA: Alberta, Brazeau area, RR 72xTWP 482 (53.032, -115.522); date collected: 2013-08-15; collector(s): McDonald, C., Aerial net | KY318731; ps_9227 | visually scored: 1, 0, 1, 1, 0, 0, 1, 2, 0, 0; RGB: 165.85, 71.06, 13.47; 208.46, 114.42, 5.98; 223.78, 148.83, 27.98; 149.91, 82.24, 16.25; 175.32, 125.5, 76.59; 141.42, 98.32, 58.76; contrasted |
| 9228, UASM370125 | *Polygonia satyrus* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-12; collector(s): McDonald, C., Aerial net | KY318732; ps_S-D | visually scored: 0, 0, 0, 1, 0, 0, 0, 2, 0, 0; RGB: 154.55, 72.53, 18.91; 197.82, 113.95, 1.28; 200.9, 128.23, 10.38; 155.14, 88.41, 23.34; 161.14, 117.33, 75.52; 119.76, 82.99, 52.01; contrasted |
| 9229, UASM370126 | *Polygonia satyrus* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-12; collector(s): McDonald, C., Aerial net | KY318733; ps_9229 | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 133.74, 49.54, 18.12; 216.31, 139.17, 30.08; 212.94, 139.26, 23.4; 147.4, 73.29, 14.19; 139.92, 91.5, 54.55; 107.59, 67.44, 40.12; contrasted |
| 9230, UASM370127 | *Polygonia satyrus* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-12; collector(s): McDonald, C., Aerial net | KY318734; ps_S-D | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 152.97, 63.41, 8.43; 204.69, 114.06, 5.83; 210.57, 130.01, 12.46; 160.87, 94.42, 26.15; 164.59, 113.4, 65.21; 125.85, 84.48, 48.89; contrasted |
| 9231, UASM370128 | *Polygonia satyrus* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-12; collector(s): McDonald, C., Aerial net | KY318735; ps_S-C | visually scored: 0, 0, 0, -, 0, 0, 0, 2, 0, 1; RGB: 132.97, 58.01, 22.92; 198.4, 118.66, 12.74; 211.02, 139.56, 24.84; 148.98, 77.74, 18.42; 134.51, 82.21, 45.3; 116.62, 71.52, 43.08; smeared |
| 9232, UASM370129 | *Polygonia satyrus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318736; ps_9232 | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 140.53, 60.16, 16.09; 201.44, 122.88, 6.86; 202.52, 138.07, 24.12; 141.51, 83.06, 22.11; 140.49, 97.29, 60.9; 99.47, 67.44, 43.34; contrasted |
| 9233, UASM370130 | *Polygonia satyrus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318737; ps_S-C | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 144, 60.59, 12.32; 205.97, 119.14, 7.52; 206.71, 127.5, 5.73; 152.91, 90.09, 26.96; 172.32, 128.56, 80.91; 120.09, 84.06, 50.91; contrasted |
| 9234, UASM370131 | *Polygonia satyrus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318738; ps_S-D | visually scored: 0, 0, 1, 2, 0, 0, 1, 2, 0, 0; RGB: 125.74, 51.85, 19.24; 210.76, 143.54, 44.74; 207.22, 138.33, 26.66; 133.6, 71.11, 16.12; 158.49, 110.66, 70.14; 115.66, 78.2, 47.87; contrasted |
| 9235, UASM370132 | *Polygonia satyrus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318739; ps_S-A | visually scored: 0, 0, 1, 1, 0, 0, 1, 2, 0, 0; RGB: 137.21, 67.12, 20.13; 196.14, 113.56, 9.47; 207.02, 133.57, 15.44; 111.22, 57.61, 18.72; 139.2, 95.27, 59.6; 95.33, 62.7, 39.68; contrasted |
| 9236, UASM370133 | *Polygonia satyrus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318740; ps_S-B | visually scored: 2, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 138.95, 63.76, 21.39; 210.41, 136.63, 24.19; 215.1, 145.47, 33.01; 151.62, 83.46, 16.74; 176.51, 127.76, 79.73; 136.34, 93.87, 57.76; contrasted |
| 9237, UASM370134 | *Polygonia satyrus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318741; ps_S-A | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 147.24, 60.35, 13.66; 203.03, 115.87, 7.02; 216.53, 142.31, 20.67; 168.3, 98.26, 23.74; 159.97, 107.8, 56.46; 118.07, 76.3, 41.41; contrasted |
| 9238, UASM370135 | *Polygonia satyrus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-23; collector(s): McDonald, C., Aerial net | NA | visually scored: 0, 0, 0, 2, 0, -, 0, ?, 0, 1; RGB: NA |
| 9239, UASM370136 | *Polygonia satyrus* | CANADA: Alberta, East Cougar Creek camp (53.542, -117.01); date collected: 2013-08-23; collector(s): McDonald, C., Bait trap | KY318742; ps_9227 | visually scored: 0, 0, 0, 1, 0, 1, 0, 2, 0, 1; RGB: 116.72, 43.9, 12.86; 192.84, 103.2, 5.18; 203.82, 123.86, 4.84; 128.49, 59.31, 15.24; 123.89, 81.35, 50.06; 101.51, 65.11, 41.46; smeared |
| 9240, UASM370137 | *Polygonia satyrus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318743; ps_S-B | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 144.6, 62.15, 12.36; 196.84, 103.55, 4.87; 204.38, 131.46, 32.61; 146.47, 74.3, 14.79; 144.18, 99.12, 61.48; 103.94, 72.11, 46.43; contrasted |

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| 9241, UASM370138 | *Polygonia satyrus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318744; ps_S-A | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 107.17, 39.48, 19.03; 193.22, 110.58, 8.7; 196.75, 120.84, 12.03; 136.74, 69.88, 12.86; 152.54, 103.88, 62.7; 110.06, 72.61, 45.24; contrasted |
| 9242, UASM370139 | *Polygonia satyrus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318745; ps_S-B | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 116.25, 42.64, 13.24; 194.69, 106.35, 9.04; 196.95, 123.26, 23.11; 128.14, 63.83, 17.1; 133.76, 91.14, 56.31; 96.57, 64.39, 38.93; contrasted |
| 9243, UASM370140 | *Polygonia satyrus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318746; ps_S-B | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 168.46, 71.92, 7.02; 202.07, 116.2, 3.15; 198.93, 124.85, 10.17; 136.07, 73.58, 14.18; 157.38, 111.74, 68.28; 115.67, 77.42, 46.47; contrasted |
| 9244, UASM370141 | *Polygonia satyrus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-48 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318747; ps_S-A | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 149.79, 84, 36.07; 194.03, 109.43, 8.91; 206.39, 134.61, 16.94; 143.38, 81.3, 25.42; 156.18, 107.97, 63.69; 116.4, 74.89, 43.55; contrasted |
| 9245, UASM370142 | *Polygonia satyrus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-49 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318748; ps_9245 | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 159.56, 70.09, 6.66; 194.16, 107.26, 2.76; 196.25, 123.35, 10.12; 152.66, 81.53, 15.43; 174.68, 126.44, 78.2; 134.38, 93.54, 55.84; contrasted |
| 9246, UASM370143 | *Polygonia satyrus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-50 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318749; ps_S-B | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 154.84, 67.08, 11; 207.31, 142.75, 52.95; 205.61, 143.76, 54.11; 178.52, 95.85, 15.28; 159.45, 109.14, 64.93; 131.08, 90.62, 55.72; contrasted |
| 9247, UASM370144 | *Polygonia satyrus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-51 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318750; ps_S-A | visually scored: 0, 0, 0, 1, 1, 1, 0, 2, 0, 1; RGB: 156.65, 74.32, 12.76; 204.57, 137.95, 30.47; 204.27, 141.31, 28.41; 154.07, 83.88, 20.66; 133.27, 87.22, 51.04; 111.48, 72.14, 44.29; smeared |
| 9248, UASM370145 | *Polygonia satyrus* | USA: Montana, Flathead Lake Biological Research station (47.876, -114.032); date collected: 2013-09-07; collector(s): Dupuis, J. R., Aerial net | KY318751; ps_9248 | visually scored: 0, 0, 0, 2, 0, 0, 1, 2, 0, 0; RGB: 111.17, 34.31, 15.85; 185.17, 90.56, 3.64; 189.7, 108.99, 5.15; 154.39, 80.59, 18.98; 138.7, 92.75, 54.88; 94.67, 63.39, 39.43; contrasted |
| 9249, UASM370146 | *Polygonia satyrus* | USA: Montana, North of Coram, 10km NW of Blankenship bridge (48.47, -114.08); date collected: 2013-09-01; collector(s): Dupuis, J. R., Aerial net | KY318752; ps_S-B | visually scored: 0, 0, 0, 2, 1, 0, 1, 2, 0, 0; RGB: 178.62, 96.18, 33.3; 221.56, 144.39, 29.31; 222.09, 150.31, 36.95; 196.22, 123.77, 39.1; 176.14, 125.88, 77.87; 143.05, 99.92, 62.27; contrasted |
| 9250, UASM370147 | *Polygonia faunus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318554; NA | visually scored: 2, 1, 1, 1, 1, 0, 1, 3, 1, 1; RGB: 150.7, 72.04, 18.5; 196.34, 116.04, 8.88; 196.44, 128.97, 16.95; 115.45, 53.97, 23.61; 105.27, 79.4, 58.67; 96.5, 74.43, 54.41; smeared |
| 9251, UASM370148 | *Polygonia faunus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318555; pf_A-Ida | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 136.86, 68.83, 26.5; 204.11, 141.84, 52.64; 211.1, 157.96, 71.88; 136, 87.98, 48.16; 154.45, 125.07, 92.24; 121.57, 97.74, 70.82; contrasted |
| 9252, UASM370149 | *Polygonia faunus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318556; pf_A-WS7 | visually scored: 0, 1, 1, 2, 1, 0, 1, 3, 1, 0; RGB: 164.75, 72.03, 16.77; 220.83, 147.85, 30.97; 222.88, 159.17, 47.81; 129.64, 60.09, 22.72; 145.12, 113.48, 82.59; 116.2, 94.26, 69.09; contrasted |
| 9253, UASM370150 | *Polygonia faunus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318557; pf_A-Ida | visually scored: 0, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 117.57, 48.11, 12.77; 188.68, 117.09, 17.63; 199.32, 132.8, 23.98; 77.21, 38.22, 17.16; 116.82, 95.47, 72.47; 83.41, 68.84, 50.93; contrasted |
| 9254, UASM370151 | *Polygonia faunus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318558; pf_A-Ida | visually scored: 2, 1, 1, 2, 0, 0, 1, 2, 1, 0; RGB: 142.27, 83.63, 43.34; 207.2, 151.43, 68; 208.71, 162.44, 89.66; 132.46, 69.96, 30.64; 151.17, 118.08, 88.19; 118.69, 92.68, 69.15; contrasted |

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| 9255, UASM370152 | *Polygonia faunus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318559; pf_A-WS8 | visually scored: 0, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 123.74, 63.56, 28.15; 191.83, 129.6, 45.75; 209.92, 152.9, 53.29; 120.61, 60.35, 26.81; 130.68, 103.65, 76.99; 102.72, 82.33, 60.69; contrasted |
| 9256, UASM370153 | *Polygonia faunus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318560; pf_A-Ida | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 136.4, 61.99, 22.8; 205.45, 123.45, 25.1; 220.22, 160.67, 61.36; 113.18, 50.93, 23.39; 121.53, 89.13, 63.23; 95.07, 71.39, 50.63; contrasted |
| 9257, UASM370154 | *Polygonia faunus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318561; pf_A-Ida | visually scored: 0, 1, 1, 1, 1, 0, 1, 0, 1, 0; RGB: 142.48, 72.44, 26.47; 200.31, 124.89, 20.95; 210.91, 149.57, 47.97; 126.77, 67.42, 27.64; 153.27, 119.77, 85.89; 126.08, 98.98, 70.58; contrasted |
| 9258, UASM370155 | *Polygonia faunus* | CANADA: Alberta, Chickakoo Lake (53.616, -114.074); date collected: 2013-05-21; collector(s): McDonald, C., Aerial net | NA | visually scored: 2, 1, 1, 1, 0, -, -, -, -, ; RGB: NA |
| 9259, UASM370156 | *Polygonia faunus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 56 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318562; pf_F-NHa3 | visually scored: 1, 1, 1, 1, 1, 0, 1, 1, 1, 1; RGB: 149.94, 100.09, 56.96; 191.2, 133.85, 41.48; 204.7, 149.13, 49.09; 130.38, 74.26, 35.82; 133.03, 102.77, 75.7; 121.64, 96.73, 72.45; smeared |
| 9260, UASM370157 | *Polygonia faunus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 57 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318563; pf_9260 | visually scored: 2, 0, 0, 2, 1, 0, 0, 0, 1, 1; RGB: 198.79, 114.35, 23.04; 227.11, 174.6, 80.63; 223.77, 157.46, 38.22; 183.93, 109.08, 31.74; 147.07, 114.93, 83.22; 128.2, 100.96, 72.47; smeared |
| 9261, UASM370158 | *Polygonia faunus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 58 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318564; pf_A-Ida | visually scored: 0, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 157.86, 99.86, 56.3; 190.71, 138.46, 79.02; 190.97, 132.93, 48.12; 128.91, 63.68, 24.68; 142.91, 113.88, 88.29; 112.37, 90.1, 69.7; contrasted |
| 9262, UASM370159 | *Polygonia faunus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 59 (52.635, -114.624); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318565; pf_A-Ida | visually scored: 0, 0, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 143.53, 78.53, 40.33; 219.04, 145.27, 41.93; 227.78, 162.24, 53.73; 131.29, 61.92, 22.41; 156.13, 121.09, 91.19; 116.62, 91.3, 67.09; contrasted |
| 9263, UASM370160 | *Polygonia faunus* | CANADA: Alberta, Devonian Bot. Garden (53.408, -113.758); date collected: 2013-06-04; collector(s): Acorn, J., Aerial net | KY318566; pf_S_Geo | visually scored: 0, 0, 0, 2, 0, 0, 1, 1, 1, 1; RGB: 154.73, 74.46, 17.27; 206.85, 147.88, 49.18; 208.79, 160.55, 73.83; 135.45, 72.85, 29.64; 132.12, 100.79, 71.52; 121.1, 94.16, 97.47; smeared |
| 9264, UASM370161 | *Polygonia faunus* | CANADA: Alberta, Devonian Bot. Garden (53.408, -113.758); date collected: 2013-06-04; collector(s): Acorn, J., Aerial net | KY318567; pf_A-Ida | visually scored: 0, 1, 0, 1, 0, -, 1, 2, 1, 0; RGB: 186.85, 110.66, 30.2; 225.05, 174.9, 87.31; 224.71, 181.98, 102.22; 146.64, 93.94, 47.77; 173.65, 135.75, 95.89; 140.94, 109.56, 78.57; contrasted |
| 9265, UASM370162 | *Polygonia faunus* | CANADA: Alberta, Chickakoo Lake (53.616, -114.074); date collected: 2013-07-30; collector(s): McDonald, C., Aerial net | KY318568; pf_A-Ida | visually scored: 0, 1, 1, 1, 2, 0, 1, 2, 1, 0; RGB: 148.97, 78.99, 29.18; 203.73, 131.9, 12.86; 210.19, 143.88, 14.72; 133.02, 60.71, 11.7; 105.44, 78.53, 57.55; 81.89, 64.04, 47.2; contrasted |
| 9266, UASM370163 | *Polygonia faunus* | CANADA: Alberta, Chickakoo Lake (53.616, -114.074); date collected: 2013-07-30; collector(s): McDonald, C., Aerial net | KY318569; pf_A-Ida | visually scored: 1, 0, 1, 2, 2, 0, 1, 3, 1, 0; RGB: 157.96, 68.39, 9.14; 218.15, 131, 10.44; 222.37, 153.22, 30.56; 150.92, 74.02, 21.39; 168.23, 131.25, 95.94; 131.53, 104.83, 76.18; contrasted |
| 9267, UASM370164 | *Polygonia faunus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | KY318570; pf_A-Ida | visually scored: 1, 1, 1, 2, 0, 0, 1, 1, 1, 0; RGB: 149.27, 65.43, 14.63; 195.11, 121.39, 10.9; 189.78, 127.6, 17.9; 95.58, 47.16, 20.21; 117.09, 88.75, 61.48; 86.91, 66.03, 46.5; contrasted |
| 9268, UASM370165 | *Polygonia faunus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | NA | visually scored: 0, 1, 1, 1, 1, 1, 1, 3, 1, 1; RGB: NA |

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| 9269, UASM370166 | *Polygonia faunus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | KY318571; pf_A-Ida | visually scored: 1, 1, 1, 2, 1, 0, 1, 3, 1, 0; RGB: 156.07, 66.99, 7.21; 195.23, 123.08, 8.54; 195.96, 135.84, 26.27; 135.55, 65.73, 16.42; 122.51, 95.31, 66.85; 97.46, 79.16, 56.78; contrasted |
| 9270, UASM370167 | *Polygonia faunus* | CANADA: Alberta, Open creek dam (52.652, -114.724); date collected: 2013-05-23; collector(s): McDonald, C., Aerial net | KY318572; pf_8301 | visually scored: 0, 1, 0, 1, 1, 0, 1, 1, 1, 0; RGB: 146.51, 66.91, 14.91; 194.4, 138.81, 39; 205.04, 167.86, 91.5; 164.32, 97.81, 25.67; 134.4, 98.55, 67.33; 114.14, 82.63, 58.13; contrasted |
| 9271, UASM370168 | *Polygonia faunus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318573; pf_A-Ida | visually scored: 2, 1, 1, 2, 2, 0, 1, 3, 1, 0; RGB: 171.39, 84.52, 6.74; 209.69, 147.67, 38.35; 221.97, 170.61, 75.09; 134.31, 74.9, 30.66; 152, 128.78, 99.86; 126.2, 107.31, 80.94; smeared |
| 9272, UASM370169 | *Polygonia satyrus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318753; NA | visually scored: 0, 0, 0, 1, 0, 0, 1, 2, 0, 0; RGB: 107.8, 38.5, 13.34; 219.06, 134.19, 21.22; 227.91, 152.01, 35.86; 134.58, 77.2, 20.76; 167.07, 119.86, 81.35; 117.92, 82.39, 54.91; contrasted |
| 9273, UASM370170 | *Polygonia faunus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318574; pf_9273 | visually scored: 1, 0, 1, 2, 1, 0, 1, 1, 1, 1; RGB: 110.05, 46.73, 21.25; 197.35, 119.15, 11.1; 200.86, 141.35, 37.34; 120.19, 52.87, 18.05; 116.91, 87.3, 58.21; 99.87, 74.45, 49.89; contrasted |
| 9274, UASM370171 | *Polygonia faunus* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318575; pf_A-Ida | visually scored: 0, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 85.79, 39.26, 18.61; 182.47, 99.05, 7.57; 193.65, 120.29, 12.1; 116.12, 52.08, 15.52; 114.19, 86.47, 59.28; 85.94, 68.62, 48.42; smeared |
| 9275, UASM370172 | *Polygonia faunus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318576; pf_F-NHa3 | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 142.11, 65.52, 17.03; 216.95, 133.44, 10.38; 216.16, 148.6, 30.2; 102, 49.34, 22.6; 106.38, 83.56, 58.98; 77.65, 64.18, 46.4; contrasted |
| 9276, UASM370173 | *Polygonia faunus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318577; pf_9276 | visually scored: 1, 1, 1, 2, 1, 0, 1, 1, 1, 0; RGB: 126.51, 64.05, 17.42; 185.87, 120.14, 11.68; 193.82, 130.52, 9.01; 97.77, 53.17, 18.32; 102.55, 81.22, 57.89; 81.91, 67.73, 49.88; contrasted |
| 9277, UASM370174 | *Polygonia faunus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318578; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, 0, 0, 1, 1, 1; RGB: 141.6, 78, 26.51; 185.51, 120.27, 25.47; 188.36, 130.63, 36.19; 110.34, 52.27, 19.92; 109.64, 79.53, 54.6; 104.22, 78.08, 55.19; contrasted |
| 9278, UASM370175 | *Polygonia faunus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318579; pf_A-Ida | visually scored: 0, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 142.51, 67.44, 21.15; 189.89, 120.69, 18.25; 187.48, 131.37, 30.59; 105.58, 59.02, 22.93; 110.43, 84.36, 61.31; 76.88, 61.61, 46.81; smeared |
| 9279, UASM370176 | *Polygonia faunus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318580; pf_9279 | visually scored: 1, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 99.69, 40.16, 18.05; 199.15, 121.14, 10.27; 204.26, 147.87, 39.1; 105.32, 51.12, 20.46; 121.2, 92.9, 65.95; 90.2, 71.44, 50.74; contrasted |
| 9280, UASM370177 | *Polygonia faunus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318581; pf_A-Ida | visually scored: 1, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 103.25, 48.02, 15.93; 149.9, 84.39, 6.99; 162.23, 100.36, 7.43; 80.24, 38.58, 17; 107.44, 98.34, 75.8; 79.39, 73.49, 55.91; contrasted |
| 9281, UASM370178 | *Polygonia faunus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318582; pf_A-Ida | visually scored: 0, 1, 1, 1, 2, 0, 1, 3, 1, 0; RGB: 97.57, 38.45, 20.21; 186.76, 105.65, 7.16; 180.97, 116.03, 16.61; 87.81, 40.13, 19.21; 88.48, 64.65, 46.4; 63.93, 50.97, 37.87; contrasted |
| 9282, UASM370179 | *Polygonia faunus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318583; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, 0, 1, 0, 1, 0; RGB: 133.98, 54.55, 14.93; 190.69, 130.1, 47.91; 194.49, 136.29, 48.07; 132.15, 67.52, 23.6; 130.85, 103.94, 80.67; 90.78, 74.77, 57.41; contrasted |

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| 9283, UASM370180 | *Polygonia faunus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-52 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318584; pf_A-Ida | visually scored: 0, 1, 1, 2, 1, 0, 1, 2, 1, 0; RGB: 144.82, 65.19, 15.23; 193.12, 115.64, 5.38; 197.09, 134.25, 20.17; 104.59, 43.76, 19.07; 122.9, 92.82, 69.29; 92.4, 72.36, 54.62; contrasted |
| 9284, UASM370181 | *Polygonia faunus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-53 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318585; pf_A-Ida | visually scored: 0, 1, 1, 2, 1, 0, 1, 2, 1, 0; RGB: 124.44, 55.49, 15.35; 195.41, 116.37, 3.21; 194.42, 133.21, 24.08; 94.9, 39.64, 15.78; 107.28, 76.73, 52.41; 77.97, 59.51, 42.78; contrasted |
| 9285, UASM370182 | *Polygonia faunus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-54 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318586; pf_F-NHa3 | visually scored: 0, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 130.62, 48.11, 12.68; 197.54, 120.2, 11.15; 186.91, 118.51, 12.3; 102.49, 48.11, 22.46; 99.84, 74.2, 52.65; 85.4, 67.9, 51.83; contrasted |
| 9286, UASM370183 | *Polygonia faunus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-55 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318587; pf_8316 | visually scored: 2, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 137.92, 52.2, 9.34; 182.75, 96.91, 4.62; 188.84, 123, 21.73; 109.97, 44.11, 13.85; 106.55, 79.08, 57.39; 77.96, 61.84, 46.65; contrasted |
| 9287, UASM370184 | *Polygonia faunus* | CANADA: British Columbia, N. Kelowna (49.951, -119.556); date collected: 2013-07-14; collector(s): Sperling, F. A. H., Aerial net | KY318588; pf_A-Ida | visually scored: 0, 1, 1, 2, 0, 0, 1, 1, 1, 0; RGB: 93.58, 40.04, 19.4; 186.36, 105.29, 13.54; 198.2, 137.65, 38.79; 81.27, 36.98, 22.38; 114.97, 95.88, 71.95; 81.46, 67.82, 50.03; contrasted |
| 9288, UASM370185 | *Polygonia faunus* | USA: Montana, Flathead Lake Biological Research Station (47.876, -114.032); date collected: 2013-09-07; collector(s): Dupuis, J. R., Aerial net | NA | visually scored: 0, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: NA |
| 9289, UASM370186 | *Polygonia faunus* | USA: Montana, Flathead Lake Biological Research Station (47.876, -114.032); date collected: 2013-09-07; collector(s): Dupuis, J. R., Aerial net | KY318589; pf_A-Ida | visually scored: 0, 1, 1, 2, 1, 0, 1, 2, 1, 0; RGB: 99.06, 35.96, 11.75; 176.65, 85.18, 2.72; 173.54, 98.76, 3.85; 86.5, 47.43, 26.2; 93.05, 71.89, 52.27; 76.75, 64.7, 50.18; contrasted |
| 9290, UASM370187 | *Polygonia faunus* | USA: Montana, North of Coram, 10km NW of Blankenship bridge (48.47, -114.08); date collected: 2013-09-01; collector(s): Dupuis, J. R., Aerial net | KY318590; pf_A-Ida | visually scored: 0, 1, 1, 2, 1, 0, 1, 2, 1, 0; RGB: 117.18, 37.91, 15.08; 191.28, 108.94, 14.6; 195.63, 127.19, 19.68; 100.32, 46.1, 24.24; 118.84, 100.7, 78.18; 91.31, 78.52, 59.22; contrasted |
| 9291, UASM370188 | *Polygonia faunus* | USA: Montana, North of Coram, 10km NW of Blankenship bridge (48.47, -114.08); date collected: 2013-09-01; collector(s): Dupuis, J. R., Aerial net | NA | visually scored: 0, 1, 1, 2, 2, 0, 1, 2, 1, 0; RGB: NA |
| 9292, UASM370189 | *Polygonia faunus* | USA: Montana, North of Coram, 10km NW of Blankenship bridge (48.47, -114.08); date collected: 2013-09-01; collector(s): Dupuis, J. R., Aerial net | NA | visually scored: 0, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: NA |
| 9293, UASM370190 | *Polygonia faunus* | USA: Montana, North of Coram, 10km NW of Blankenship bridge (48.47, -114.08); date collected: 2013-09-01; collector(s): Dupuis, J. R., Aerial net | KY318591; pf_A-Ida | visually scored: 0, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 117.79, 40.66, 13.66; 187.84, 111.99, 4.61; 198.65, 139.12, 28.7; 103.01, 47.83, 20.29; 109.48, 91.56, 70.24; 80.21, 69.3, 53.2; contrasted |
| 9294, UASM370191 | *Polygonia faunus* | USA: Montana, Flathead Lake Biological Research Station (47.876, -114.032); date collected: 2013-09-07; collector(s): Dupuis, J. R., Aerial net | KY318592; pf_A-Ida | visually scored: 0, 1, 1, 1, 0, 0, 1, 0, 1, 0; RGB: 95.83, 44.57, 21.87; 185.06, 104.66, 7.15; 181.71, 118.85, 18.1; 73.23, 33.37, 16.79; 101.94, 82.66, 59.95; 78.51, 65.86, 47.72; contrasted |
| 9295, UASM370192 | *Polygonia faunus* | USA: Montana, Flathead Lake Biological Research Station (47.876, -114.032); date collected: 2013-09-07; collector(s): Dupuis, J. R., Aerial net | KY318593; pf_A-Ida | visually scored: 0, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 100.71, 43.6, 13.68; 196.5, 122.71, 18.82; 191.28, 132.61, 36.12; 64.1, 38.03, 24.75; 123.44, 107.03, 84.39; 84.01, 72.99, 56.21; contrasted |
| 9296, UASM370193 | *Polygonia gracilis gracilis* | CANADA: British Columbia, South of Tumbler Ridge (55.114, -121.019); date collected: 2013-08-09; collector(s): Dupuis, J. R., Aerial net | KY318686; pg_G-A | visually scored: 2, 1, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 111.9, 42.34, 14.64; 194.17, 103.72, 5.17; 199.88, 128.62, 17.85; 132.89, 63.51, 19.78; 137.18, 118.11, 96.94; 102.9, 87.02, 69.94 |

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| 9297, UASM370194 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318687; pg_9297 | visually scored: 2, 1, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 125.34, 61.49, 28.49; 201.83, 138.01, 37.16; 207.44, 148.06, 46.54; 154.23, 92.84, 30.98; 137.65, 112.88, 89.14; 98.79, 80.32, 63.13 |
| 9298, UASM370195 | *Polygonia gracilis gracilis* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-56 (54.963, -112.01); date collected: 2013-08-29; collector(s): McDonald, C., Aerial net | KY318692; NA | visually scored: 2, 1, 0, 1, 2, 0, 0, 3, 0, 0; RGB: 124.91, 54.07, 16.61; 172.23, 85.52, 5.94; 192.58, 113.17, 11.67; 119.59, 53.45, 15.44; 132.36, 112.09, 93.61; 98.57, 82.25, 66.59 |
| 9299, UASM370196 | *Polygonia gracilis gracilis* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-57 (54.963, -112.01); date collected: 2013-08-29; collector(s): McDonald, C., Aerial net | NA | visually scored: 2, 1, 0, 1, 2, 0, 0, 0, 0, 0; RGB: NA |
| 9300, UASM370197 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | NA | visually scored: 0, 0, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 176.16, 109.7, 57.55; 229.06, 166.1, 79.97; 229.96, 172.85, 81.28; 177.68, 104.06, 32.92; 156.22, 129.24, 101.83; 123.53, 99.21, 75.16 |
| 9301, UASM370198 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, East of Valemont (52.843, -119.256); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318688; pg_9301 | visually scored: 2, 0, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 157.4, 72.05, 22.68; 215.91, 135.45, 26.94; 229.87, 172.44, 61.51; 168.91, 95.23, 25.68; 161.52, 134.35, 104.7; 133.6, 108.34, 81.99 |
| 9302, UASM370199 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318689; pg_G-A | visually scored: 1, 0, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 135.04, 63.92, 23.99; 204.61, 125.56, 23.39; 221.25, 163.67, 52.55; 170.35, 108.96, 41.24; 174.61, 145.56, 114.58; 138.02, 111.36, 84.83 |
| 9303, UASM370200 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Dupuis, J. R., Aerial net | KY318690; pg_G-B | visually scored: 2, 0, 0, 1, 2, 0, 0, 0, 0, 0; RGB: 128.57, 60.94, 19.73; 178.75, 99.63, 12.35; 206.38, 146.3, 48.62; 142.89, 78.23, 23.82; 165.89, 138.95, 110.54; 130.89, 106.03, 81.47 |
| 9304, UASM370201 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Brunet, B., Aerial net | KY318694; pg_G-B | visually scored: 2, 0, 0, 1, 2, 0, 1, 3, 0, 0; RGB: 154.25, 86.26, 27.38; 208.66, 149.12, 58.08; 228.05, 181.65, 88.87; 181.98, 123.07, 56.81; 189.37, 163.9, 132.85; 152.96, 126.04, 97.12 |
| 9305, UASM370202 | *Polygonia gracilis gracilis* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Brunet, B., Aerial net | KY318695; pg_G-A | visually scored: 2, 0, 0, 2, 2, 1, 0, 0, 0, 0; RGB: 134.65, 61.31, 25.26; 198.96, 115.93, 27.18; 214.63, 151.49, 45.24; 157.44, 80.67, 21.66; 172.94, 146.22, 116.41; 132.78, 106.24, 80.14 |
| 9306, UASM370203 | *Polygonia gracilis zephyrus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318696; pg_G-A | visually scored: 2, 0, 0, 1, 0, 1, 0, 1, 0, 0; RGB: 177.63, 79.68, 7.82; 229.25, 152.35, 16.03; 237.77, 185.34, 59; 201.93, 124.6, 24.22; 147.61, 123.62, 98.04; 108.01, 91.13, 71.98 |
| 9307, UASM370204 | *Polygonia gracilis zephyrus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318697; pg_9301 | visually scored: 3, 0, 0, 1, 2, 0, 0, 1, 0, 0; RGB: 108.15, 46.61, 21.49; 169.98, 81.92, 5.11; 187.23, 119.85, 18.92; 128.6, 64.65, 14.06; 118.98, 100.15, 78.44; 83.78, 70.98, 55.35 |
| 9308, UASM370205 | *Polygonia gracilis zephyrus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318691; pg_9308 | visually scored: 2, 0, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 115.5, 53.07, 22.25; 187.42, 114.63, 19.35; 195.3, 130.98, 20.97; 124.74, 71.67, 28.02; 130.29, 111.62, 90.13; 92.33, 78.48, 62.17 |
| 9309, UASM370206 | *Polygonia gracilis gracilis* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-23; collector(s): McDonald, C., Aerial net | KY318693; pg_9308 | visually scored: 2, 1, 0, 1, 2, 0, 1, 0, 0, 0; RGB: 140.29, 64.66, 17.7; 185.37, 105.87, 16.69; 212.17, 139.81, 20.3; 131.29, 61.05, 17.67; 111.8, 94.11, 74.95; 78.51, 66.41, 52.34 |
| 9310, UASM370207 | *Polygonia comma* | USA: New York, Ithaca, Mundey Flower Grdn (42.45, -76.468); date collected: 2013-07-20; collector(s): Dupuis, J. R., Aerial net | KY318754; NA | visually scored: NA; RGB: NA |

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| 9311, UASM370208 | *Polygonia interrogationis* | USA: New York, Ithaca, Monkey Run (42.463, -76.426); date collected: 2013-07-20; collector(s): Dupuis, J. R., Aerial net | KY318755; NA | visually scored: NA; RGB: NA |
| 9312, UASM370209 | *Polygonia progne* | CANADA: Alberta, Edmonton, Windsor Park (53.519, -113.521); date collected: 2013-05-04; collector(s): Sperling, F. A. H.; Sperling, J., Aerial net | KY318655; pp_P-A | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 163.87, 86.81, 24.37; 185.16, 104.77, 15.81; 199.37, 134.24, 39.35; 122.75, 58.17, 25.03; 108.04, 76.28, 53.84; 90.28, 66.24, 47.92 |
| 9313, UASM370210 | *Polygonia progne* | CANADA: Alberta, Edmonton, Windsor Park (53.519, -113.521); date collected: 2013-05-04; collector(s): Sperling, F. A. H., Aerial net | KY318656; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 163.9, 90.5, 16.56; 196.88, 128.72, 29.36; 198.56, 125.03, 15.28; 117.45, 56.29, 22.3; 114.9, 88.78, 65.7; 98.55, 77.04, 57.47 |
| 9314, UASM370211 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-05-19; collector(s): Sperling, F. A. H., Aerial net | NA | visually scored: 3, 1, 0, 0, 2, 1, 1, 0, 0, 1; RGB: NA |
| 9315, UASM370212 | *Polygonia progne* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-05-19; collector(s): Sperling, F. A. H., Aerial net | KY318657; pp_P-A | visually scored: 2, 1, 0, 0, 2, 1, 0, 1, 0, 1; RGB: 161.74, 80.79, 5.31; 189.53, 129.76, 27.83; 207.01, 151.82, 49.86; 132.27, 75.73, 34.48; 133.42, 101.97, 75.49; 117.79, 90.4, 66.59 |
| 9316, UASM370213 | *Polygonia progne* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.087); date collected: 2013-05-18; collector(s): Sperling, F. A. H., Aerial net | KY318658; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 0, 0, 0, 1; RGB: 164.16, 87.34, 20.93; 198.04, 131.88, 42.02; 212.04, 157.82, 71.89; 144.88, 85.69, 37.5; 115.4, 84.86, 58.41; 94.13, 70.9, 50.52 |
| 9317, UASM370214 | *Polygonia progne* | CANADA: Alberta, West of Stauffer (52.174, -114.643); date collected: 2013-05-18; collector(s): Acorn, J., Aerial net | KY318659; pp_P-A | visually scored: 2, 1, 1, 0, 2, 1, 0, 1, 0, 1; RGB: 169.75, 98.22, 27.72; 198.89, 133.97, 38.33; 208.54, 140.91, 30.96; 127.68, 61.07, 24.79; 117.75, 93.09, 69.14; 98.7, 79.28, 59.54 |
| 9318, UASM370215 | *Polygonia progne* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 60 (52.635, -114.624); date collected: 2013-05-23; collector(s): McDonald, C., Aerial net | KY318660; pp_P-A | visually scored: 3, 1, 0, 0, 2, 1, 0, 1, 0, 1; RGB: 181.61, 121.12, 54.12; 194.32, 126.47, 45.09; 201.8, 138.16, 40.28; 165.5, 100.89, 37; 142.94, 113.22, 87.94; 123.76, 98.54, 76.41 |
| 9319, UASM370216 | *Polygonia progne* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 61 (52.635, -114.624); date collected: 2013-05-22; collector(s): Dupuis, J. R., Aerial net | KY318661; pp_9319 | visually scored: 2, 1, 1, 1, 2, -, 0, 1, 0, 1; RGB: 183.3, 91, 6.25; 205.43, 131.02, 21.56; 217.25, 154.68, 48.01; 120.97, 54.84, 23.45; 127.24, 95.55, 71.89; 106.47, 80.99, 61.18 |
| 9320, UASM370217 | *Polygonia progne* | CANADA: Alberta, Edmonton (53.506, -113.613); date collected: 2013-05-21; collector(s): Anweiler, G., Bait trap | NA | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: NA |
| 9321, UASM370218 | *Polygonia progne* | CANADA: Alberta, Edmonton (53.506, -113.613); date collected: 2013-05-21; collector(s): Anweiler, G., Bait trap | KY318662; pp_P-A | visually scored: 3, 0, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 171.75, 80.4, 8.78; 187.64, 114.84, 9.36; 210.59, 151.83, 45.59; 138.41, 65.81, 23.66; 118.53, 90.35, 67.12; 106.06, 81.91, 61.34 |
| 9322, UASM370219 | *Polygonia progne* | CANADA: Alberta, Northeast of Peace River (56.253, -117.288); date collected: 2013-05-24; collector(s): Dupuis, J. R., Aerial net | KY318663; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 0, 0, 0, 1; RGB: 169.02, 80.89, 10.78; 192.94, 104.96, 11.48; 211.23, 140.34, 31.07; 120.82, 65.22, 29.94; 123.37, 96.96, 73.09; 100.84, 80.4, 61.6 |
| 9323, UASM370220 | *Polygonia progne* | CANADA: Alberta, Kaufman Hill (56.248, -117.279); date collected: 2013-07-14; collector(s): Dupuis, J. R., Aerial net | KY318664; pp_9323 | visually scored: 3, 1, 1, 0, 2, 1, 1, 1, 0, 1; RGB: 179.59, 84.48, 1.62; 193.01, 119.66, 15.39; 196.56, 132.24, 33.05; 124.99, 53.53, 15.84; 117.47, 92.61, 70.4; 103.34, 82.58, 63.19 |
| 9324, UASM370221 | *Polygonia progne* | CANADA: Alberta, Grimshaw near Shaftsbury Ferry (56.095, -117.572); date collected: 2013-07-17; collector(s): Dupuis, J. R., Aerial net | KY318665; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 153.59, 93.1, 30.81; 188.8, 127.85, 41.81; 208.6, 161.59, 88.42; 83.25, 40.23, 22.58; 90.41, 73.75, 57.73; 80.52, 66.47, 51.89 |

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| 9325, UASM370222 | *Polygonia progne* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | KY318666; pp_P-A | visually scored: 3, 1, 1, 0, 2, 1, 1, 1, 0, 1; RGB: 176.83, 78.71, 3.23; 198.54, 105.59, 4.94; 197.23, 110.6, 7.24; 96.82, 38.28, 17.33; 116.87, 97.06, 78.26; 100.04, 83.89, 67.47 |
| 9326, UASM370223 | *Polygonia progne* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-02; collector(s): McDonald, C., Aerial net | KY318667; pp_8380 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 154.9, 76.33, 14.43; 188.03, 100.59, 4.61; 190.33, 129.07, 36.21; 130.25, 62, 21.65; 94.77, 75.01, 59.93; 83.31, 67.07, 53.86 |
| 9327, UASM370224 | *Polygonia progne* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 62 (52.635, -114.624); date collected: 2013-08-11; collector(s): Acorn, J., Aerial net | KY318668; pp_P-B | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 162.09, 63.02, 2.7; 193.76, 115.03, 20.64; 196.93, 107.55, 4.87; 87.9, 33.88, 16.49; 91.18, 75.41, 59.62; 73.31, 61.85, 49.2 |
| 9328, UASM370225 | *Polygonia progne* | CANADA: Alberta, Muir Lake (53.623, -113.958); date collected: 2013-08-14; collector(s): McDonald, C., Aerial net | KY318669; pp_P-B | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 175.75, 84.65, 2.33; 196.97, 123.61, 19.61; 209.38, 135.19, 15.61; 120.85, 75.08, 41.79; 104.13, 84.9, 66.99; 90.22, 74.5, 59.48 |
| 9329, UASM370226 | *Polygonia progne* | CANADA: Alberta, Chickakoo Lake (53.616, -114.074); date collected: 2013-08-14; collector(s): McDonald, C., Aerial net | NA | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: NA |
| 9330, UASM370227 | *Polygonia progne* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-09; collector(s): McDonald, C., Aerial net | KY318670; pp_P-A | visually scored: 2, 1, 1, 1, 2, 1, 1, 1, 0, 1; RGB: 163.11, 85.2, 14.87; 189.42, 120.88, 23.84; 192.02, 113.78, 14.31; 99.22, 51.62, 30.59; 89.5, 71.57, 54.61; 80.04, 65.43, 50.69 |
| 9331, UASM370228 | *Polygonia progne* | CANADA: Alberta, Pigeon Lake, Itaska (53.071, -114.078); date collected: 2013-08-10; collector(s): Sperling, F. A. H., Aerial net | KY318671; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 167.02, 65.1, 13.21; 194.72, 124.58, 13.26; 206.55, 140.56, 26.26; 90.49, 43.88, 23.48; 105.51, 86.04, 66.3; 78.41, 65.75, 51.78 |
| 9332, UASM370229 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-57 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318672; pp_P-A | visually scored: -, -, -, -, -, -, -, -, -, ; RGB: NA |
| 9333, UASM370230 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-58 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318673; pp_P-A | visually scored: 2, 1, 1, 1, 2, 1, 1, 1, 0, 1; RGB: 167.33, 79.7, 4.72; 191.67, 107.11, 2.34; 200.91, 132.97, 19.69; 115.05, 58.15, 27.56; 92.37, 75.15, 56.9; 72.28, 60.03, 46.25 |
| 9334, UASM370231 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-59 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318674; pp_8391 | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 164.31, 81.35, 3.58; 183.4, 122.52, 14.95; 196.77, 135.27, 17.82; 103.37, 46.07, 18.14; 102.83, 81.36, 60.07; 78.03, 63.8, 48.22 |
| 9335, UASM370232 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-60 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318675; pp_P-D | visually scored: 3, 1, 1, 0, 2, 1, 1, 0, 0, 1; RGB: 177.64, 83.46, 4.68; 197.06, 115.06, 2.87; 199.49, 123.49, 14.44; 119.78, 50.33, 19.71; 99.47, 80.75, 63.69; 81.29, 68.09, 54.58 |
| 9336, UASM370233 | *Polygonia progne* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2010-06-05; collector(s): Dupuis, J. R., Aerial net | KY318676; pp_P-A | visually scored: 2, 1, 1, 1, 2, -, 1, 0, 0, 1; RGB: 165.61, 71.17, 7.29; 198.66, 118.5, 14.34; 208.7, 129.81, 19.69; 104.9, 47.86, 21.87; 113.28, 86.76, 63.8; 96.16, 74.63, 55.45 |
| 9337, UASM370234 | *Polygonia progne* | CANADA: Alberta, NorthWest of Delburne, RR 251 (52.229, -113.453); date collected: 2013-08-16; collector(s): Dupuis, J. R., Aerial net | KY318677; pp_9337 | visually scored: 2, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 188.96, 114.48, 23.49; 205.48, 135.26, 31.54; 210.93, 149.1, 35.47; 109.73, 51.8, 19.16; 112.09, 92.61, 72.58; 87.9, 73.44, 57.1 |
| 9338, UASM370235 | *Nymphalis vaualbum* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 63 (52.635, -114.624); date collected: 2013-08-11; collector(s): Acorn, J., Aerial net | KY318756; NA | visually scored: NA; RGB: NA |

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| 9339, UASM370236 | *Nymphalis vaualbum* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-61 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318757; NA | visually scored: NA; RGB: NA |
| 9340, UASM370237 | *Nymphalis californica* | USA: Oregon, Santiam Pass HWY 20 (44.424, -121.849); date collected: 2001-07-25; collector(s): Sperling, F. A. H., Aerial net | KY318758; NA | visually scored: NA; RGB: NA |
| 9341, UASM370238 | *Aglais milberti* | CANADA: Alberta, Ram Trail (52.339, -115.742); date collected: 2012-08-15; collector(s): Dupuis, J. R., Reared from larva | KY318759; NA | visually scored: NA; RGB: NA |
| 9342, UASM370239 | *Aglais milberti* | CANADA: Alberta, Moose Mountain nr Bragg Creek (50.883, -114.758); date collected: 2001-07-07; collector(s): Sperling, F. A. H., Aerial net | KY318760; NA | visually scored: NA; RGB: NA |
| 9343, UASM370240 | *Nymphalis antiopa* | CANADA: Alberta, Hasse Lake (53.494, -114.176); date collected: 2013-08-12; collector(s): McDonald, C., Aerial net | KY318761; NA | visually scored: NA; RGB: NA |
| 9344, UASM370241 | *Nymphalis antiopa* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.327); date collected: 2011-09-18; collector(s): Sperling, F. A. H., Aerial net | KY318762; NA | visually scored: NA; RGB: NA |
| 9345, UASM370242 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-62 (54.963, -112.01); date collected: 2013-08-29; collector(s): McDonald, C., Aerial net | KY318678; pp_9345 | visually scored: 2, 1, 1, 1, 2, 1, 1, 1, 0, 1; RGB: 147.89, 57.71, 7.81; 171.73, 83.9, 2.14; 193.92, 118.77, 16.74; 68.9, 29.99, 16.83; 83.69, 65.96, 50.41; 64.65, 53.66, 42.23 |
| 9346, UASM370243 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-63 (54.963, -112.01); date collected: 2013-08-29; collector(s): McDonald, C., Aerial net | KY318679; pp_8350 | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 160.07, 65.7, 5.21; 190.01, 107.56, 13.12; 203.54, 132.84, 34.79; 97.85, 38.86, 18.08; 99.91, 83.7, 68.33; 81.28, 69.45, 65.77 |
| 9347, UASM370244 | *Polygonia progne* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-64 (54.963, -112.01); date collected: 2013-08-29; collector(s): McDonald, C., Aerial net | KY318680; pp_P-A | visually scored: 3, 1, 1, 1, 2, 1, 1, 0, 0, 1; RGB: 161.05, 74.57, 8.65; 195.9, 117.96, 5.52; 209.67, 144.48, 26.73; 115.26, 52.46, 18.61; 109.56, 88.47, 68.37; 84.02, 69.6, 54.85 |
| 9348, UASM370245 | *Polygonia faunus* | CANADA: Alberta, Opal Natural Area, west side (53.987, -113.319); date collected: 2013-07-31; collector(s): McDonald, C., Aerial net | KY318594; pf_9348 | visually scored: 1, 1, 1, 1, 1, 0, 0, 3, 1, 1; RGB: 147.31, 57.54, 11.1; 202.24, 113.15, 4.04; 203.46, 137.64, 22.16; 115.82, 48.28, 20.47; 112.31, 88.86, 65.92; 96.71, 77.59, 57.88; smeared |
| 9349, UASM370246 | *Polygonia faunus* | CANADA: Alberta, Chickakoo Lake (53.616, -114.074); date collected: 2013-05-21; collector(s): McDonald, C., Aerial net | KY318595; pf_F-NHa3 | visually scored: 0, 1, 0, 1, 1, 0, 0, 1, 1, 1; RGB: 174.04, 84.09, 12.62; 198.97, 126.33, 36.77; 215.04, 152.56, 46.2; 152.2, 95.43, 42.66; 150.89, 117.43, 85.2; 131.68, 102.79, 75.37; smeared |
| 9350, UASM370247 | *Polygonia faunus* | CANADA: Alberta, North of Fawcett, HWY 44 nr. Hubert Grazing Reserve (54.588, -114.159); date collected: 2013-08-13; collector(s): McDonald, C., Aerial net | KY318596; pf_A-Ida | visually scored: 0, 0, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 132.62, 55.83, 12.39; 187.66, 98.85, 2.63; 185.17, 105.47, 6.26; 94.27, 39.43, 14.28; 135.98, 110.16, 83.29; 98.08, 81.57, 61.66; contrasted |
| 9351, UASM370248 | *Polygonia faunus* | CANADA: Alberta, North of Fawcett, HWY 44 nr. Hubert Grazing Reserve (54.588, -114.159); date collected: 2013-08-13; collector(s): McDonald, C., Aerial net | NA | visually scored: 0, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: NA |
| 9352, UASM370249 | *Polygonia gracilis gracilis* | CANADA: British Columbia, Kettle Valley (49.794, -119.345); date collected: 2013-07-14; collector(s): Sperling, F. A. H., Aerial net | KY318698; pg_G-A | visually scored: 2, 1, 1, 1, 2, -, 1, 0, -, 0; RGB: 137.01, 70.65, 31.28; 191.76, 133.17, 70.54; 198.57, 143.01, 78.05; 149.47, 90.55, 48.36; 158.63, 120.01, 88.32; 139.53, 107.61, 80.64 |

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| 9353, UASM370250 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-05-19; collector(s): Sperling, F. A. H., Aerial net | KY318597; pf_A-Ida | visually scored: 1, 1, 1, 2, 0, 0, 1, 3, 1, 0; RGB: 148.85, 73.47, 16.04; 199.63, 141.93, 39.71; 203.69, 156.97, 70.94; 123.53, 61.31, 23.48; 113.31, 83.91, 57.71; 85.88, 65.61, 45.9; contrasted |
| 9354, UASM370251 | *Polygonia faunus* | CANADA: Alberta, Lloyd Creek Natural area, near Haley Swamp (52.939, -114.288); date collected: 2013-05-18; collector(s): Sperling, F. A. H., Aerial net | KY318598; pf_A-WS9 | visually scored: 1, 1, 1, 2, 1, 0, 1, 0, 1, 1; RGB: 146.69, 64.86, 19.75; 186.45, 106.38, 16.59; 208.09, 143.42, 39.76; 136.32, 65.36, 21.07; 120.77, 92.08, 65.41; 102.78, 79.85, 57.14; smeared |
| 9355, UASM370252 | *Polygonia faunus* | CANADA: Alberta, Pigeon Lake, Itaska, Audubon Bay (53.071, -114.085); date collected: 2013-05-18; collector(s): Sperling, F. A. H., Aerial net | KY318599; pf_A-Ida | visually scored: 0, 1, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 170.99, 113.22, 49.51; 209.32, 152.77, 57.52; 222.7, 172.75, 74.05; 152.19, 84.69, 24.7; 145.24, 112.35, 77.88; 108.12, 84, 58.63; contrasted |
| 9356, UASM370253 | *Polygonia faunus* | CANADA: Alberta, Pigeon Lake, Itaska, Audubon Bay (53.071, -114.086); date collected: 2013-05-18; collector(s): Sperling, F. A. H., Aerial net | NA | visually scored: 1, 0, 0, 2, 0, 0, 0, 1, 1, 1; RGB: NA |
| 9357, UASM370254 | *Polygonia faunus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 53 (52.635, -114.624); date collected: 2013-05-23; collector(s): McDonald, C., Aerial net | NA | visually scored: 0, 1, 0, 1, 0, 0, 1, 3, 1, 0; RGB: NA |
| 9358, UASM370255 | *Polygonia faunus* | CANADA: Alberta, West of Rimbey, Township Road 6-4 c.5km S Hwy 64 (52.635, -114.624); date collected: 2013-05-23; collector(s): McDonald, C., Aerial net | KY318600; pf_A-Ida | visually scored: 1, 1, 1, 2, 1, 0, 1, 1, 1, 0; RGB: 145.31, 76.63, 29.56; 192.01, 121.57, 29.49; 195.49, 125, 26.1; 130.39, 55.81, 20.59; 117.16, 84.65, 60.63; 103.55, 77.96, 57; smeared |
| 9359, UASM370256 | *Polygonia faunus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318601; pf_F-NHa3 | visually scored: 1, 0, 1, 1, 0, 0, 1, 3, 1, 0; RGB: 137.27, 57.33, 9.96; 196.07, 122.37, 23.45; 199.07, 131.54, 33.07; 107.5, 48.6, 14.98; 117.77, 87.77, 64.85; 82.09, 65.62, 50.76; contrasted |
| 9360, UASM370257 | *Polygonia faunus* | CANADA: Alberta, South of Hinton, HWY 40, 20km S of Hinton, Gregg River (53.208, -117.551); date collected: 2013-08-22; collector(s): McDonald, C., Aerial net | KY318602; pf_A-Ida | visually scored: 0, 1, 0, 1, 1, 0, 1, 3, 1, 0; RGB: 127.49, 60.68, 18.61; 203.75, 122.67, 15.04; 214.11, 145.18, 25.04; 137.3, 64.67, 11.81; 119.78, 94.21, 68.69; 107.06, 86.32, 66.49; contrasted |
| 9361, UASM370258 | *Polygonia faunus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318603; pf_A-Ida | visually scored: 2, 1, 1, 1, 2, 0, 1, 3, 1, 0; RGB: 140.34, 69.67, 23.18; 189.67, 106.88, 8.63; 193.73, 126.96, 20.25; 134.96, 58.27, 16.12; 116.76, 95.11, 73.68; 88.5, 73.31, 57.15; contrasted |
| 9362, UASM370259 | *Polygonia faunus* | CANADA: British Columbia, South of Chetwynd, Rocky Ck FSR (55.675, -121.645); date collected: 2013-08-10; collector(s): Dupuis, J. R., Aerial net | KY318604; pf_A-Ida | visually scored: 0, 1, 0, 2, 1, 0, 1, 1, 1, 1; RGB: 114.79, 50.31, 19.23; 173.28, 97.72, 6.32; 188.69, 128.45, 32.98; 122.89, 61.04, 15.8; 112.51, 90.39, 68.6; 88.15, 71.88, 54.85; smeared |
| 9363, UASM370260 | *Polygonia faunus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-65 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318605; pf_F-NHa3 | visually scored: 0, 0, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 161.29, 76.52, 5.85; 208.37, 147.99, 35.47; 221.89, 172.39, 68.66; 143.98, 71.83, 18.03; 147.62, 118.86, 86.57; 107.21, 88.21, 63.61; contrasted |
| 9364, UASM370261 | *Polygonia faunus* | CANADA: Alberta, North of Lac la Biche (lake), AB RR141 Km 20-66 (54.963, -112.01); date collected: 2013-08-28; collector(s): McDonald, C., Aerial net | KY318606; pf_9364 | visually scored: 0, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 137.66, 49.6, 12.96; 191.66, 109.99, 9.76; 206.62, 134.76, 16.66; 95.55, 39.9, 19.81; 119.36, 87.75, 60.34; 83.71, 65.08, 46.04; contrasted |
| 9365, UASM370262 | *Polygonia faunus* | CANADA: Alberta, Buck Mountain (53.052, -114.74); date collected: 2010-06-02; collector(s): Dupuis, J. R., Aerial net | KY318607; pf_A-Ida | visually scored: 1, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 134.71, 66.67, 34.46; 200.36, 139.46, 58.37; 214.24, 169.77, 96.38; 125.03, 70.83, 41.71; 148.11, 111.86, 79.14; 122.12, 92.79, 66.59; contrasted |
| 9366, UASM370263 | *Polygonia faunus* | CANADA: Alberta, Northwest of Delburne, RR 251 (52.229, -113.453); date collected: 2013-08-16; collector(s): Dupuis, J. R., Aerial net | KY318608; pf_F-NHa3 | visually scored: 1, 1, 1, 1, 1, 0, 1, 3, 1, 0; RGB: 124.41, 63.15, 17.69; 200.17, 143.38, 52.54; 211.42, 154.2, 49.18; 108.95, 72.94, 46.64; 108.68, 83.65, 59.51; 88.05, 71.98, 53.8; contrasted |

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|---|---|---|---|---|
| 9367, UASM370264 | *Polygonia faunus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Brunet, B., Aerial net | KY318609; pf_A-Ida | visually scored: 0, 1, 1, 2, 0, 0, 1, 0, 1, 0; RGB: 112.47, 48.44, 21.6; 206.71, 144.53, 70.26; 206.39, 146.28, 57.43; 111.48, 69.79, 44.86; 137.7, 106.69, 78.08; 105.83, 83.38, 61.7; contrasted |
| 9368, UASM370265 | *Polygonia faunus* | CANADA: British Columbia, North of Gambel Creek (51.86, -119.299); date collected: 2013-05-12; collector(s): Brunet, B., Aerial net | KY318610; pf_A-Ida | visually scored: 1, 1, 1, 2, 1, 0, 1, 1, 1, 1; RGB: 141.28, 66.78, 22.72; 200.51, 128.75, 29.36; 194.25, 138.66, 45.79; 114.39, 58.51, 31.62; 118.44, 92.88, 65.41; 106.76, 85.2, 60.86; smeared |
| mtDNA only; EW12-7 | *Polygonia faunus* | CANADA, British Columbia, km 6 Santa Rosa road nr Rossland | JX134831; pf_AR-OreBCo | |
| mtDNA only; EW18-13 | *Polygonia faunus* | USA: Washington, Silver Star MtNymphalis | JX134773; pf_RA-Was | |
| mtDNA only; EW21-12 | *Polygonia faunus* | CANADA: Quebec, Reserve faunique de Laurentides | JX134760; pf_F-Que2 | |
| mtDNA only; EW22-2 | *Polygonia faunus* | CANADA: Quebec, Reserve faunique de Assinica | JX134756; pf_F-Que1 | |
| mtDNA only; EW22-3 | *Polygonia faunus* | USA: New Hampshire, Scott's Bog (Pittsfield) | JX134759; pf_F-NHa2 | |
| mtDNA only; EW22-4 | *Polygonia faunus* | USA: New Hampshire, Scott's Bog (Pittsfield) | JX134758; pf_F-NHa1 | |
| mtDNA only; EW22-5 | *Polygonia faunus* | USA: New Hampshire, Scott's Bog (Pittsfield) | JX134785; pf_F-NH13 | |
| mtDNA only; EW23-17 | *Polygonia faunus* | CANADA: Alberta, South of Ft. McMurray | JX134794; pf_A-Alb2 | |
| mtDNA only; EW23-9 | *Polygonia faunus* | CANADA: Alberta, Opal | JX134786; pf_A-Alb1 | |
| mtDNA only; EW29-8 | *Polygonia faunus* | USA: California, Dorrington | JX134757; pf_R-Cal1 | |
| mtDNA only; EW39-11 | *Polygonia faunus* | USA: California, Sugar Pine | JX134798; pf_R-Cal2 | |
| mtDNA only; EW39-13 | *Polygonia faunus* | USA: Colorado, Roaring River | JX134803; pf_H-Col1 | |
| mtDNA only; EW39-15 | *Polygonia faunus* | USA: Utah, Red Cloud Loop | JX134811; pf_H-Uta2 | |
| mtDNA only; EW40-10 | *Polygonia faunus* | USA: Utah, Red Cloud Loop | JX134806; pf_HA_WS2 | |
| mtDNA only; EW40-11 | *Polygonia faunus* | USA: Utah, Red Cloud Loop | JX134835; pf_H-Uta3 | |
| mtDNA only; EW40-13 | *Polygonia faunus* | USA: Utah, Red Cloud Loop | JX134807; pf_H-Uta1 | |
| mtDNA only; EW40-14 | *Polygonia faunus* | USA: Utah, Red Cloud Loop | JX134838; pf_H-Uta4 | |
| mtDNA only; EW40-18 | *Polygonia faunus* | USA: Colorado, Crested Butte | JX134836; pf_H-Col4 | |
| mtDNA only; EW40-19 | *Polygonia faunus* | USA: Colorado, Roaring River | JX134860; pf_H-Col5 | |
| mtDNA only; EW40-23 | *Polygonia faunus* | USA: Colorado, Turquoise Lake | JX134837; pf_H-Col3 | |
| mtDNA only; EW40-4 | *Polygonia faunus* | USA: Montana, Granite Co. | JX134804; pf_AFRH-WS1 | |
| mtDNA only; EW40-8 | *Polygonia faunus* | USA: California, Sugar Pine | JX134805; NA | |
| mtDNA only; EW40-9 | *Polygonia faunus* | USA: California, Sugar Pine | JX134834; pf_R-Cal4 | |
| mtDNA only; EW41-13 | *Polygonia faunus* | USA: Utah, Red Cloud Loop | JX134884; pf_H-Uta5 | |
| mtDNA only; EW41-6 | *Polygonia faunus* | USA: Colorado, Roaring River | JX134840; pf_H-Col7 | |
| mtDNA only; EW41-7 | *Polygonia faunus* | USA: Colorado, Crested Butte | JX134849; pf_H-Col6 | |
| mtDNA only; EW42-16 | *Polygonia faunus* | USA: Colorado, Rampart Range | JX134821; pf_H-Col2 | |
| mtDNA only; EW42-3 | *Polygonia faunus* | USA: Montana, Trail to Stuart Peaks | JX134883; pf_A-Mon4 | |
| mtDNA only; EW42-9 | *Polygonia faunus* | USA: Montana, Trail to Stuart Peaks | JX134830; pf_A-Mon1 | |
| mtDNA only; EW43-15 | *Polygonia faunus* | USA: Washington, Blue Mountains | JX134847; pf_A-WS3 | |

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|---|---|---|---|---|
| mtDNA only; EW43-19 | *Polygonia faunus* | USA: Washington, Blue Mountains | JX134858; pf_A-BCoWas | |
| mtDNA only; EW43-5 | *Polygonia faunus* | USA: Montana, Crazy mts., Half moon Camp area | JX134856; pf_A-Mon3 | |
| mtDNA only; EW43-6 | *Polygonia faunus* | USA: Montana, Crazy mts., Half moon Camp area | JX134853; pf_A-Mon2 | |
| mtDNA only; EW48-13 | *Polygonia faunus* | USA: Idaho, Idaho co. | JX134888; pf_A-Ida | |
| mtDNA only; EW50-7 | *Polygonia faunus* | USA: Georgia, Cooper Creek | JX134891; pf_S-Geo | |
| mtDNA only; EW10-10 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639457; pg_G-B | |
| mtDNA only; EW10-11 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639458; ps_9212 | |
| mtDNA only; EW10-12 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639459; pg_G-A | |
| mtDNA only; EW10-13 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639460; pg_Z_EW10_13 | |
| mtDNA only; EW10-14 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639461; pg_Z_EW10_14 | |
| mtDNA only; EW10-9 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639462; pg_G-A | |
| mtDNA only; EW11-4 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639463; pg_Z_EW11_4 | |
| mtDNA only; EW11-9 | *Polygonia gracilis zephyrus* | USA: Oregon, Jackson Co. Mt. Ashland Rd | FJ639464; pg_8382 | |
| mtDNA only; EW14-10 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Skagit Valley, SE of Hope | FJ639465; pg_G-B | |
| mtDNA only; EW14-11 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Skagit Valley, SE of Hope | FJ639466; pg_G-A | |
| mtDNA only; EW14-12 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Goatfel FSR nr Yahk | FJ639467; pg_G-A | |
| mtDNA only; EW14-14 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Goatfel FSR nr Yahk | FJ639468; pg_G-B | |
| mtDNA only; EW14-15 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Goatfel FSR nr Yahk | FJ639469; pg_G-A | |
| mtDNA only; EW15-1 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Skagit Valley, SE of Hope | FJ639470; pg_Z_EW15_1 | |
| mtDNA only; EW15-15 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Santa Rosa rd nr Rossland | FJ639471; pg_Z_EW15_15 | |
| mtDNA only; EW15-4 | *Polygonia gracilis zephyrus* | USA: Washington, Silver Star MtNymphalis | FJ639473; pg_Z_EW15_4 | |
| mtDNA only; NW63-11 | *Polygonia gracilis zephyrus* | CANADA: British Columbia, Vale Mountain (Valemont?) | FJ639474; pg_Z_NW63_11 | |
| mtDNA only; NW74-5 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | FJ639475; pg_G_NW74_6 | |
| mtDNA only; NW74-6 | *Polygonia gracilis zephyrus* | USA: Wyoming, Big Piney, Forest Road 10046 | AY248797; pg_G_NW74_6 | |
| mtDNA only; EW15-3 | *Polygonia gracilis gracilis* | USA: Washington, Silver Star MtNymphalis | FJ639472; NA | |
| mtDNA only; EW19-7 | *Polygonia gracilis gracilis* | USA: Colorado, San Juan Co, Cascade Creek | FJ639456; pg_G-A | |
| mtDNA only; EW21-14 | *Polygonia gracilis gracilis* | CANADA: Quebec, Assinica | FJ639414.1 ; NA | |
| mtDNA only; EW22-8 | *Polygonia gracilis gracilis* | USA: New Hampshire, Scott's Bog (Pittsfield) | FJ639415.1 ; NA | |
| mtDNA only; EW22-9 | *Polygonia gracilis gracilis* | CANADA: Quebec, Reserve faunique de Laurentides | FJ639416; pg_8382 | |
| mtDNA only; EW19-9 | *Polygonia progne* | CANADA: Alberta, Edmonton, McTaggart Sanctuary | FJ639425; pp_P-B | |
| mtDNA only; EW21-16 | *Polygonia progne* | CANADA: Quebec, Reserve faunique de Ashuapmushuan | FJ639476; pp_P-A | |
| mtDNA only; EW21-2 | *Polygonia progne* | USA: New Hampshire, Scott's Bog (Pittsfield) | FJ639426; pp_EW21_2 | |
| mtDNA only; EW21-3 | *Polygonia progne* | USA: Virginia, THR | AY248795; NA | |
| mtDNA only; EW21-4 | *Polygonia progne* | CANADA: Quebec, Reserve faunique de Ashuapmushuan | FJ639427; pp_P-A | |
| mtDNA only; EW22-15 | *Polygonia progne* | USA: New Hampshire, Scott's Bog (Pittsfield) | FJ639428; pp_EW22_15 | |

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|---|---|---|---|---|
| mtDNA only; EW22-16 | *Polygonia progne* | USA: New Hampshire, Scott's Bog (Pittsfield) | FJ639429; pp_P-A | |
| mtDNA only; EW22-17 | *Polygonia progne* | CANADA: Quebec, Reserve faunique de Ashuapmushuan | FJ639430; pp_EW22_17 | |
| mtDNA only; EW22-18 | *Polygonia progne* | CANADA: Quebec, Reserve faunique de Assinica | FJ639431; pp_P-B | |
| mtDNA only; EW22-19 | *Polygonia progne* | CANADA: Quebec, Reserve faunique de Assinica | FJ639432; pp_P-B | |
| mtDNA only; EW22-20 | *Polygonia progne* | USA: New Hampshire, Scott's Bog (Pittsfield) | FJ639433; pp_P-A | |
| mtDNA only; EW22-21 | *Polygonia progne* | USA: New Hampshire, Scott's Bog (Pittsfield) | FJ639434; pp_P-A | |
| mtDNA only; NW63-10 | *Polygonia progne* | USA: Wisconsin, Riveredge Nature Center | FJ639435; pp_NW15-10 | |
| mtDNA only; EW11-10 | *Polygonia satyrus* | USA: Idaho, Bonneville Co. Little Elk Creek | FJ639436; ps_S_EW11_10 | |
| mtDNA only; EW11-11 | *Polygonia satyrus* | USA: Oregon, Benton Co., MacDonald Forest | FJ639437; ps_9212 | |
| mtDNA only; EW11-12 | *Polygonia satyrus* | USA: Oregon, Benton Co., MacDonald Forest | FJ639438; ps_S-A | |
| mtDNA only; EW15-10 | *Polygonia satyrus* | CANADA: British Columbia, Pend-d'Oreille FSR, SE of Trail | FJ639439; ps_S_EW15_10 | |
| mtDNA only; EW15-11 | *Polygonia satyrus* | CANADA: British Columbia, Pend-d'Oreille FSR, SE of Trail | FJ639440; ps_S_EW15_11 | |
| mtDNA only; EW15-12 | *Polygonia satyrus* | CANADA: British Columbia, Pend-d'Oreille FSR, SE of Trail | FJ639441; ps_S_EW15_12 | |
| mtDNA only; EW15-13 | *Polygonia satyrus* | CANADA: British Columbia, Pend-d'Oreille FSR, SE of Trail | FJ639442; ps_S-B | |
| mtDNA only; EW15-14 | *Polygonia satyrus* | CANADA: British Columbia, Santa Rosa rd nr Rossland | FJ639443; ps_S_EW15_14 | |
| mtDNA only; EW15-16 | *Polygonia satyrus* | CANADA: British Columbia, Santa Rosa rd nr Rossland | FJ639444; ps_Z_EW15_15 | |
| mtDNA only; EW15-9 | *Polygonia satyrus* | CANADA: British Columbia, Skagit Valley, SE of Hope | FJ639445; ps_S-B | |
| mtDNA only; EW16-6 | *Polygonia satyrus* | CANADA: British Columbia, Skagit Valley, SE of Hope | FJ639446; ps_S_EW16_6 | |
| mtDNA only; EW16-7 | *Polygonia satyrus* | CANADA: British Columbia, Skagit Valley, SE of Hope | FJ639447; ps_S-C | |
| mtDNA only; EW23-13 | *Polygonia satyrus* | CANADA: Alberta, South of Ft. McMurray | FJ639448; ps_S-C | |
| mtDNA only; EW23-14 | *Polygonia satyrus* | CANADA: Alberta, Ft. McKay | FJ639449; ps_S-A | |
| mtDNA only; NW65-9 | *Polygonia satyrus* | USA: Washington, Blue Mountains | FJ639450; ps_S-B | |
| mtDNA only; NW74-8 | *Polygonia satyrus* | USA: Oregon, Benton Co. | FJ639451; ps_S_NW74_8 | |
| mtDNA only; NW74-9 | *Polygonia satyrus* | USA: Oregon, Benton Co. | AY248796; ps_S_NW74_9 | |

Table S2 (next three pages). Diagnostic characters of *Polygonia* butterflies published in North American field guides, with an emphasis on western North America. If an author designated a character as being primary, or highly diagnostic, it is indicated with a grey background. VHW: ventral hindwing, VFW: ventral forewing, DHW: dorsal hindwing, DFW: dorsal forewing.

| | *Polygonia faunus* (Edwards, 1862) | | | | | | | | *P. satyrus* (Edwards, 1869) | | | | | | *P. progne* (Cramer, 1776) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Publication | Row of greenish spots on VFW | **Two rows green spots** | **DHW row of subM spots, separated** | Very irregular margins | **2 blk spots, inner DFW margin** | **dark spot on DHW** | **comma reduced, barbed or not** | **DFW inner spots fused** | **DHW englarged subM spots; merge** | **Comma curved, barbed** | **Triangular spot on DHW** | **2 blk spots, inner DFW margin** | Golden orange dorsally | Fine, dark striations VWs | DHW black margin expanded | **DHW small subM spots** | VHW one tone, VFW two-toned | VWs striated dark grey | **Comma is wide angle "L", thin** | 1 blk spot on DFW margin | **dark spot on DHW** | **V subMar light grey "v"s** |
| Alberta Butterflies (Bird *et al.* 1995) | + | | + | | | | | | + | | | | + | | + | + | + | | | | | |
| Butterflies of Manitoba (Klassen *et al.* 1989) | + | | + | + | | | | | | + | | | + | + | | + | | + | + | | | |
| Butterflies through Binoculars - The West (Glassberg 2001) | + | | + | + | + | + | | | + | | + | + | | | | + | + | + | + | + | - | |
| Butterflies of Cascadia (Pyle 2002) | | + | | + | | | + | | + | + | + | | + | + | | | | | | | | |
| Butterflies of Saskatchewan (Hooper 1973) | + | | | | + | + | + | | + | + | | + | + | | | | + | + | + | | | |
| Butterflies of North America (Brock and Kaufman 2003) | + | | + | + | | + | | | + | | + | + | | | | + | + | + | + | + | - | |
| Butterflies of British Columbia (Acorn 2006) | + | | + | | | + | | | | + | | + | + | | | | | | | | | + |
| Western Butterflies (Opler 1999) | + | | | + | | | | | | | | + | | + | | | + | | | | | |
| The Butterflies of Oregon (Dornfeld 1980) | | + | + | | | | + | | | + | | | | + | | | | | | | | |
| Butterflies of British Columbia (Guppy and Shepard 2001) | + | | | | | | | | | | | | + | | | | + | | + | | | |
| The butterflies of North America (Scott 1986) | + | | + | + | | | | | + | | + | + | + | | | + | | | | + | | |
| The Butterflies of Colorado: Nymphalidae (Fisher 2006) | + | | | + | | | | | + | | | | + | | | | | | | | | |
| The Guide to Butterflies of Oregon and Washington (Neill 2001) | + | | | | | | | | | | | | | + | | | + | | | | | |
| Butterflies of Rocky Mountain National Park (Angel 2005) | + | | | | | + | | | | | | | | | | | | | | | | |
| Butterflies of North Dakota (Royer 1988) | + | | + | + | | | | | + | | | + | | | | + | | | | | | |
| Butterflies of Grand Teton and Yellowstone National Parks (Poole 2009) | + | | + | + | | | + | + | + | + | + | + | + | | | + | + | | + | | | |

+/- denotes presence/absence; < indicates faint character; **bold** indicates visually scored character used here

| | | *P. gracilis gracilis* (Grote and Robinson, 1867) | | | | | | | *P. g. zephyrus* (Edwards, 1870) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Publication | separate *P. g. gracilis* and *P. g. zephyrus* | VHW high contrast b/w Bas/Mar | two morphs, grey and brown | **Comma tapered at both ends** | **DHW enclosed subM spots** | **DHW subM spots run together** | **Triangual DHW spot** | **DHW two black spots, faint third** | VHW contrast b/w Bas/Mar | two morphs, grey and brown | **DHW subM spots merge** | **Triangual DHW spot** | Broken row of green spots | **Comma unbarbed curve, "L"** | **2 blk spots basal front edge DFW** | **1 blk patch DHW edge spot below** | **3 spots inward fron DFW mar** | **Comma abruptly curved** | **V subMar spots sml, blk, ylw ring** |
| Alberta Butterflies (Bird *et al.* 1995) | Y | + | + | | | | | | + | + | + | | | | | | | | |
| Butterflies of Manitoba (Klassen *et al.* 1989) | | + | | + | + | | | | | | | | | | | | | | |
| Butterflies through Binoculars - The West (Glassberg 2001) | N | + | | | | + | - | | | | | | | | | | | | |
| Butterflies of Cascadia (Pyle 2002) | Y | | | | | | | | + | | + | - | + | + | | | | | |
| Butterflies of Saskatchewan (Hooper 1973) | Y | + | | + | | | | + | + | | + | - | | + | + | + | + | | |
| Butterflies of North America (Brock and Kaufman 2003) | N | + | | | | + | - | | | | | | | | | | | | |
| Butterflies of British Columbia (Acorn 2006) | N | + | | + | | | | | | | | | | | | | | | |
| Western Butterflies (Opler 1999) | N | | | | | | | | + | | | | | | | | | + | |
| The Butterflies of Oregon (Dornfeld 1980) | Y | | | | | | | | | | + | | | | | | + | | + |
| Butterflies of British Columbia (Guppy and Shepard 2001) | Y | + | | | | | | | + | | | | + | | | | | | |
| The butterflies of North America (Scott 1986) | Y | + | | | | | | | + | | + | | | | | | | | |
| The Butterflies of Colorado: Nymphalidae (Fisher 2006) | N | | | | | | | | + | | + | | | | | | | | |
| The Guide to Butterflies of Oregon and Washington (Neill 2001) | N | | | | | | | | + | | + | | | | | | | | + |
| Butterflies of Rocky Mountain National Park (Angel 2005) | N | | | | | | | | + | | | - | | + | | | | | |
| Butterflies of North Dakota (Royer 1988) | Y | + | + | | | + | | | | | | | | | | | | | |
| Butterflies of Grand Teton and Yellowstone National Parks (Poole 2009) | | + | | + | | | | | | | | | | | | | | | |

+/- denotes presence/absence; < indicates faint character; **bold** indicates visually scored character used here

| | *P. oreas* (Edwards, 1869) | | | | | | | | | *P. interrogationis* (Fabricius, 1798) | | | | | | *P. comma* (Harris, 1842) | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Publication | DHW with white marginal hair | VHW slight separation b/w basal/mar | Heavily striated VW | Mar band not darker near Mar | Triangular spot on DHW | subMar spots form band | V subMar light grey "v"s | VHW very dark | comma sharp 90o, tapered | Longer tails | Comma is "?" | Small horizontal bar VFW | FW hooked | DFW 2 inner spot | DFW trailing spots divided | VHW postmedian line very jagged | VFW apex two-toned | Small horizontal bar VFW | Comma hooked at both ends | DFW top spot reduced | DFW spots enclosed | DHW spot |
| Alberta Butterflies (Bird *et al.* 1995) | + | + | | | | | | | | + | | | | | | | | | | | | |
| Butterflies of Manitoba (Klassen *et al.* 1989) | | | | | | | | | | | | | | | | | | | + | + | | |
| Butterflies through Binoculars - The West (Glassberg 2001) | | + | | + | | + | | | | | + | + | | | | + | + | - | | | | |
| Butterflies of Cascadia (Pyle 2002) | | | | | | | | | | | | | | | | | | | | | | |
| Butterflies of Saskatchewan (Hooper 1973) | | | | | | | | | | + | + | | | + | + | | | | + | + | + | + |
| Butterflies of North America (Brock and Kaufman 2003) | | + | + | + | < | | | | | + | + | + | | | | + | | - | | | | |
| Butterflies of British Columbia (Acorn 2006) | | | | | | | + | | | | | | | | | | | | | | | |
| Western Butterflies (Opler 1999)* | | | | | | | | + | | | | | | | | | | | | | | |
| The Butterflies of Oregon (Dornfeld 1980) | | | + | | | + | | + | + | | | | | | | | | | | | | |
| Butterflies of British Columbia (Guppy and Shepard 2001) | + | + | | | | | | + | | | | | | | | | | | | | | |
| The butterflies of North America (Scott 1986)* | | | | | | + | | + | | | + | | + | | | + | | | + | | + | |
| The Butterflies of Colorado: Nymphalidae (Fisher 2006) | | | | | | + | | + | | | + | | | | | | | | | | | |
| The Guide to Butterflies of Oregon and Washington (Neill 2001) | | | | | | | | | | | | | | | | | | | | | | |
| Butterflies of Rocky Mountain National Park (Angel 2005) | | | | | | | | | | | | | | | | | | | | | | |
| Butterflies of North Dakota (Royer 1988) | | | | | | | | | | | + | | + | | | | | | + | + | + | |
| Butterflies of Grand Teton and Yellowstone National Parks (Poole 2009) | | | | | | + | | + | | | | | | | | | | | | | | |

+/- denotes presence/absence; < indicates faint character; bold indicates visually scored character used here; *treats *P. oreas* as subspecies of *P. progne*

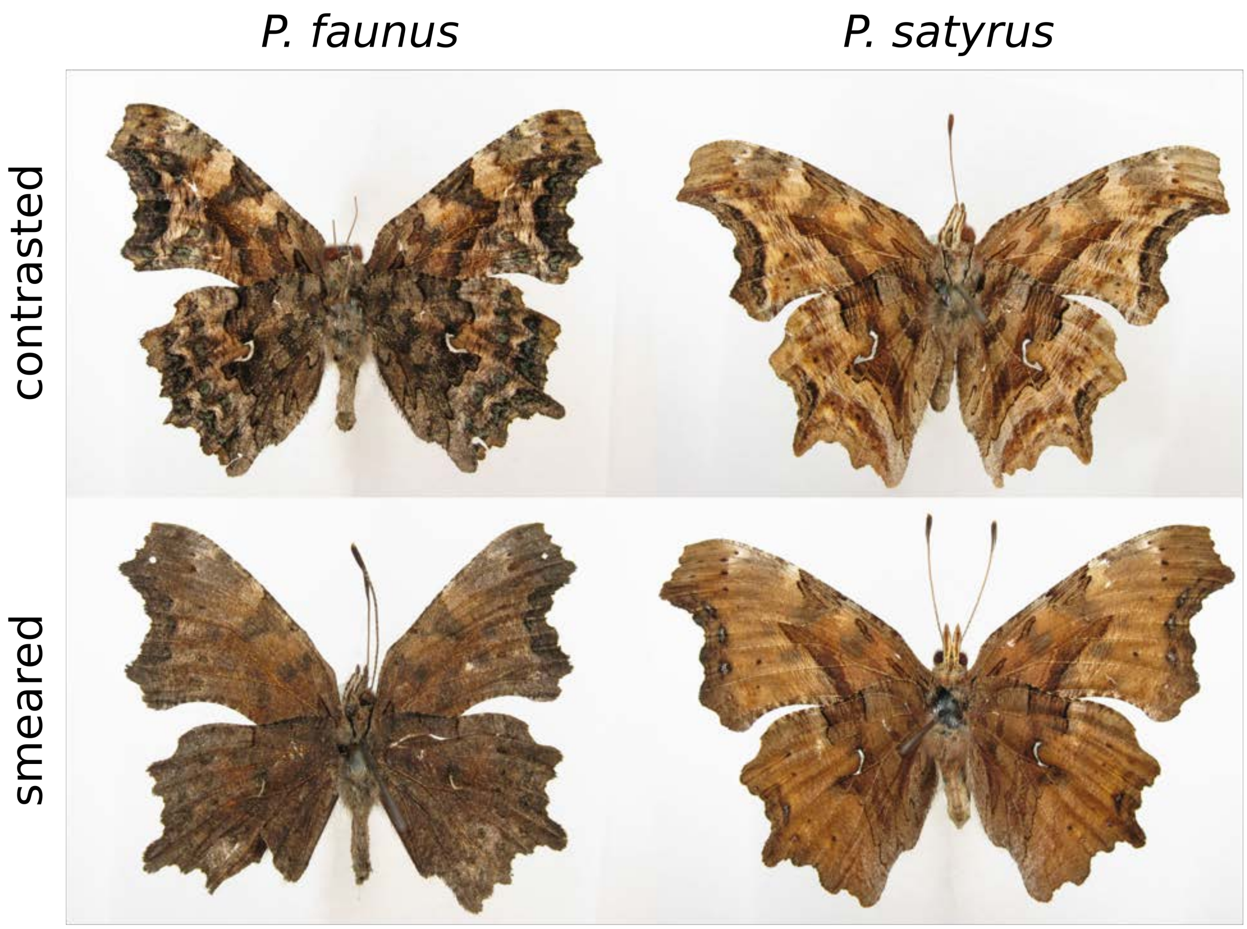


Figure S1. Examples of smeared and contrasted morphs of *P. faunus* and *P. satyrus* used in this study.

Figure S2 (next page). Consensus tree from ML analysis of unique COI haplotypes. Dots on nodes indicate high support: >80% SH-aLRT and >95% ultra-fast bootstrap in ML analysis, and >0.8 posterior probability in Bayesian analysis. Major clades are named (as in Figure 4) in grey text, and grey circles indicate *P. g. zephyrus* individuals.


nv_9338
nv_9339
Nymphalis spp.
nc_9340
na_9343
na_9344
pp_8343
pp_8351
pp_8363
pp_8381
pp_9317
pp_9327
pp_9328
progne_EW22-18
progne_EW19-9
progne_EW22-19
progne_EW21-16
P. progne
pp_8353
pp_8345
pp_8387
pp_9319
pp_8379
pp_9324
pp_8357
pp_8354
pp_9312
pp_8380
pp_8349,pp_8364,pp_8365,pp_8369,pp_8373,pp_8374,pp_8376,pp_8384,pp_9313,pp_9316,pp_9318,pp_9321,pp_9322,pp_9325,pp_9330,pp_9331,pp_9332,pp_9333,pp_9336,pp_9346,pp_9347,progne_EW21-4,EW22-20,EW22-21
P-A
pp_8368
pp_8346
pp_8359,pp_8367,pp_8391,pp_9334
pp_8361
pp_8362
pp_9337
pp_8344
pp_8348
pp_9323
pp_8350
progne_EW21-2
pp_8352
pp_8372
progne_EW22-15
progne_EW22-16
pp_8360
pp_9315
pp_9345
pp_8347
pp_8371
progne_EW22-17
pp_8370
pp_8377
progne_63-10
pp_8355
pp_8378
pg_8382
gracilis_EW22-9
zephyrus_EW11-9
P. gracilis
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pg_8392
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zephyrus_EW10-10
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zephyrus_EW14-10
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G-A
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zephyrus_EW19-7
zephyrus_EW11-4
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gracilis_NW74-6
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satyrus_NW74-8
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S-B
satyrus_EW15-13
ps_9218
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ps_9203
ps_9229
ps_9248
P. satyrus
satyrus_EW15-14
satyrus_EW15-9
ps_8397
satyrus_EW11-10
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S-A
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satyrus_EW15-11
satyrus_EW15-16
satyrus_EW16-6
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pg_9298
G-X
faunus_EW40_8
faunus_EW39_11
faunus_EW29_8
faunus_EW40_9
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pf_8322
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faunus_EW39_13
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A-Ida
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faunus_EW22_2
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pf_9264
faunus_EW12_7
faunus_EW42_9
faunus_EW43_6
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A-WS3
pf_8327
pf_9279
faunus_EW43_5
pf_8386
pf_9273
pf_9348
faunus_EW40_13
faunus_EW41_13
P. faunus
faunus_EW43_19
pf_8307
pf_9276
faunus_EW21_12
faunus_EW41_6
faunus_EW40_14
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F-NHA3
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pf_8316
pf_8332
faunus_EW22_4
faunus_EW22_3
pf_9359
pf_9363
pf_8335
pf_8339
pf_8328
faunus_EW40_18
faunus_EW40_19
pf_9263
faunus_EW50_7
faunus_EW23_9
pf_9260
faunus_EW39_15
pf_8301
pf_9270
am_9341
am_9342
Aglais milberti
0.02

Figure S3 (next page). Consensus tree from ML analysis of 2,572 SNP dataset. Node support values are SH-aLRT/ultra-fast bootstrap.

0.06

Figure S4 (next page). Fifty percent majority rule consensus from BI analysis of 121,513 bp alignment of GBS data (including flanking sequence) with posterior probability values for node support.

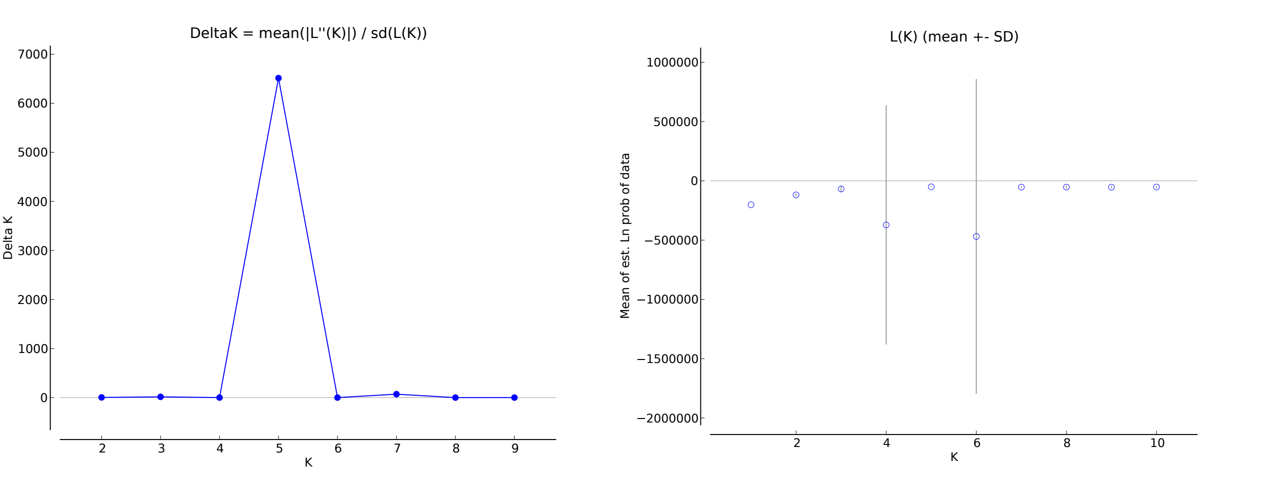

0.07

Figure S5 (next page). Fifty percent majority rule consensus from BI analysis of 2,572 SNPs from GBS data with posterior probability values for node support.


0.0020

Figure S6 (next 6 pages). $\Delta K$ and Ln Pr($X$|$K$) results, and STRUCTURE results for all values of $K$, for all individuals, *P. faunus*, *P. satyrus*, *P. gracilis* & *P. progne*, *P. gracilis*, and *P. progne*.

# STRUCTURE results for all individuals




## Major modes for the uploaded data:

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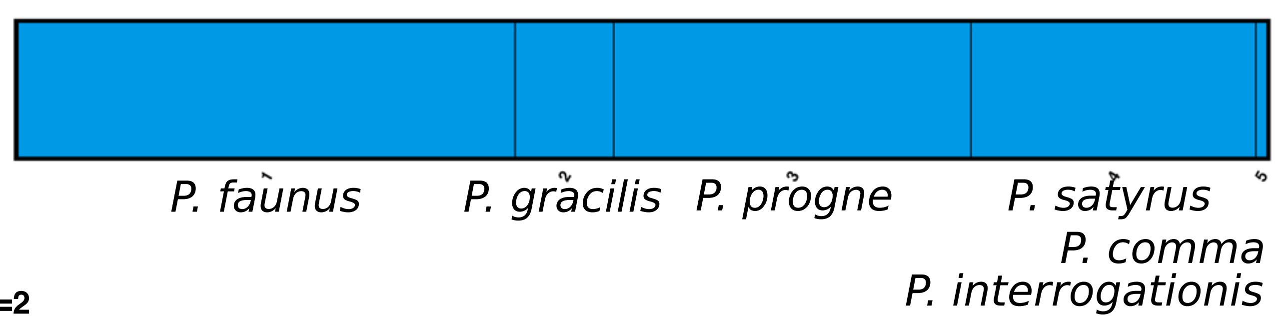
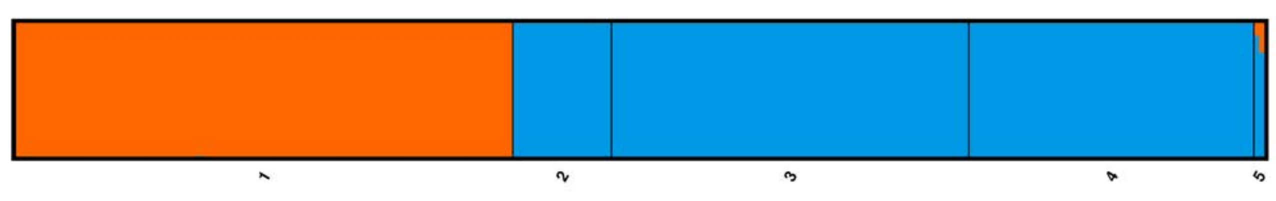



K=2



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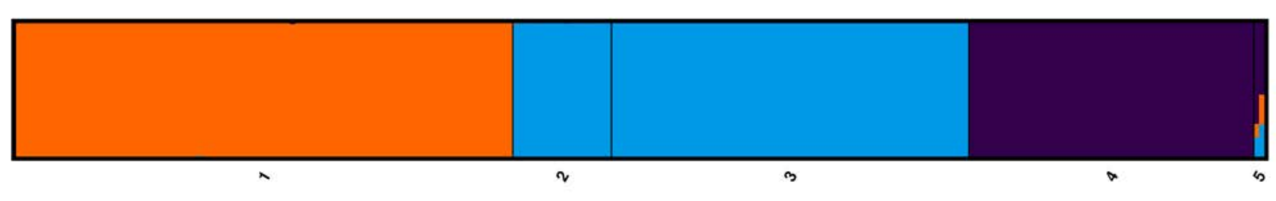

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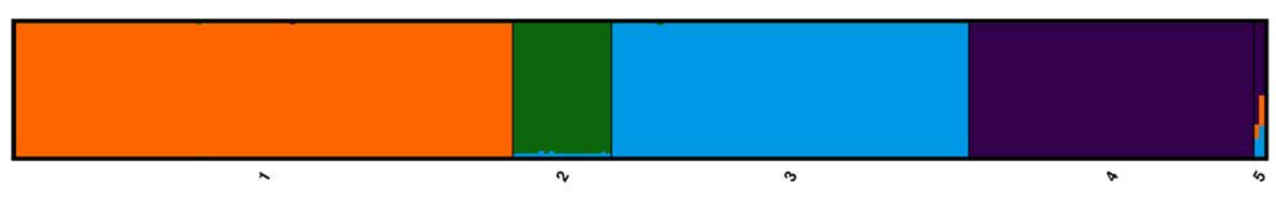

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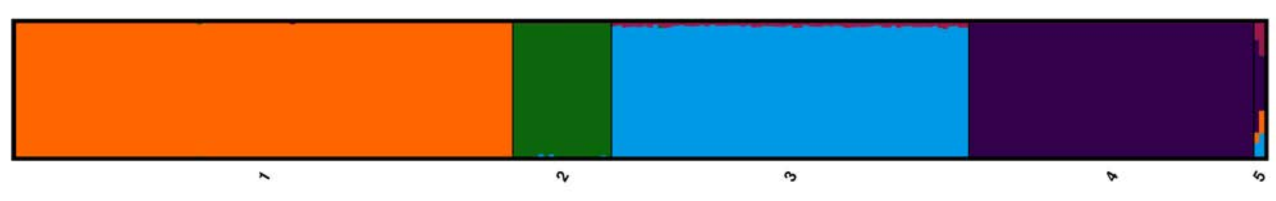

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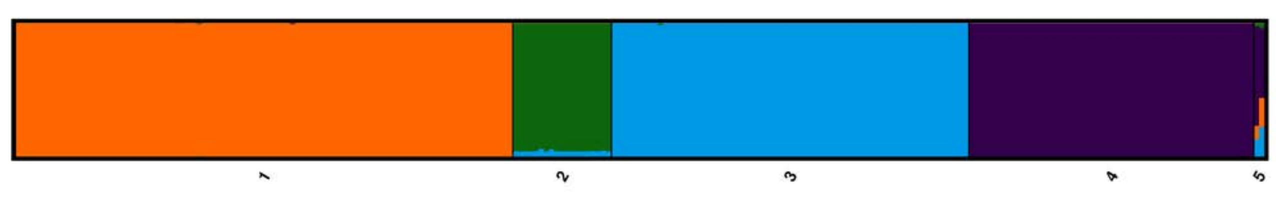

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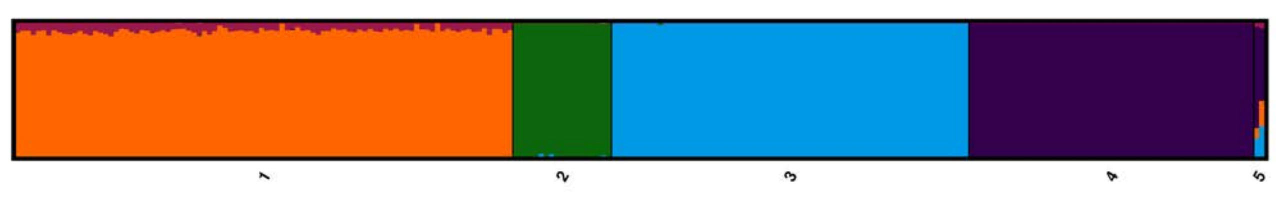

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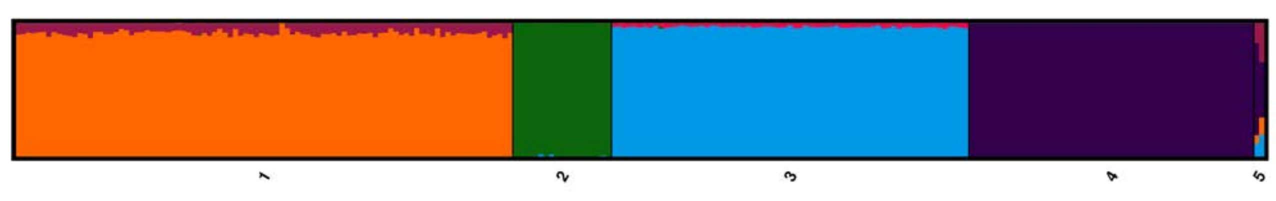

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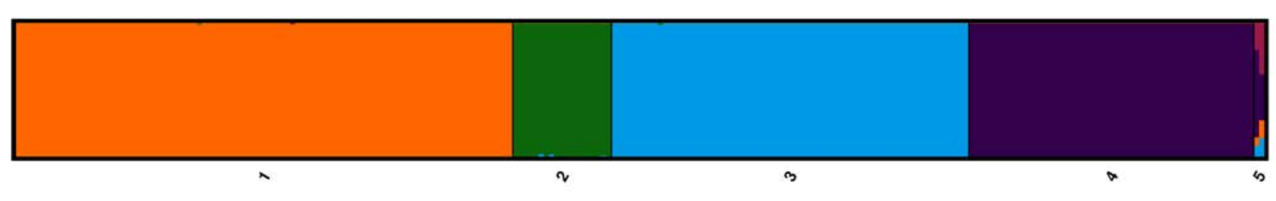

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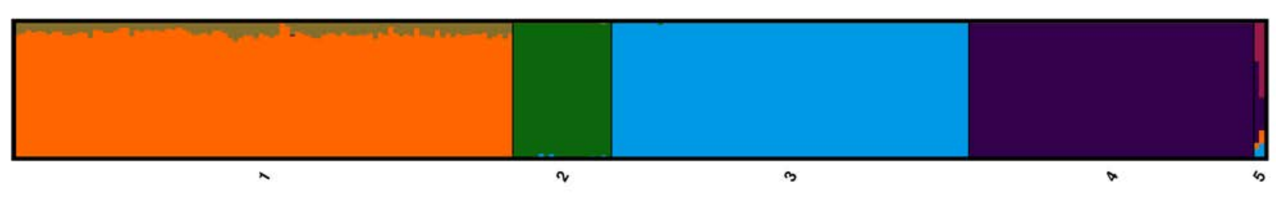

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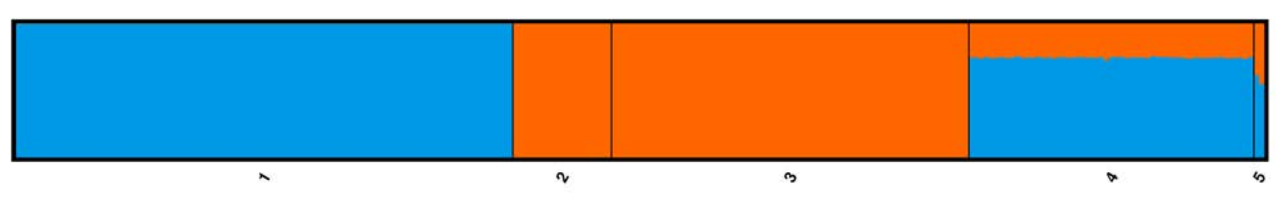

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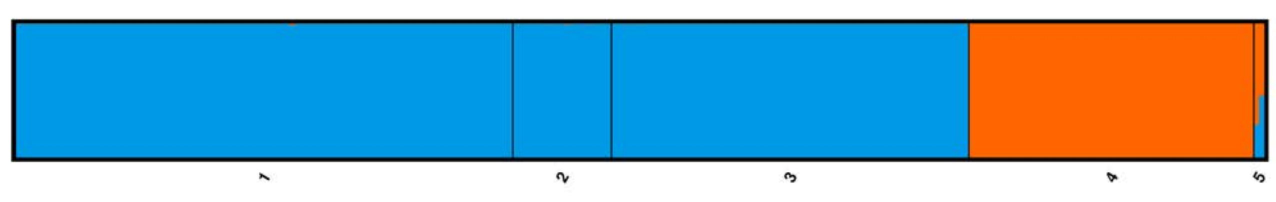

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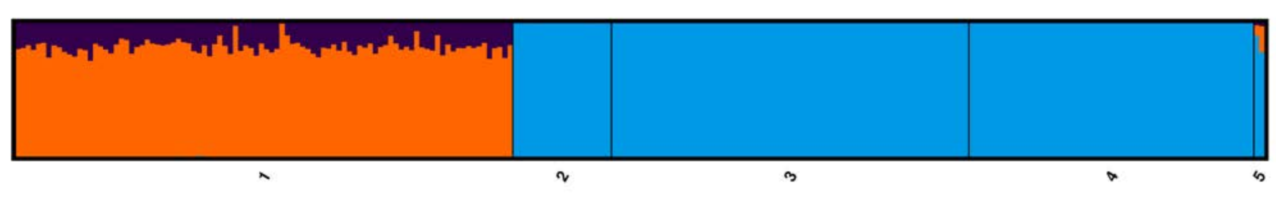

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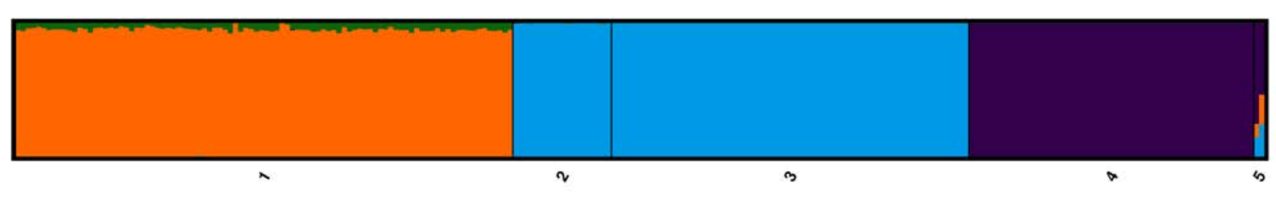

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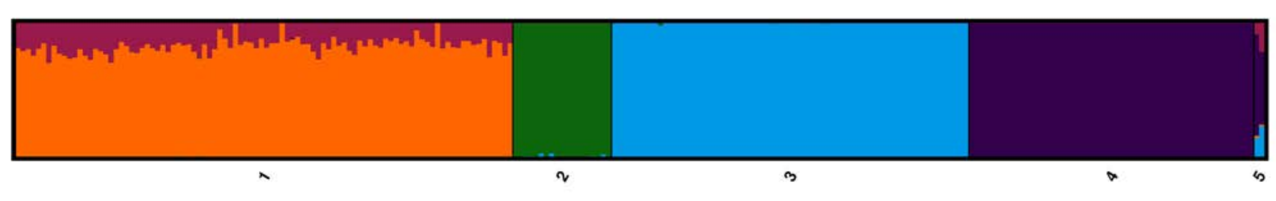

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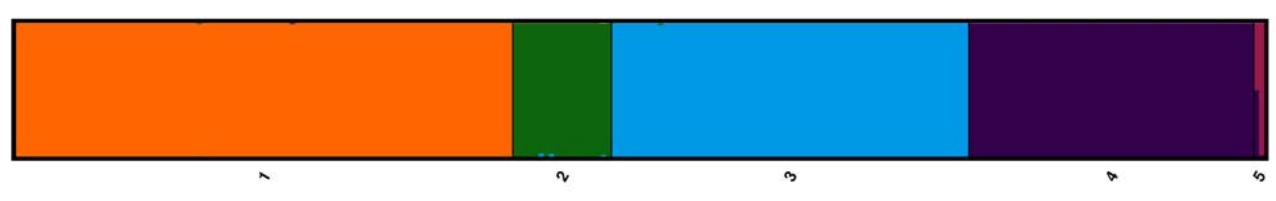

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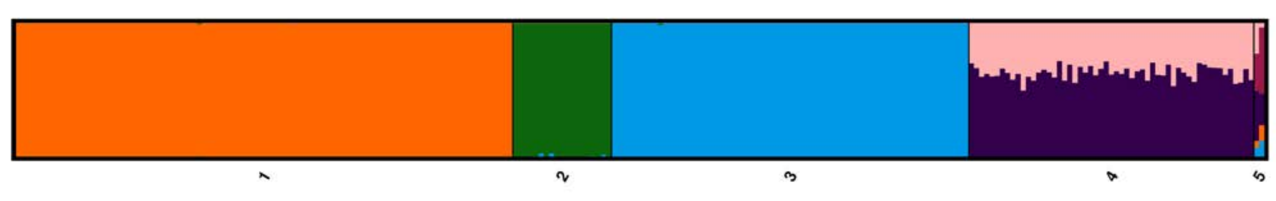

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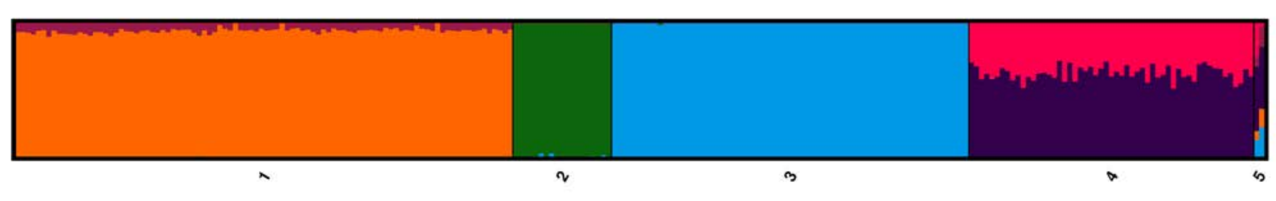

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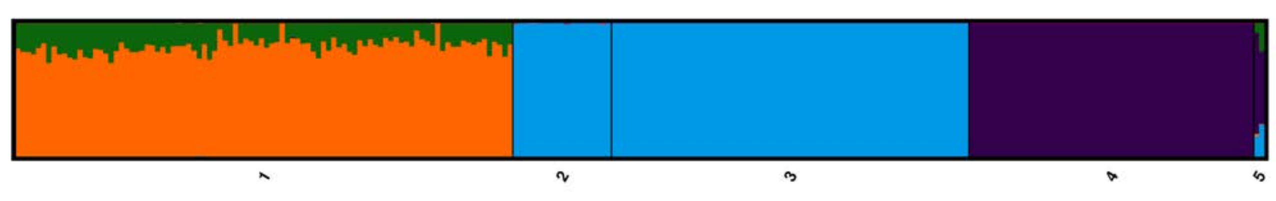

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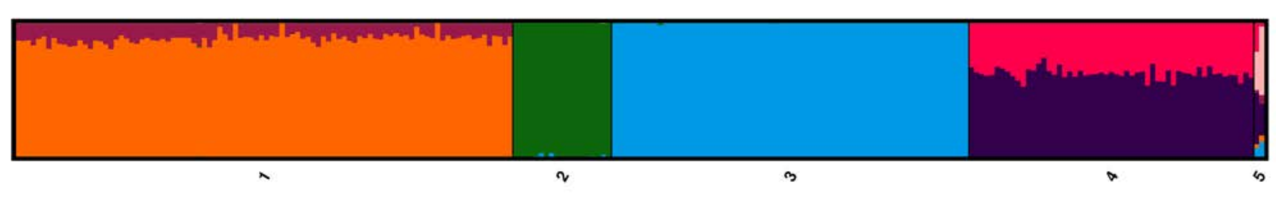

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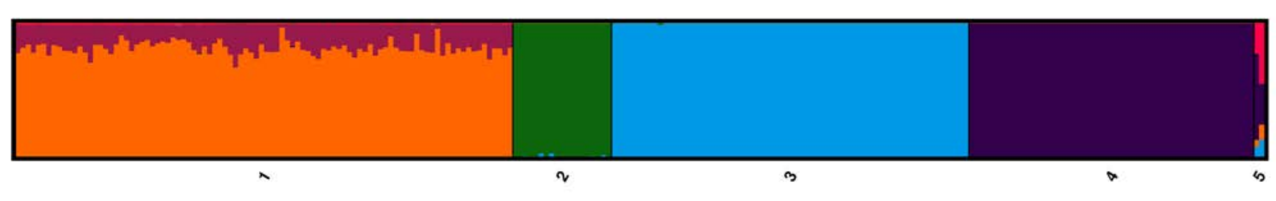

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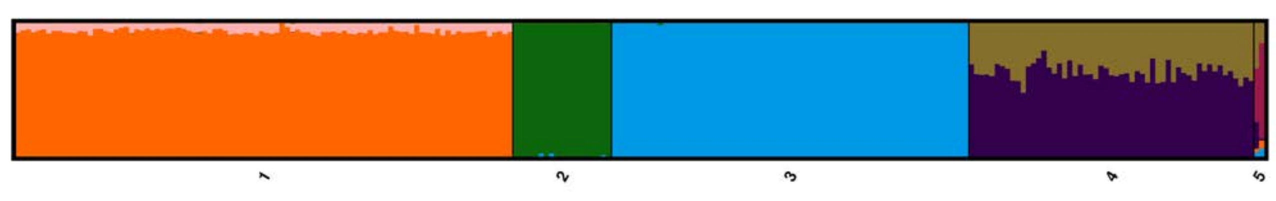
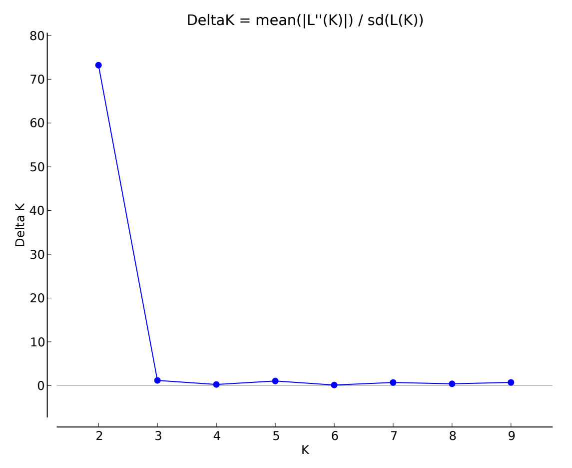
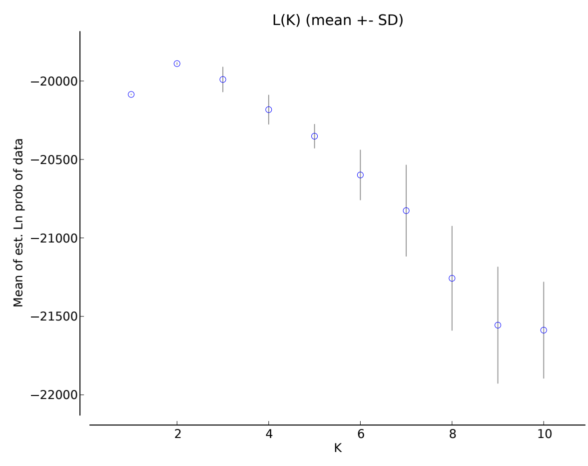


## Division of runs by mode:

| | |
|---|---|
| K=1 | 10/10 |
| K=2 | 6/10, 3/10, 1/10 |
| K=3 | 8/10, 2/10 |
| K=4 | 6/10, 4/10 |
| K=5 | 8/10, 2/10 |
| K=6 | 6/10, 2/10, 2/10 |
| K=7 | 6/10, 3/10, 1/10 |
| K=8 | 7/10, 3/10 |
| K=9 | 6/10, 4/10 |
| K=10 | 7/10, 3/10 |

STRUCTURE results for *P. faunus*




## Major modes for the uploaded data:

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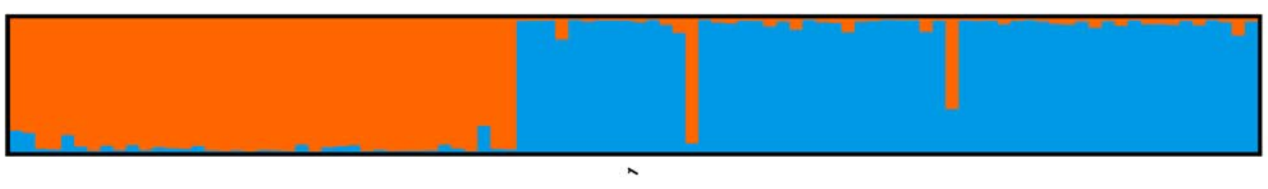
K=2



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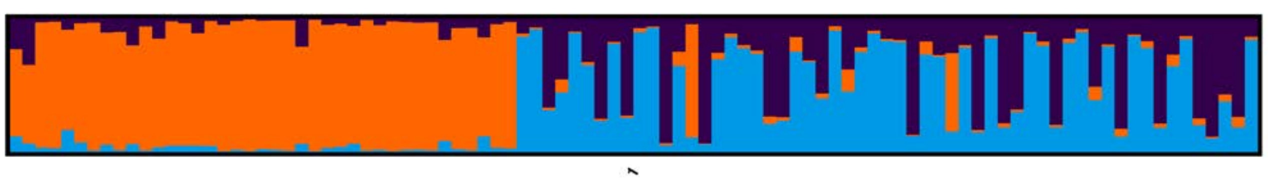

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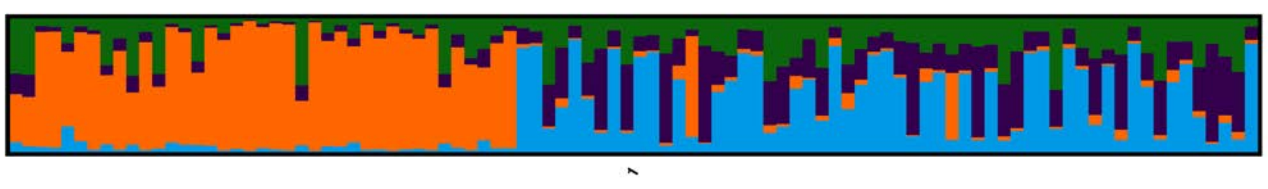

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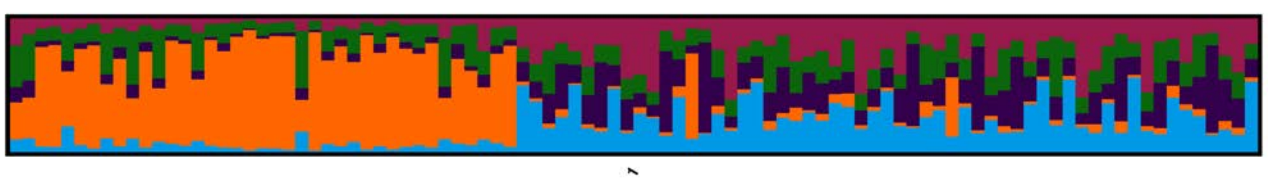

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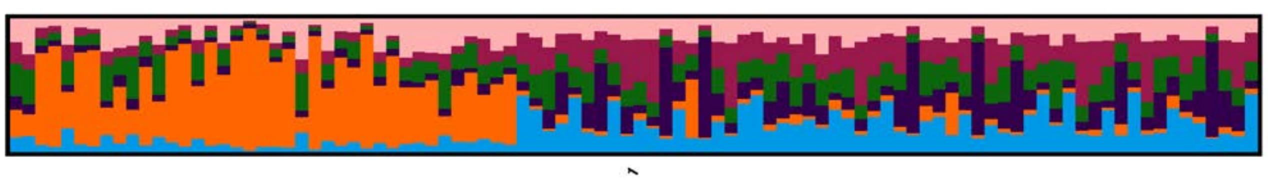

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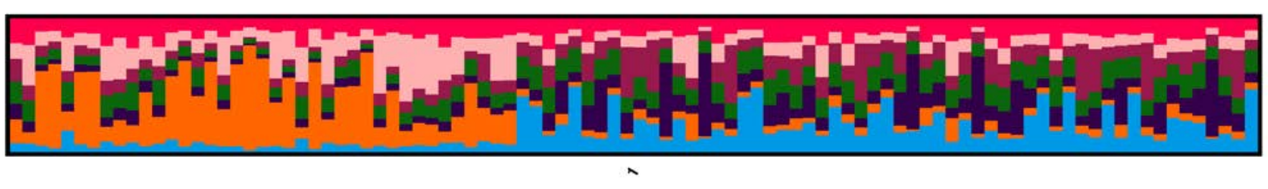

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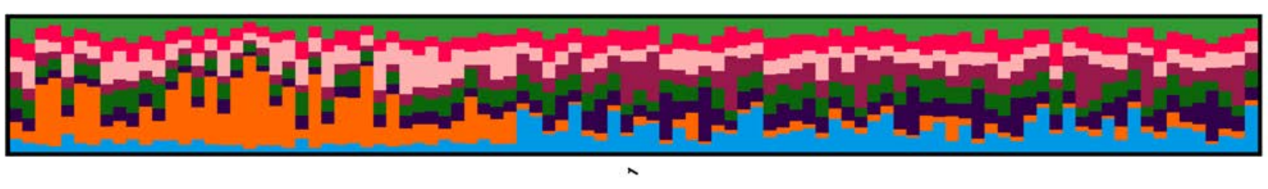

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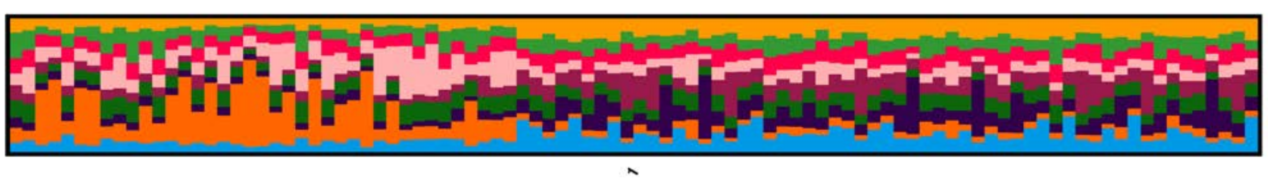

K=10

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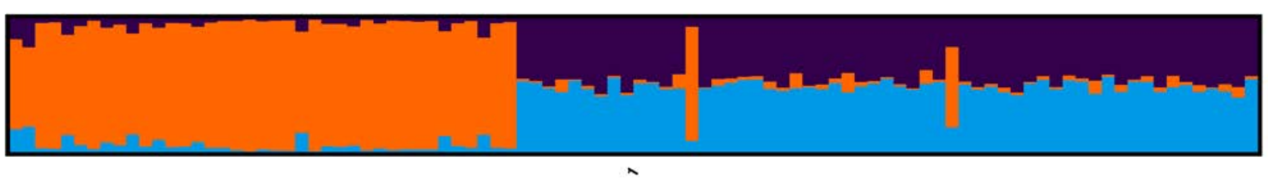

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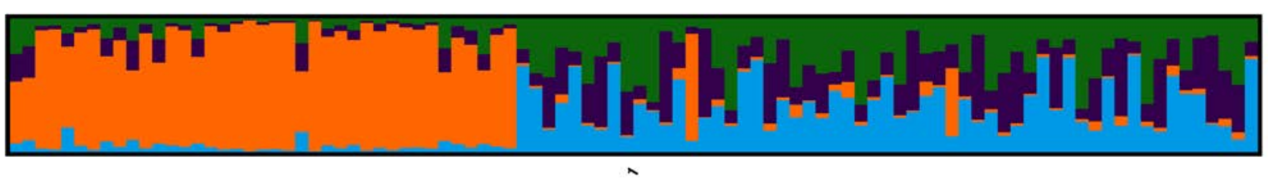

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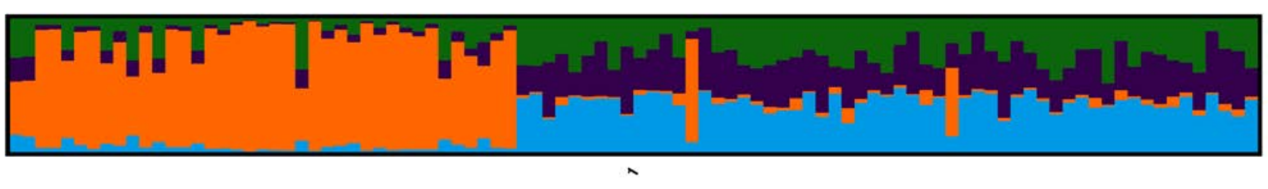

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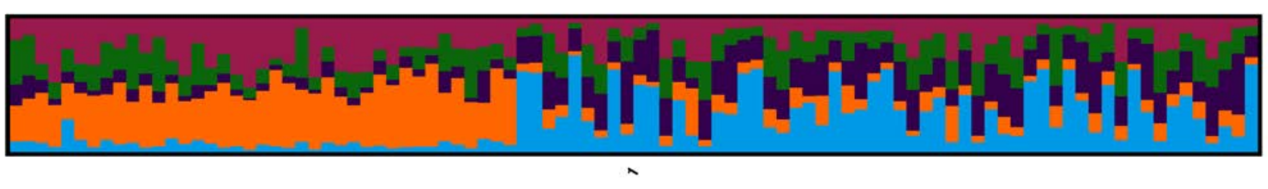

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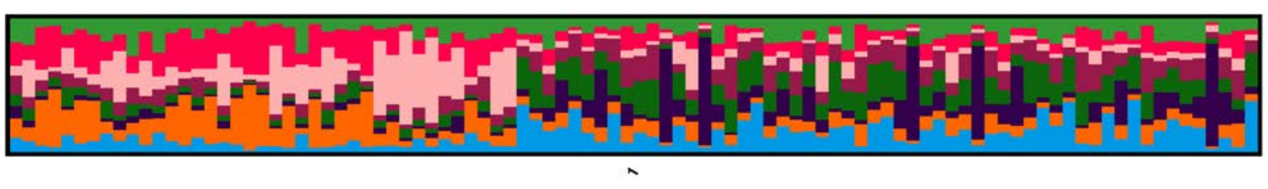
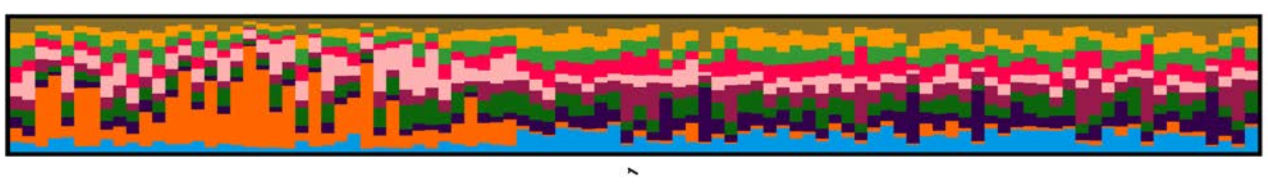


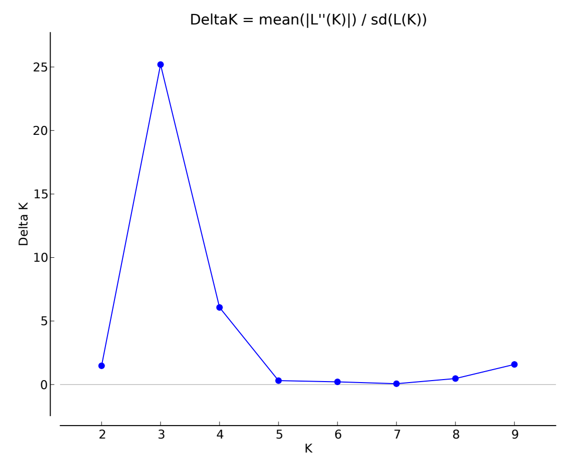
K=10 MinorCluster1

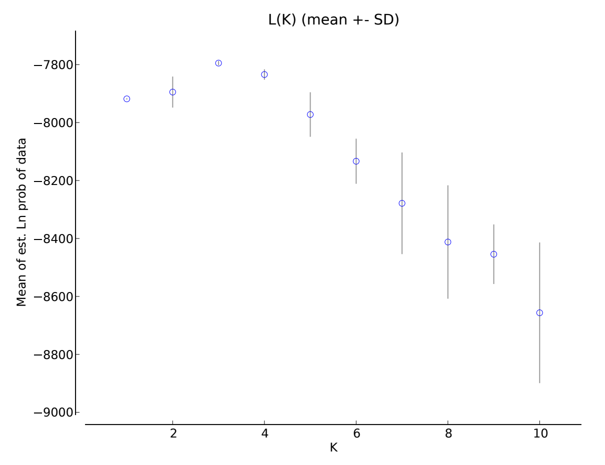


## Division of runs by mode:

K=1 10/10
K=2 10/10
K=3 9/10, 1/10
K=4 4/10, 4/10, 2/10
K=5 9/10, 1/10
K=6 10/10
K=7 10/10
K=8 9/10, 1/10
K=9 10/10
K=10 7/10, 3/10

# STRUCTURE results for *P. satyrus*



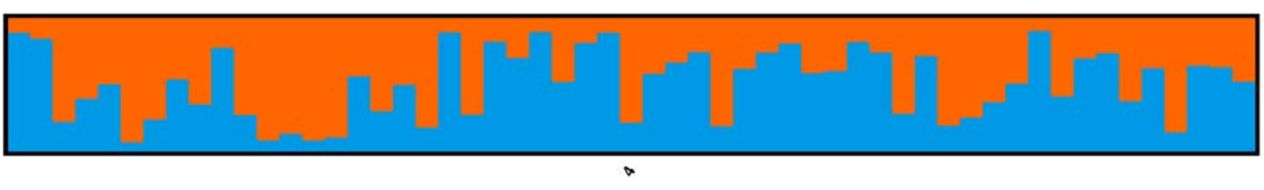




## Major modes for the uploaded data:

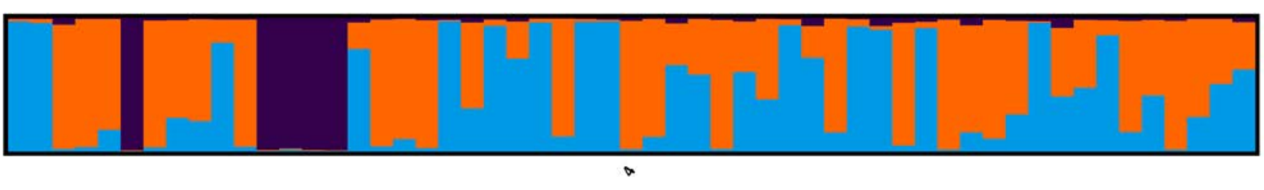
**K=1**

*P. satyrus*

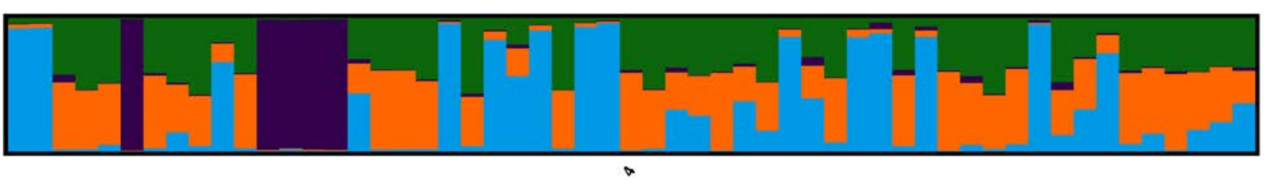
**K=2**



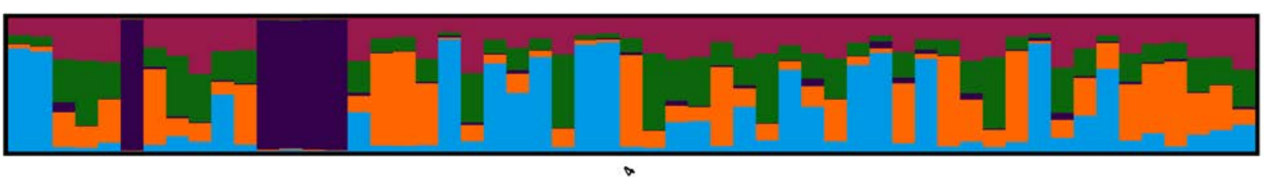
**K=3**



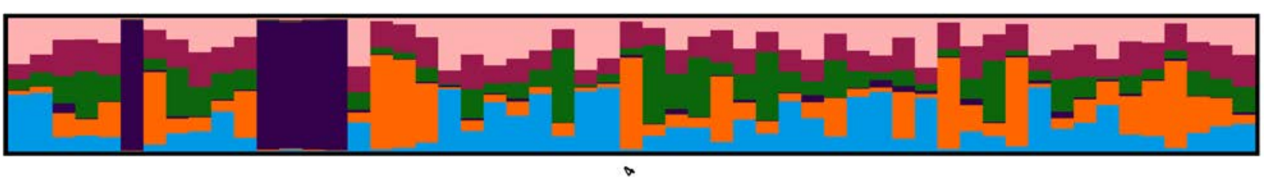
**K=4**



**K=5**

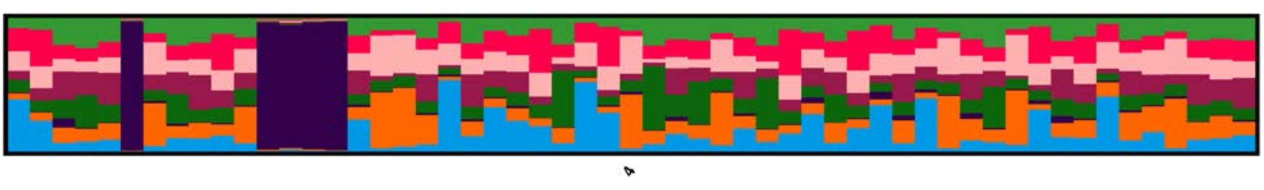


**K=6**

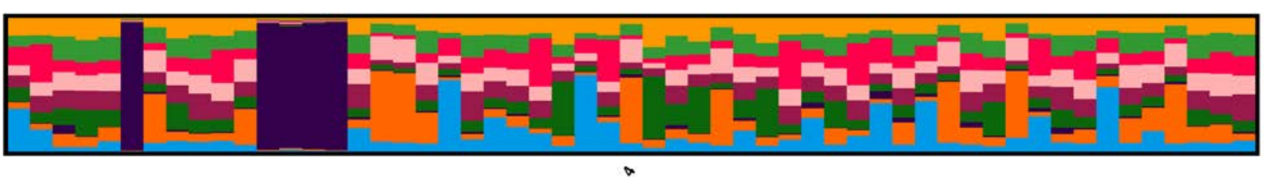


**K=7**

**K=8**



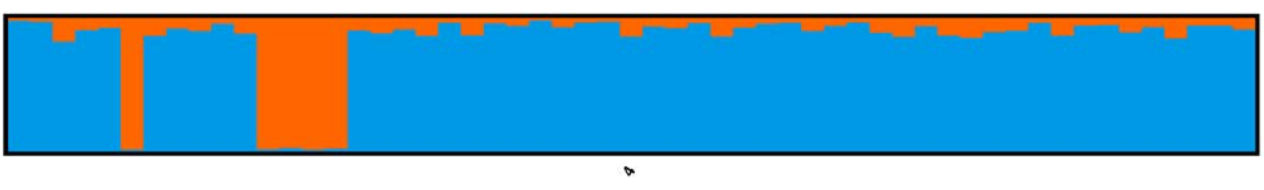
**K=9**



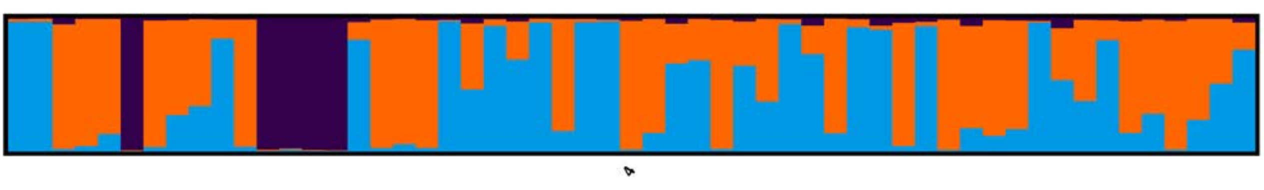
**K=10**

## Minor modes for the uploaded data:

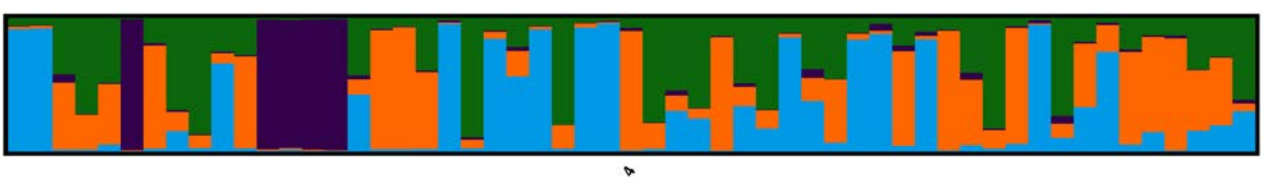
**K=2 MinorCluster1**



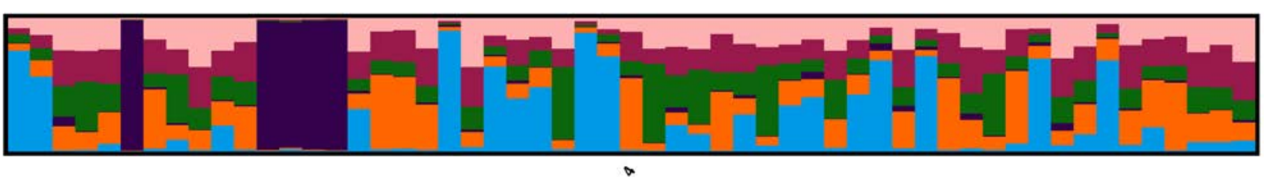
**K=3 MinorCluster1**



**K=4 MinorCluster1**



**K=6 MinorCluster1**

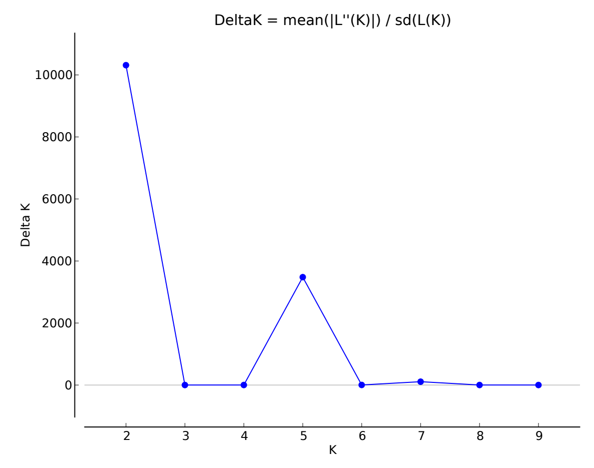


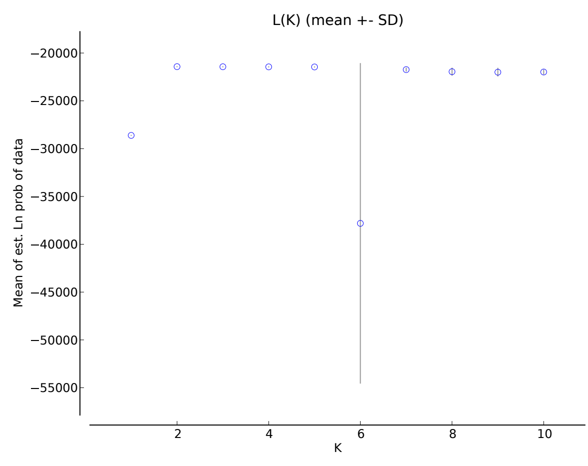
**K=10 MinorCluster1**

## Division of runs by mode:

| | |
|---|---|
| K=1 | 10/10 |
| K=2 | 7/10, 3/10 |
| K=3 | 6/10, 4/10 |
| K=4 | 5/10, 5/10 |
| K=5 | 10/10 |
| K=6 | 6/10, 4/10 |
| K=7 | 10/10 |
| K=8 | 10/10 |
| K=9 | 10/10 |
| K=10 | 7/10, 3/10 |

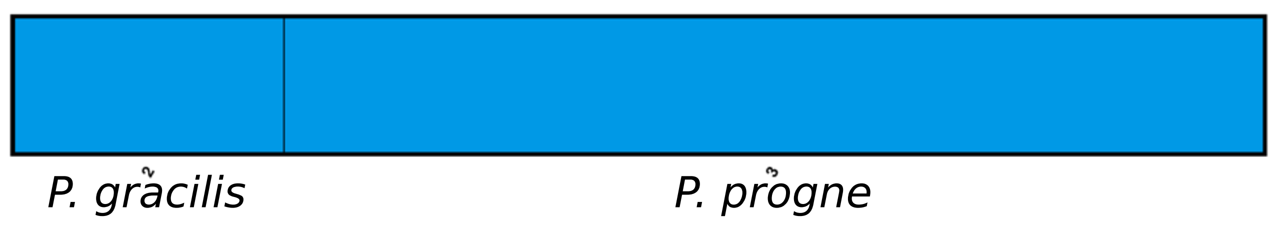
# STRUCTURE results for *P. gracilis* and *P. progne*




## Major modes for the uploaded data:

**K=1**




**K=2**

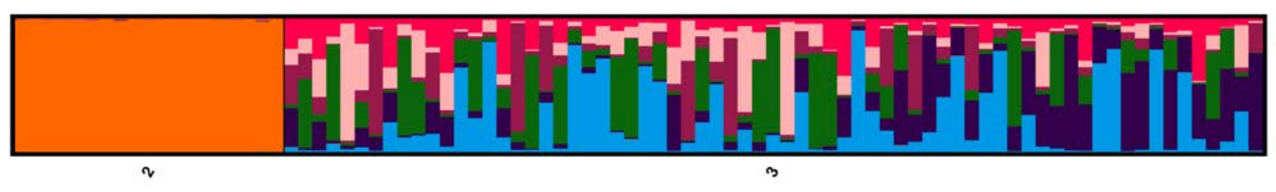
**K=3**

**K=4**

**K=5**

**K=6**

**K=7**



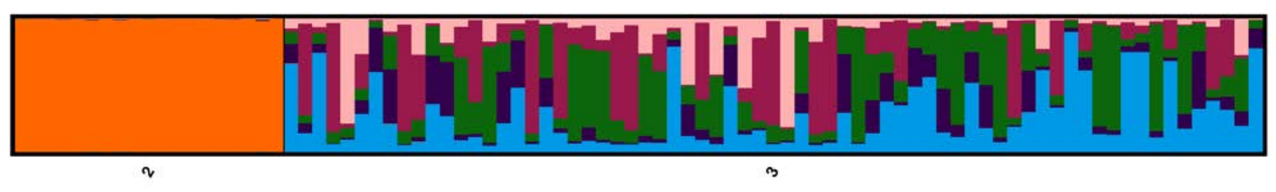
**K=8**

**K=9**

**K=10**

## Minor modes for the uploaded data:

**K=6 MinorCluster1**



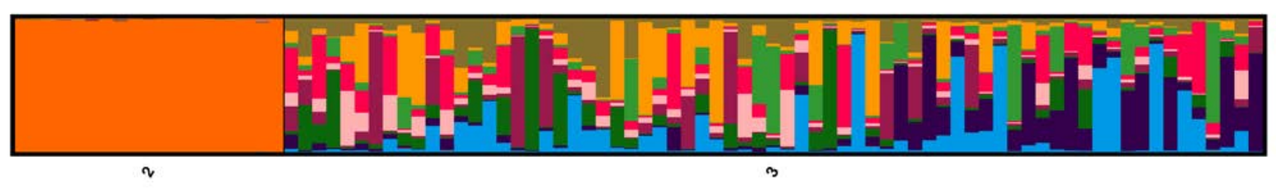
**K=7 MinorCluster1**

**K=7 MinorCluster2**

**K=8 MinorCluster1**

**K=9 MinorCluster1**

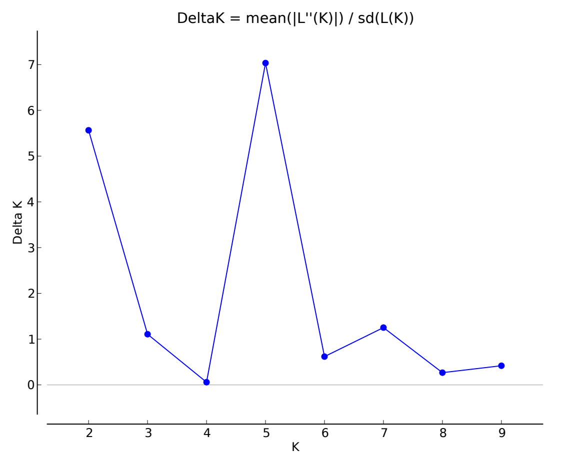
**K=10 MinorCluster1**

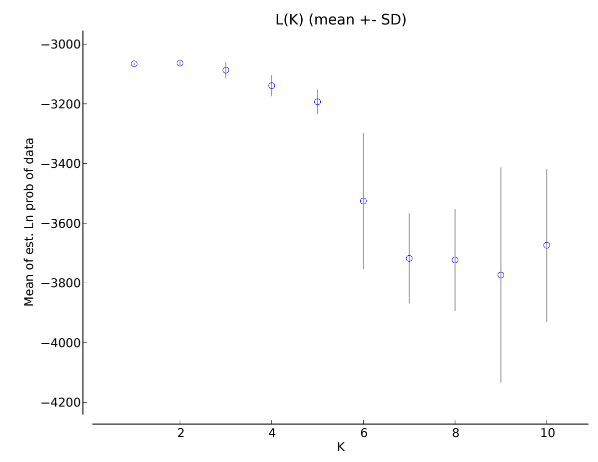


## Division of runs by mode:

**K=1 10/10**
**K=2 10/10**
**K=3 10/10**
**K=4 10/10**
**K=5 10/10**
**K=6 6/10, 4/10**
**K=7 5/10, 4/10, 1/10**
**K=8 7/10, 3/10**
**K=9 9/10, 1/10**
**K=10 8/10, 2/10**

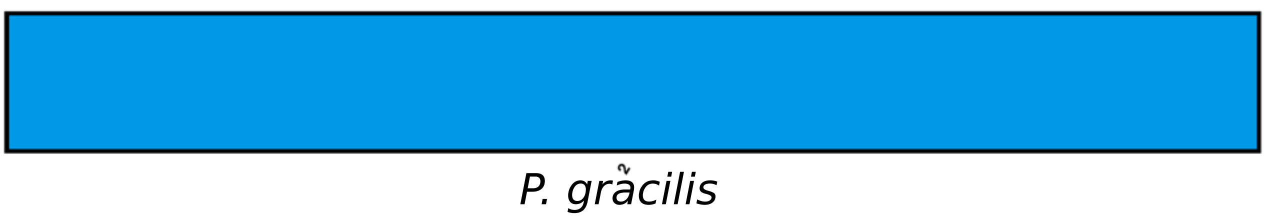
# STRUCTURE results for *P. gracilis*



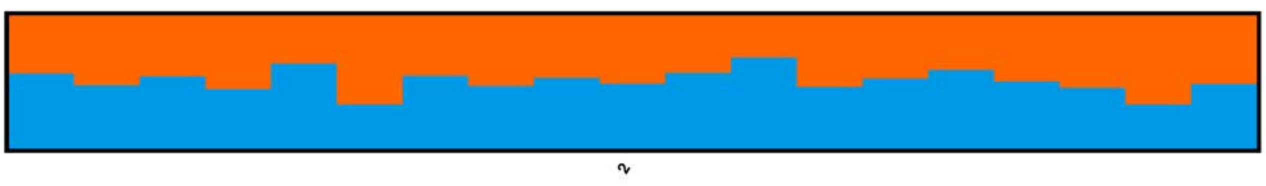




## Major modes for the uploaded data:

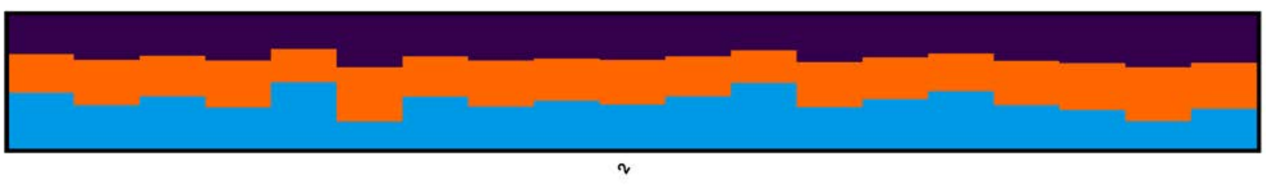
K=1



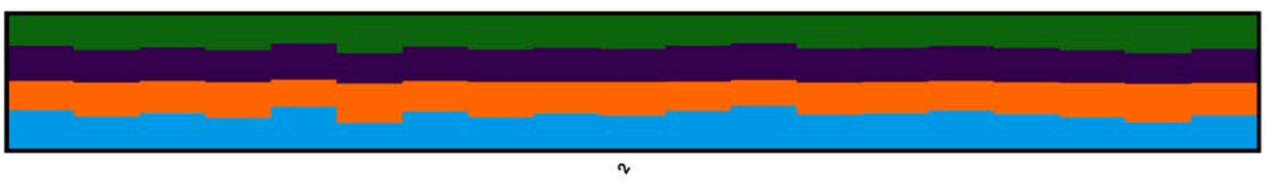

K=2



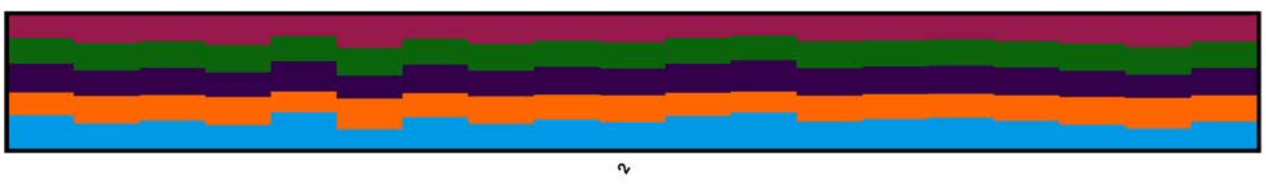
K=3



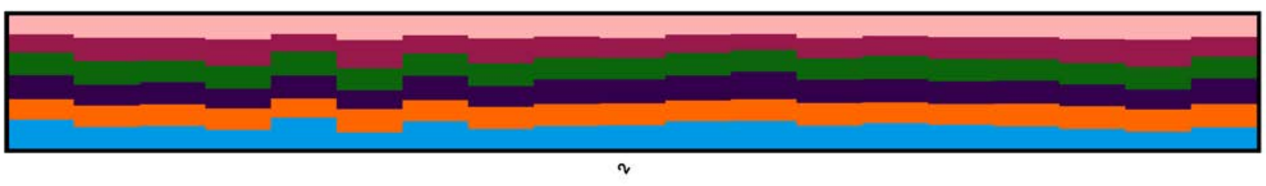
K=4



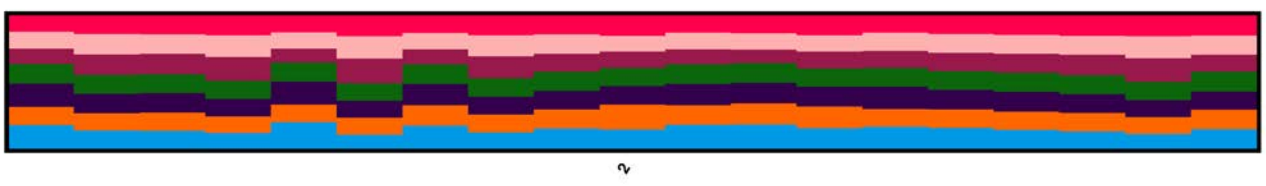
K=5



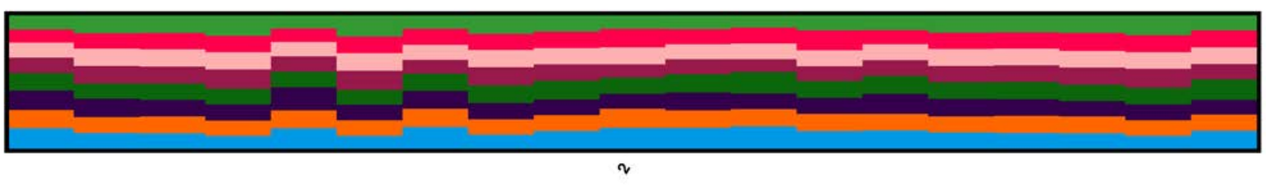
K=6



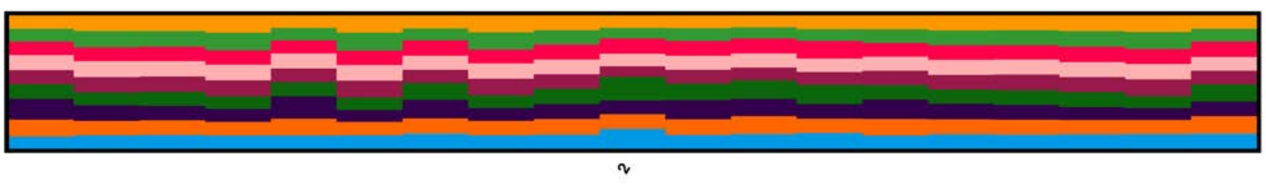
K=7



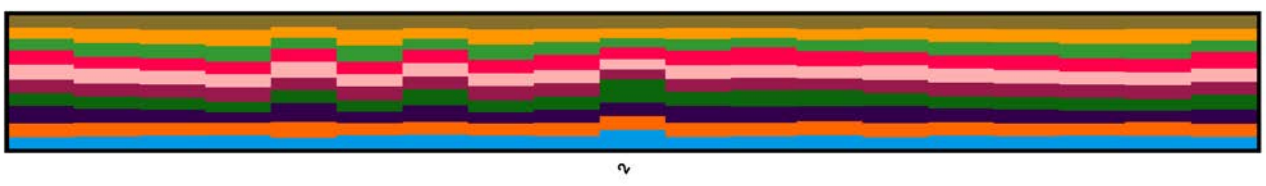
K=8



K=9

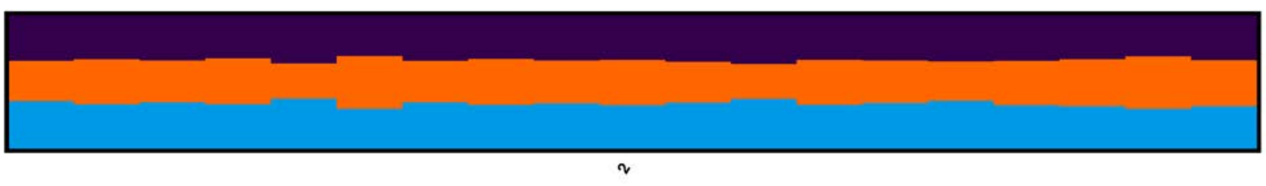


K=10

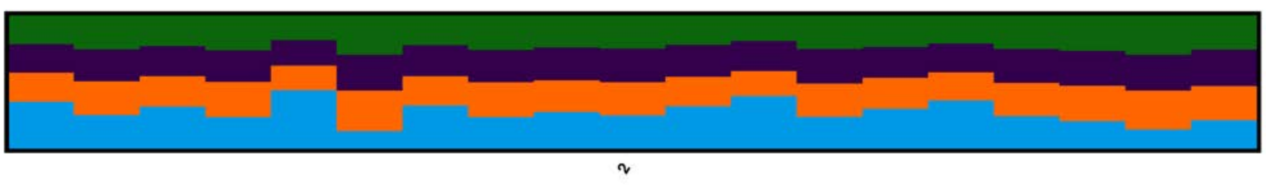


## Minor modes for the uploaded data:

K=3 MinorCluster1



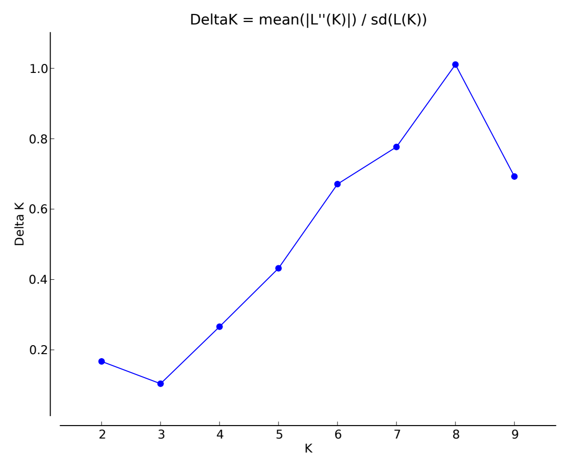
K=4 MinorCluster1

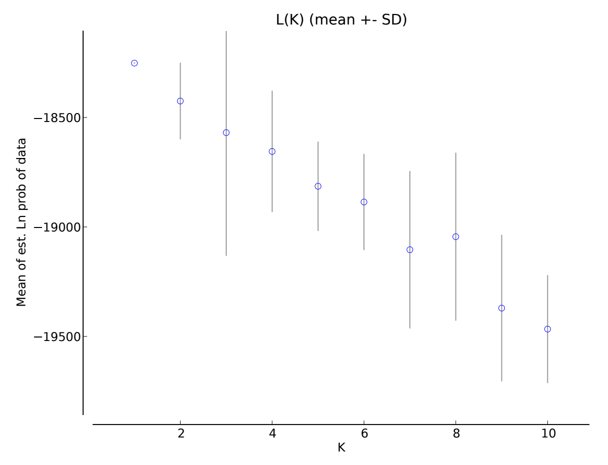


## Division of runs by mode:

| | |
|---|---|
| K=1 | 10/10 |
| K=2 | 10/10 |
| K=3 | 8/10, 2/10 |
| K=4 | 7/10, 3/10 |
| K=5 | 10/10 |
| K=6 | 10/10 |
| K=7 | 10/10 |
| K=8 | 10/10 |
| K=9 | 10/10 |
| K=10 | 10/10 |

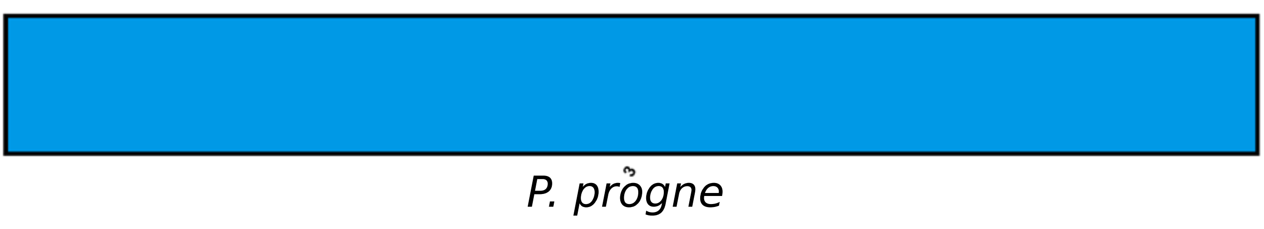
# STRUCTURE results for *P. progne*



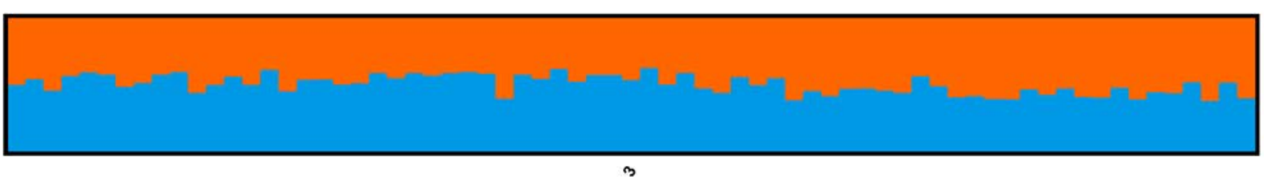




## Major modes for the uploaded data:

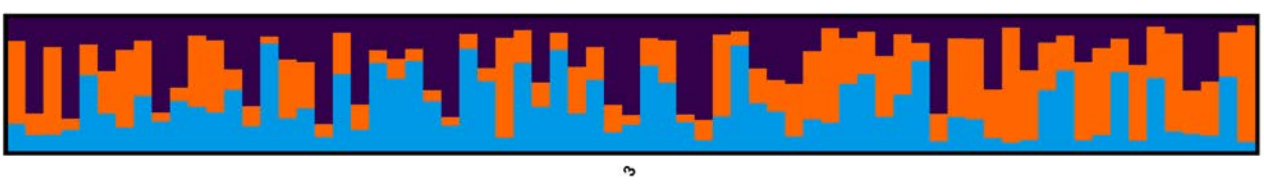
K=1



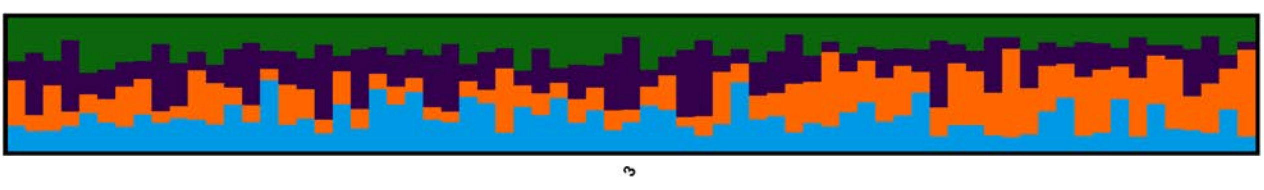

K=2



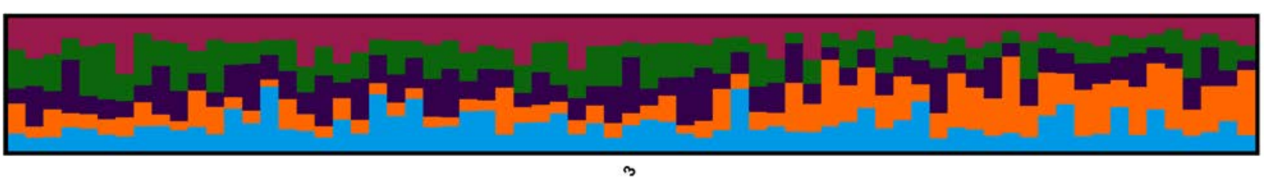
K=3



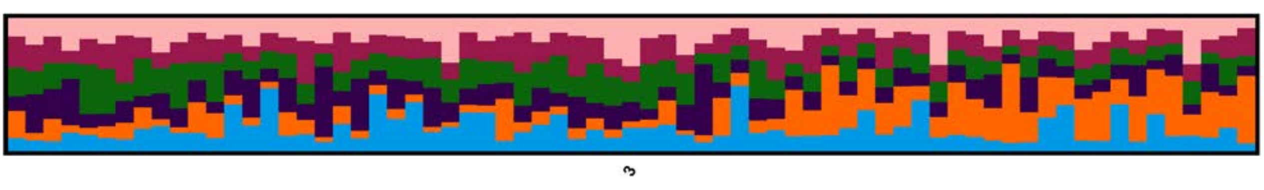
K=4



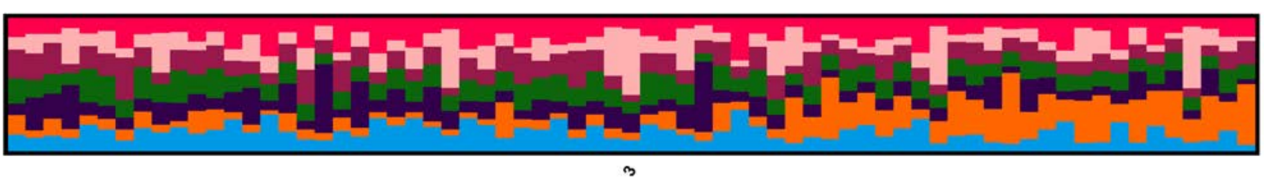
K=5



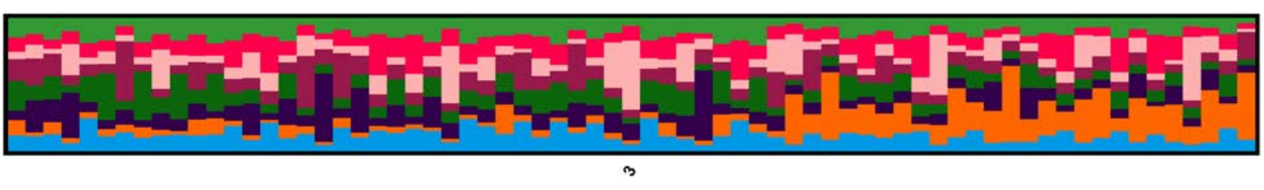
K=6



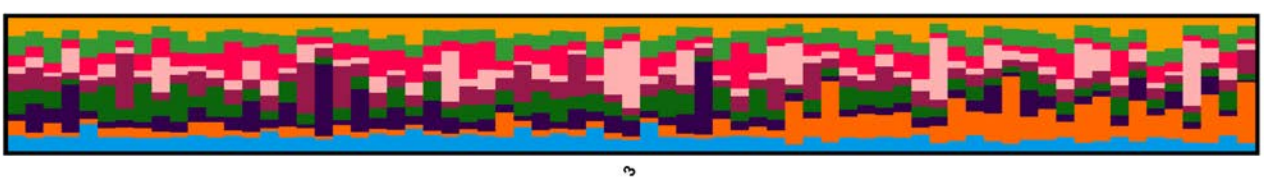
K=7



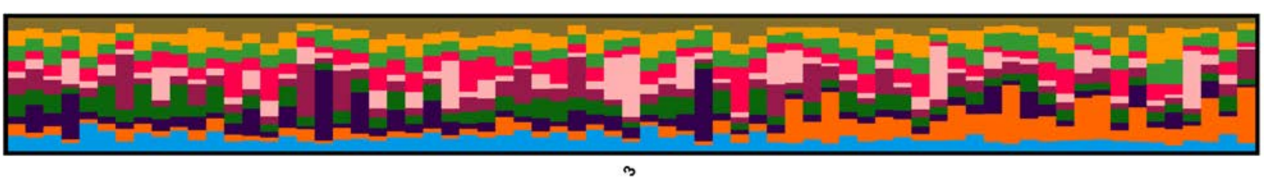
K=8



K=9

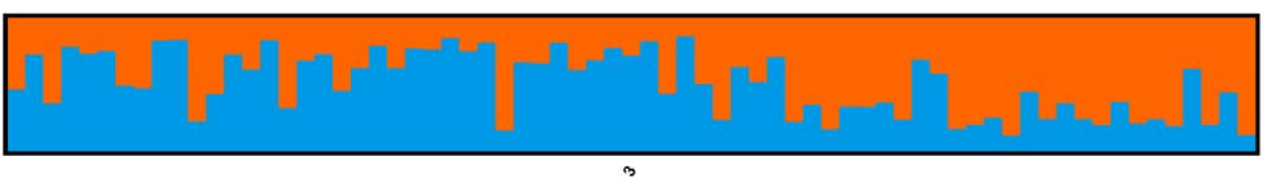


K=10

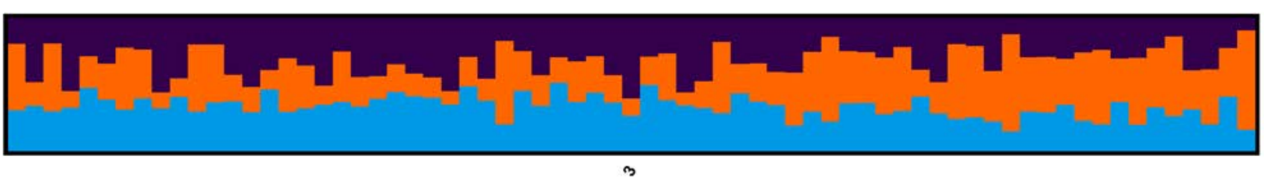


## Minor modes for the uploaded data:

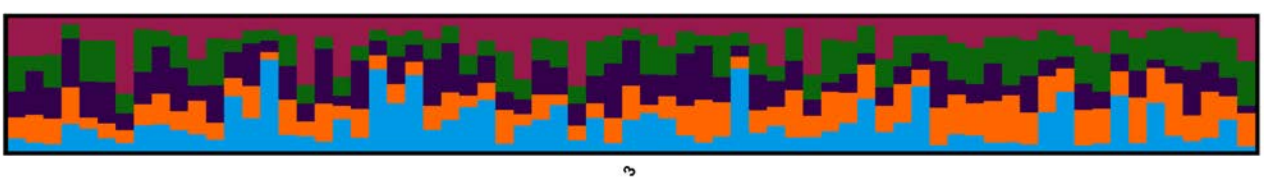
K=2 MinorCluster1



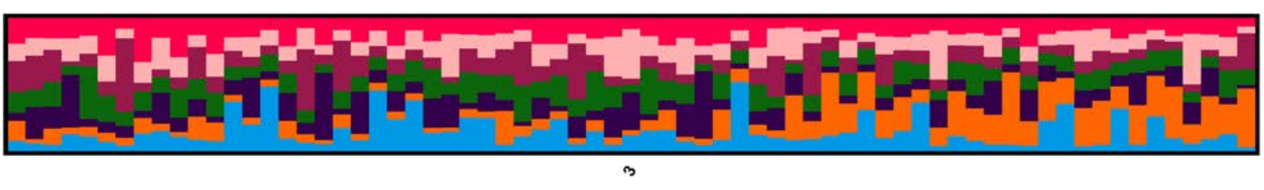
K=3 MinorCluster1



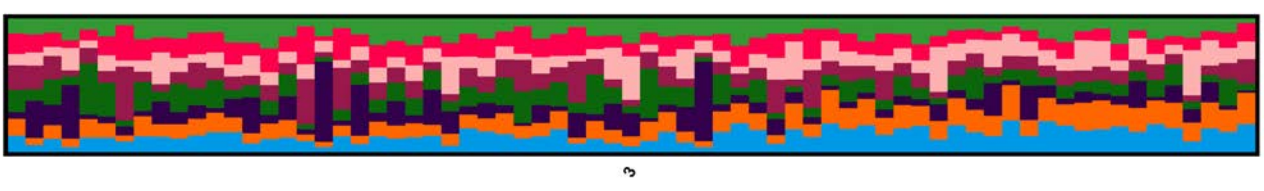
K=5 MinorCluster1



K=7 MinorCluster1



K=8 MinorCluster1



## Division of runs by mode:

| | |
|---|---|
| K=1 | 10/10 |
| K=2 | 8/10, 2/10 |
| K=3 | 7/10, 3/10 |
| K=4 | 10/10 |
| K=5 | 9/10, 1/10 |
| K=6 | 10/10 |
| K=7 | 7/10, 3/10 |
| K=8 | 7/10, 3/10 |
| K=9 | 10/10 |
| K=10 | 10/10 |

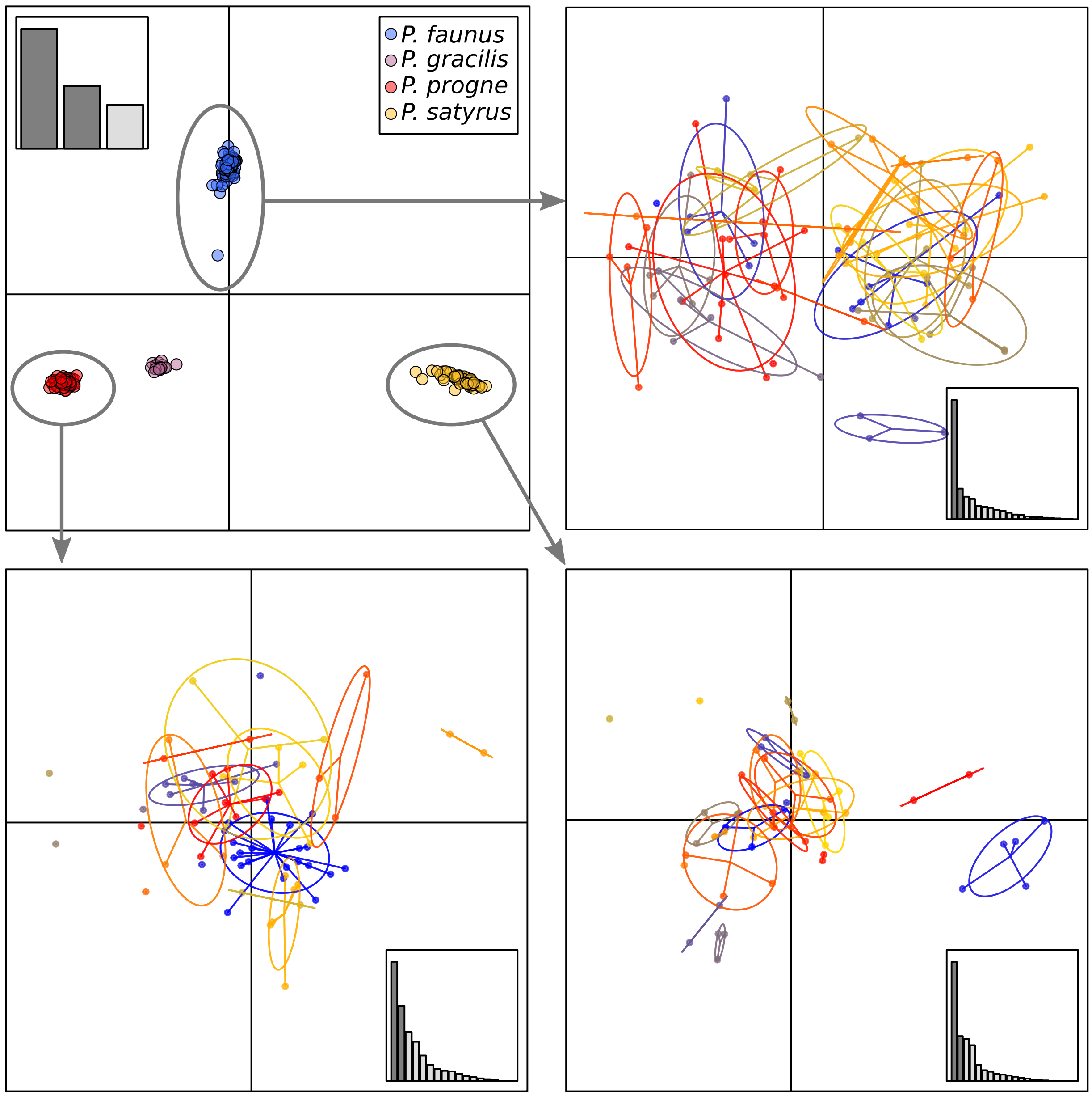



Figure S7. Discriminant analysis of principal components (DAPC) on 961 SNPs. Top left shows overall DAPC, and remaining panels are individual species colored by localities. *Polygonia gracilis* did not have adequate sampling across localities to analyze by itself. Insets show relative explanatory power of individual discriminant functions.

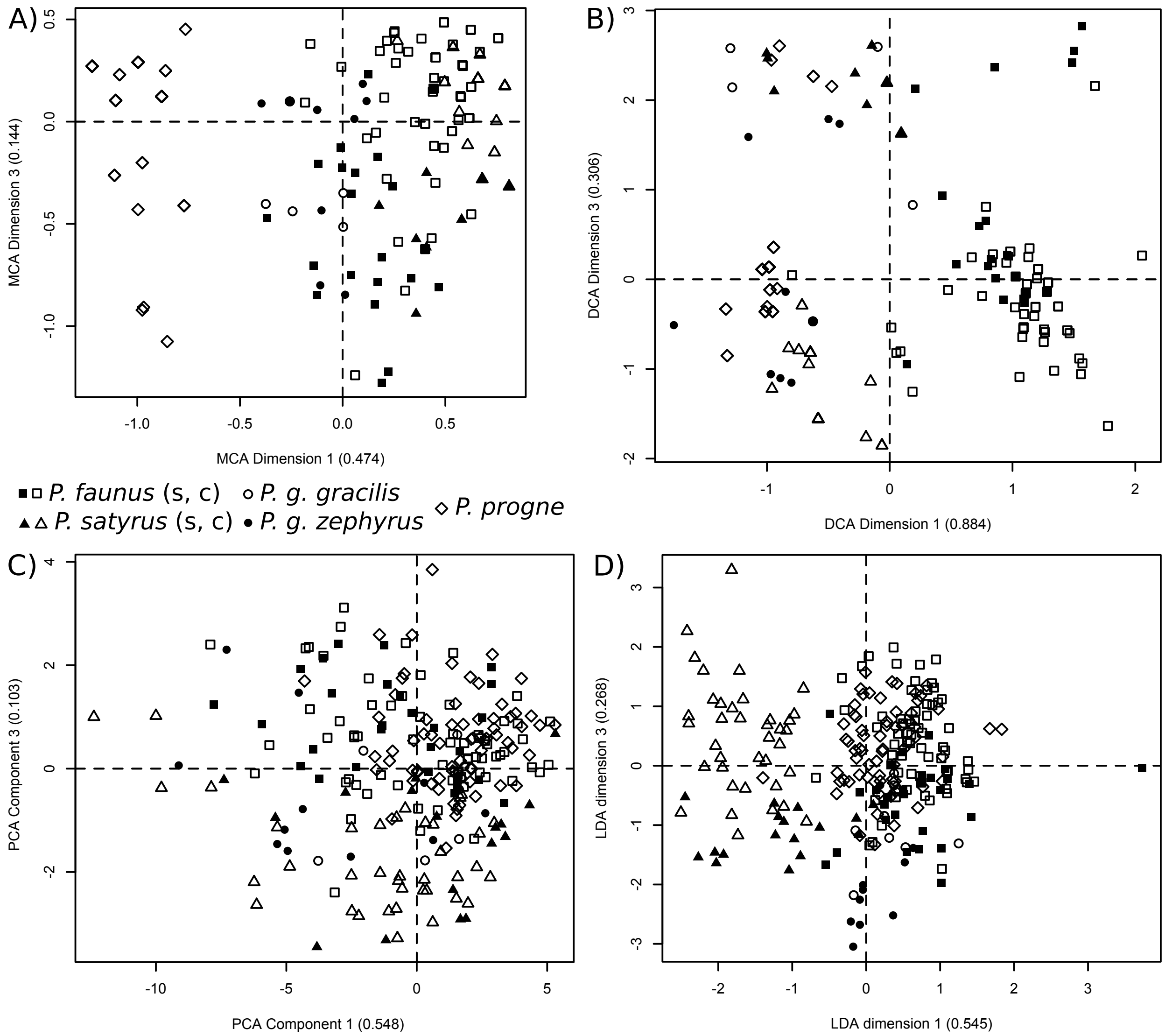



Figure S8. Multivariate analyses of wing characters, displaying third axes of each analysis. Multiple correspondence analysis (A) and discriminant correspondence analysis (B) of visually scored characters, and principal components analysis (C) and linear discriminant analysis (D) of RGB characters. Decimal values on axis labels indicate proportion of total variation explained (MCA, PCA) and the remaining proportion of variation (DCA, LDA). Species inclusion for each specimen was determined by membership in SNP clusters.

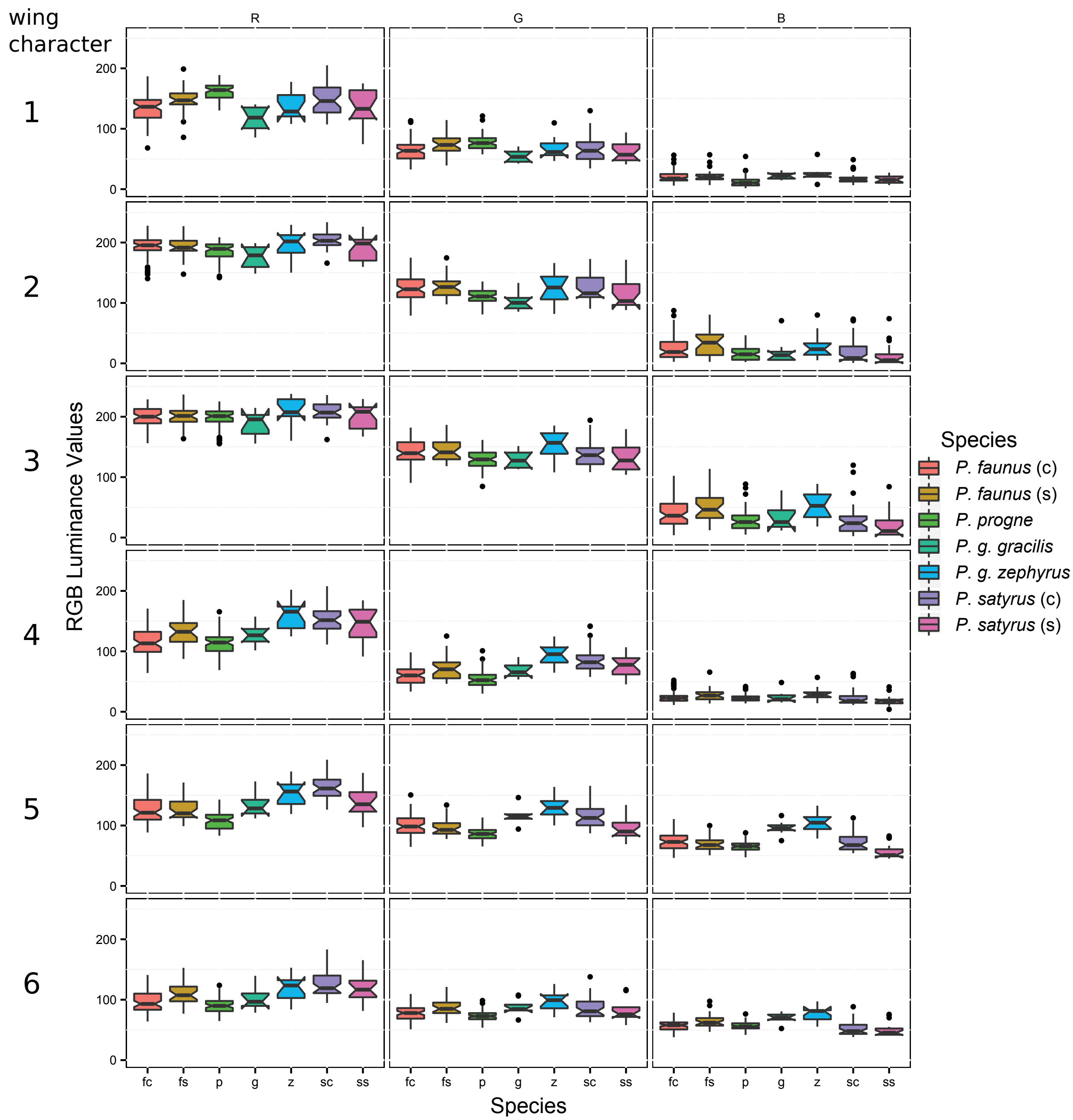



Figure S9. Central tendency statistics of mean R, G, and B values of 6 wing characters across 248 *Polygonia* butterflies. The upper and lower boundaries of each box are 3rd and 1st quartiles, respectively; vertical span of the notches correspond to the upper and lower 95% confidence bounds; the mean is represented by black lines bisecting each box. The upper and lower fences (whiskers) indicate the maximum and minimum values in the dataset, while outliers - those outside 1.5 times the interquartile range above the 4$^{th}$ quartile or below the first quartile - are shown as black points.

Table S3. Summary of pairwise luminance mean values with non-overlapping confidence intervals in pairwise comparisons.

| Species | *P. faunus* (c) | *P. faunus* (s) | *P. progne* | *P. g. gracilis* | *P. g. zephyrus* | *P. satyrus* (c) | *P. satyrus* (s) |
|---|---|---|---|---|---|---|---|
| *P. faunus* (c) | **26** | | | | | | |
| *P. faunus* (s) | 2 | **24** | | | | | |
| *P. progne* | 7 | 4 | **40** | | | | |
| *P. g. gracilis* | 3 | 5 | 7 | **24** | | | |
| *P. g. zephyrus* | 7 | 6 | 11 | 2 | **37** | | |
| *P. satyrus* (c) | 4 | 4 | 6 | 5 | 4 | **26** | |
| *P. satyrus* (s) | 3 | 3 | 5 | 2 | 7 | 3 | **23** |